Health and illness in the community

(HIC - 211)

Handout

Community medicine and public health department

Level : 2

2025-2026

Prepared

By

Prof. Dr. Fouad Metry Atia

Dr. Tasneem Mohammed Ali

Overall aim

This block aims to provide students with foundations of health and illness in the community; understanding of health and its determinants, comprehensive care through primary health care approach, the principles of epidemiology, management and communication. The course also emphasizes the primary health domains including: Occupational & environmental health, mother and child health, disease prevention & health promotion, and nutrition.

At the end of this block, the students will be able to:

By the end of the course, a student shall be able to:

1. Describe the concepts of health, Define health and disease &describe the spectrum of health
2. Understand the basic determinants of health and disease& Mention the three interacting ecological factors affecting the occurrence of disease
3. Recognize the economic, psychological, social, and cultural factors that interfere with wellbeing.
4. Explain theories of disease causation & Identify levels of prevention
5. Explain different methods for prevention and control in general and for some communicable and non-communicable diseases.
6. Describe the factors that influence the size and composition of the population and demographic transition
7. Describe what is epidemiology and its uses and the meaning of epidemic, endemic and pandemic
8. Identify the epidemiology of common communicable diseases
9. Describe Epidemiology of selected non-communicable diseases
10. Describe concepts of screening and evaluation of a screening test
11. Understand the concept of occupational health and health hazards
12. Describe preventive program for occupational health
13. Recognize the impact of environment on health
14. Identify the nature, health effect, sources environmental risks & preventive measures.
15. Define nutrition and its commonly used term, understand the role of nutrition in health & identify nutrient sources, functions, requirements.

16. Recognize the malnutrition problems
17. List the methods of nutritional assessment of individuals and populations
18. Describe the major components and objectives of a comprehensive school health program
19. Identify health problems and health needs of adolescent age group.
20. Describe the complementary role of maternal and child health describe the MCH program.
21. Define fundamental terms of communication and health education
22. Know methods of health education, identification of behavior related health problems, identification of appropriate health messages, steps in planning and evaluation of health education program
23. Identify the communication process and barriers of communication
24. Identify management functions: planning, implementation and evaluation
25. Define the principles of leadership and team building
26. Identify the characteristics of effective teams & describe team process
27. Define PHC, List the elements of PHC
28. Identify the different types of health systems in Egypt
29. Recognize & assess quality of health care.
30. Apply preventive and control measures for communicable and non communicable diseases of special public health concern.
31. Draw Egypt's population pyramid and interpret the information obtained from the pyramid
32. Calculate and interpret vital rates including fertility
33. Calculate and interpret measures of morbidity and mortality
34. Investigate of an epidemic.
35. Calculate sensitivity, specificity, and predictive values.
36. Calculate relevant vital indices, and identify health problems related to the health of mothers and children as well as risk factors
37. Plan a session for health education
38. Carry out a health education session and/or conduct a counseling session and assess performance (self/ peer) & apply principles of communication skills in dealing with different situations.
39. Apply Certifications of cause of death and Classification of diseases

40. Apply management programs in different health care services for improving its performance & outcome indicators
41. Apply different quality tools in problem solving and decision making.
42. Apply methods of assessment of nutritional status
43. Communicate effectively with clients and colleagues.
44. Do appropriate counseling, health education and communication strategies in different health care settings using behavioral change models.
45. Show leadership and act as a team manager in motivating the community served.
46. Collaborate with colleagues in a teamwork during field visits, class discussion as well as solving problem

Contents

TOPIC	PAGE
Concept of health	1
Concept of disease	5
Concept of Epidemiology and infectious cycle	9
Concept of prevention	20
Host susceptibility, immunity and vaccination	25
Meningitis	32
Leprosy	37
Tuberculosis	42
Epidemiology of water and food borne diseases	51
Typhoid	53
Human Immunodeficiency Virus (HIV)	58
Viral hepatitis	63
Tetanus	73
Epidemiology of Zoonosis (rabies)	78
Healthcare associated infection	83
General epidemiology of non-communicable diseases	88
Coronary heart disease	93
Diabetes mellitus	96
Cancer	101
Primary health care	108
Basic Nutrition Concepts	113
Malnutrition and some nutritional disorders	119
Assessment of Nutritional Status	125
Environmental health	128
School health	140
Adolescent Health	146
Communication and health education	153
Occupational health	157
Health Care System in Egypt	173
Maternal and child health (MCH)	178
Management	184

Concept of health

By the end of the lecture the student will be able to:

Define health and describe dimension of health.

Recognize the spectrum of health.

Define well-being, describe objective and subjective components of the wellbeing and define quality of life.

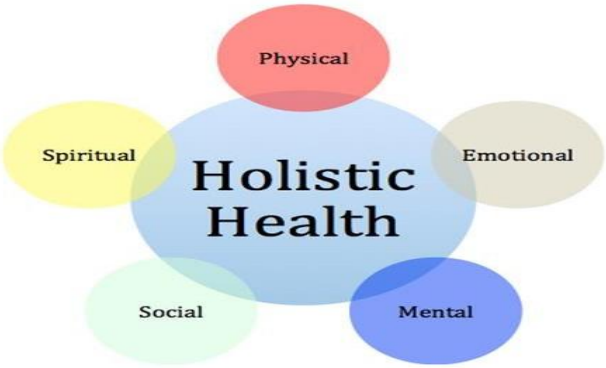
Describe the determinants of health on the individual, the family, and the community levels.

Definition of health:

WHO definition of health: "a state of complete physical, mental and social well- being and not merely the absence of disease or Infirmity"

Dimensions of health:

As per the definition of health , health is three dimensional—the physical, the mental and the social. Non-medical dimensions which can be included are spiritual, emotional, vocational and political dimensions.



Physical dimension:

A person is said to be physically healthy, when all the organs and systems in the body are functioning perfectly at their optimum capacity.

All the systems function normally.

The physical health of an individual can be assessed by the history of having taken treatment or hospitalization for any illness, thorough clinical examination, anthropometric measurements, biochemical and laboratory investigations.

Mental dimension:

Mental health is not merely an absence of mental illness.

Ability to think clearly and coherently. This deals with sound socialization in communities.

Mental health is a state of balance between the individual and the surrounding world, a state of harmony between oneself and others,
The mental health of an individual can be assessed by his behavior, and attitude.

Social dimension:

It refers the ability to make and maintain relationships with other people or communities.
It states that harmony and integration within and between each individuals and other members of the society.
Social dimension of health includes the level of social skills one possesses, social functioning and the ability to see oneself as a member of a larger society**.**

Spiritual dimension:

This is a holistic dimension .
A person is said to be spiritually healthy, when he possesses 'Sound mind in a sound body', with the knowledge of philosophy, leading a simple life with a very high level of thinking .
Thus, spiritual is 'something' which transcends physiology and psychology.

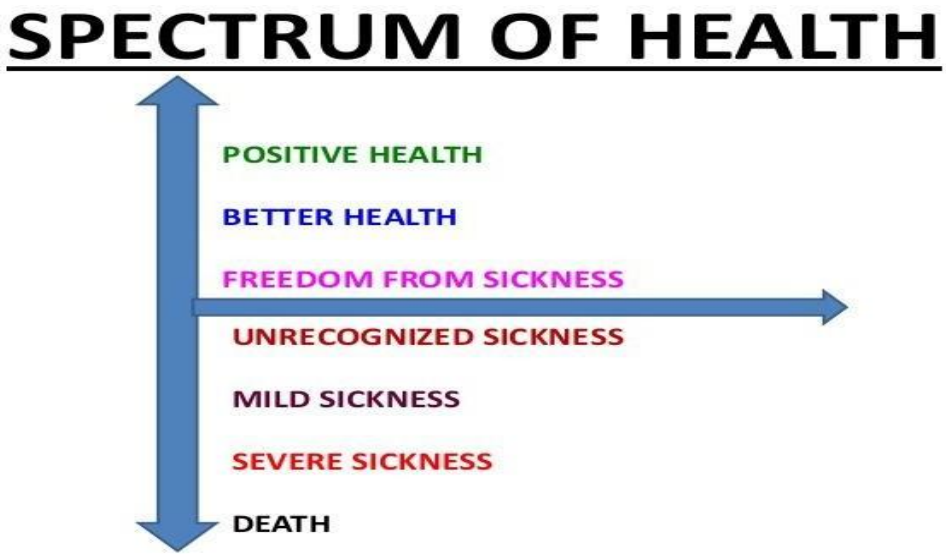
Emotional dimension:

This is rather difficult to differentiate from mental health.
However, a person is said to be emotionally healthy, when he does not loose temper or does not develop tension and has self-control.

Vocational dimension

An individual is said to be healthy vocationally, when he is capable of earning sufficiently to lead the life successfully

Spectrum of health:



Health and disease lie along a continuum & there is no single cut of point. The lowest point on health-disease spectrum is death and the highest composes the WHO definition of positive health.

Health fluctuates within range of optimal wellbeing to various level of dysfunction. The transition from optimal health to illness is often gradual & where one state ends other states begins.

The spectrum of health emphasizes that individual health state is not static, it's a dynamic phenomenon & a process of continuous change.

Concept of wellbeing:

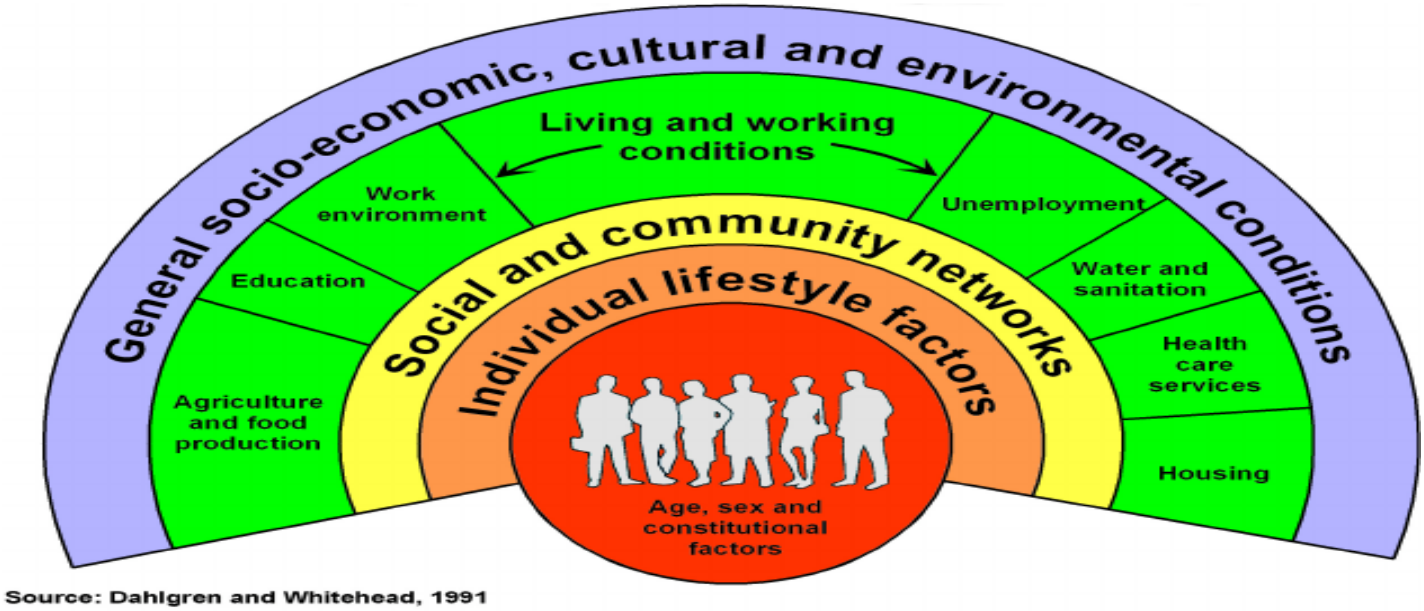
Well-being includes the presence of positive emotions and moods (e.g., contentment, happiness), the absence of negative emotions (e.g., depression, anxiety), satisfaction with life, fulfillment and positive functioning. In simple terms, well-being can be described as judging life positively and feeling good.

Different aspects of well-being that include the following:

- Physical well-being.
- Economic well-being.
- Social well-being.
- Development and activity.
- Emotional well-being.

- Psychological well-being.
- Life satisfaction.
- Domain specific satisfaction.
- Engaging activities and work.

Determinants of health:



Source: Dahlgren and Whitehead, 1991

The determinants of health include

- ***The social and economic environment***
- ***The physical environment***
- ***An individual person's characteristics and behaviors***

The context of people's lives determines their health. Therefore, blaming individuals for having poor health or crediting them for having good health is inappropriate. Individuals are unlikely to be able to directly control many of the determinants of health.

These determinants include:

- **Income and social status –** Higher income and social status are linked to better health. The greater the gap between the richest and poorest people, the greater the differences in health.
- **Education –** Low education levels are linked with poor health, more stress, and lower self- confidence**.**

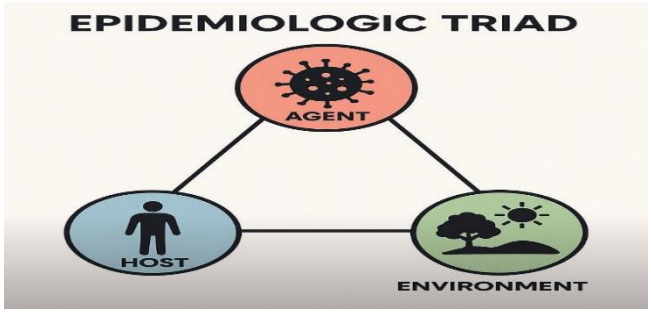
- **Physical environment –** Safe water and clean air, healthy workplaces, and safe houses, communities, and roads all contribute to good health.
- **Employment and working conditions –** People who are employed are healthier, particularly those who have more control over their working conditions.
- **Social support networks –** Greater support from families, friends, and communities is linked to better health.
- **Culture –** Customs, traditions, and the beliefs of the family and community all affect health.
- **Genetics –** Inheritance plays a part in determining lifespan, healthiness, and the likelihood of developing certain illnesses.
- **Personal behavior and coping skills—**Healthful habits, such as balanced eating and physical activity; harmful habits, such as smoking and excessive drinking; and how life's stresses and challenges are handled all affect health.
- **Health services –** Access and use of services that prevent and treat disease influences health.
- **Gender –** Men and women experience different types of diseases at different ages.

Concept of disease

Dis = opposite, Ease = comfort; health

Webster defines disease as 'A condition, in which body health is impaired and performance of vital functions in the body is interrupted.' In other words, disease is a physiological or psychological dysfunction of the body. Illness is a subjective feeling of not being well. Sickness is a state of social dysfunction (i.e. inability to perform his 'Social role').

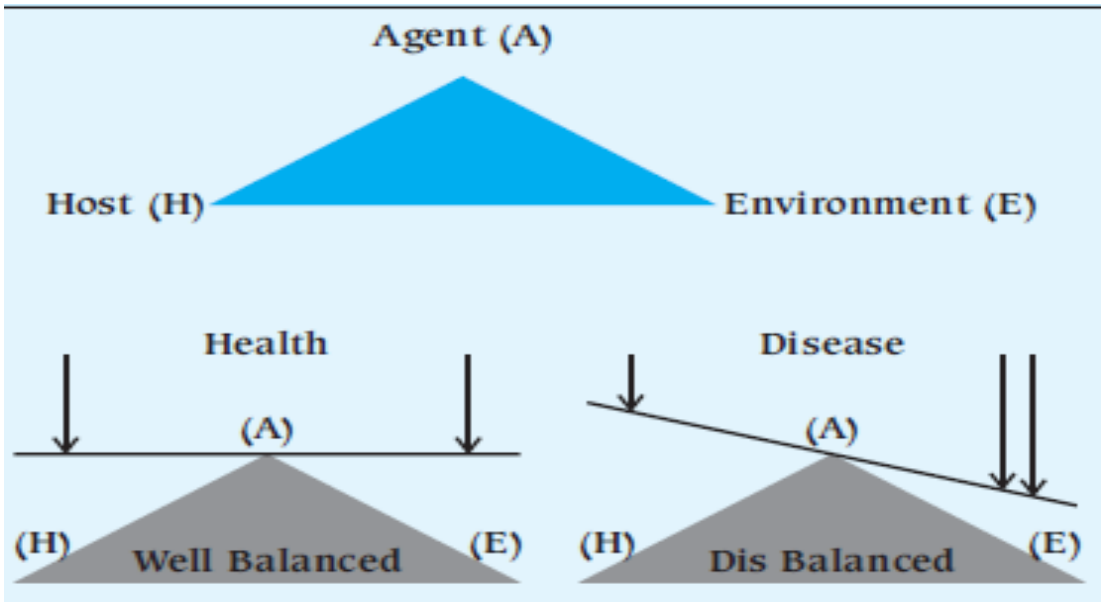
Epidemiological triad:



The epidemiological triad theory hypothesizes that there are 3 important determinants of the state of health or disease in a human being, namely:

- **The agent factors:** Related to the various characteristics of the "agent" which causes the disease.

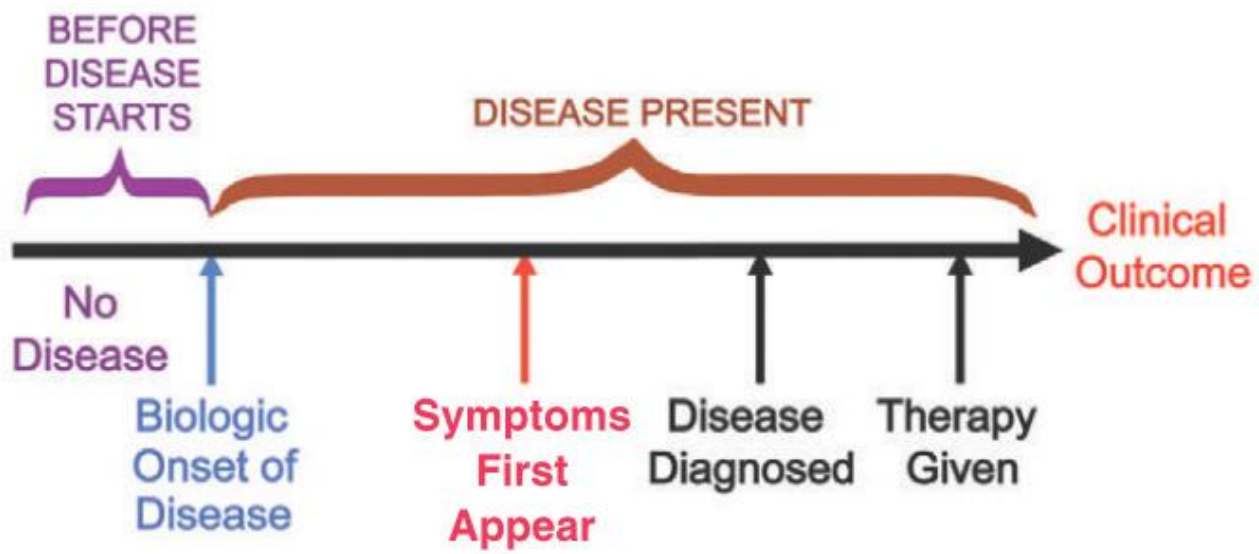
- **The host factors:** Which relate to various characteristics of the human being himself.
- **Environmental factors:** Which relate to the various characteristics of the environment in which the human being is living.



Natural history of a disease or health problem:

- **No disease**
- **Preclinical phase** (biological onset of disease = pathological phase)
- **Clinical phase**
  - Symptoms (individual feeling)
  - Signs (physician diagnoses)
  - Treatment (physician recommends treatment, such as drugs, surgery, or other therapies)
- **Outcome phase**
  - Complete recovery (cure)
  - Recovery, but with remaining pathology or handicap
  - Death

The process of disease progress is shown in the following diagram.



Stages of disease:

- The five periods of disease (sometimes referred to as stages or phases) include
  - the incubation period,
  - Prodromal stage,
  - illness,
  - decline, and
  - convalescence periods.

The incubation period

The incubation period is the period between the successful entrance of the organism and the onset of the first symptom is called 'Incubation period. Incubation periods can vary from a day or two in acute disease to months or years in chronic disease, depending upon the pathogen.

Factors involved in determining the length of the incubation period are diverse, and can include

- strength of the pathogen,
- strength of the host immune defenses,
- site of infection,
- type of infection, and
- the size infectious dose received.

During this incubation period, the patient is unaware that a disease is beginning to develop.

The prodromal stage

The prodromal period occurs after the incubation period. During this phase, the pathogen continues to multiply and the host begins to experience general signs and symptoms of illness, which typically result from activation of the immune system, such as fever, pain, soreness, swelling, or inflammation. Usually, such signs and symptoms are too general to indicate a particular disease

Illness

Following the prodromal period is the period of illness, during which the signs and symptoms of disease are most obvious and severe.

Decline

The period of illness is followed by the period of decline, during which the number of pathogen particles begins to decrease, and the signs and symptoms of illness begin to decline. However, during the decline period, patients may become susceptible to developing secondary infections because their immune systems have been weakened by the primary infection.

Convalescence period

The final period is known as the period of convalescence. During this stage, the patient generally returns to normal functions, although some diseases may inflict permanent damage that the body cannot fully repair. Infectious diseases can be contagious during all five of the periods of disease.

Which periods of disease are more likely to associated with transmissibility of an infection depends upon the disease, the pathogen, and the mechanisms by which the disease develops and progresses.

Concept of Epidemiology

By the end of the lecture the student will be able to :

- Define infectious, contagious, communicable disease .
- Define common terms used in field of epidemiology .
- Identify components of infection cycle.

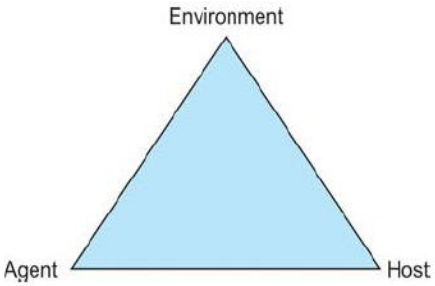
Epidemiology is a scientific study of factors and conditions related to disease as they occur in people. The word epidemiology is derived from epi (in, on,upon); demos (people) and logos (science).

Definition of Epidemiology

Epidemiology has been defined as, 'The study of **distribution** and **determinants** of health-related states or **events** in specified **populations** and application of this study to the control of health problems' (John M Last, 1988).

The Epidemiological Triad

The occurrence and manifestations of any disease, whether communicable or noncommunicable, are determined by the interactions between the agent, the host and the environment, which together constitute the epidemiological triad.



- **Agent factors**: disease agents may be classified into:
  - Biological: e.g. bacteria, viruses, rickettsia
  - Nutritional: e.g. increase of decrease of nutrients
  - Chemical: e.g. toxins, dyes, CO
  - Physical: e.g. radiation, heat
  - Mechanical e.g. moving parts of machinery.
- **Host factors:** These may be classified into:
  - Immunity and resistance
  - Personal habits as smoking, drinking
  - Age sex, race

- **Environmental factors:** These may be classified into:
    - Physical as weather, geography, geology
    - Socio-economic as income, occupation, education
    - Cultural: e.g. traditions, beliefs, behaviors
    - Biological factors: micro and macro organisms

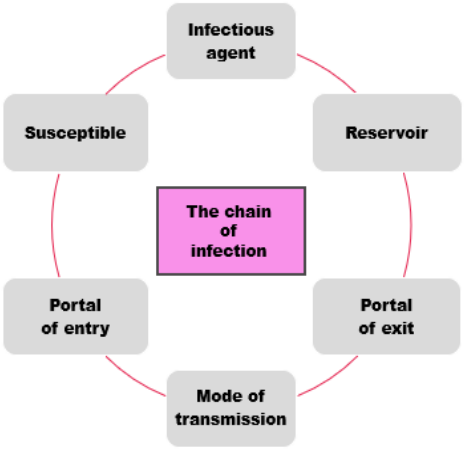
Level of Disease in the Community

- **Sporadic** refers to a disease that occurs infrequently and irregularly.
- **Endemic** refers to the constant presence and/or usual prevalence of a disease or infectious agent in a population within a geographic area.
- **Epidemic** refers to an increase, often sudden, in the number of cases of a disease above what is normally expected in that population in that area.
- **Outbreak** carries the same definition of epidemic but is often used for a more limited geographic area.
- **Pandemic** refers to an epidemic that has spread over several countries or continents, usually affecting a large number of people.

Important definitions

- **Contamination**: Presence of infectious organisms on inanimate objects (clothes, utensils, furniture, vehicles of transmission like water, milk, food, blood), making them impure.
- **Infection**: Successful entry, development, and/or multiplication of pathogenic organisms in a living host (human or animal). May or may not lead to disease.
- **Infectious Disease:** A disease resulting directly from infection.
- **Contagious Disease**: A disease spread person-to-person via close physical contact (e.g., scabies, STDs, ophthalmia neonatorum).
- **Communicable Disease:** Illness due to an infectious agent or its toxic products, transmitted directly or indirectly from a reservoir to a susceptible host.
- **Zoonotic Disease (Zoonosis):** A disease naturally transmitted between vertebrate animals and humans, in either direction.

Chain of Infection (infectious cycle)



Transmission occurs when the **agent** leaves its **reservoir** or host through a **portal of exit**, is conveyed by some **mode of transmission**, and enters through an appropriate **portal of entry** to infect a **susceptible host**. This sequence is sometimes called the chain of infection.

I. Microbiologic agent:

The microorganisms responsible for the causation of infectious diseases are classified into bacteria, viruses, rickettsia, fungi, protozoa, and Parasites.

- **Bacteria**:
  - **Cocci** e.g. pneumococci, meningococci, staphylococci, streptococci.
  - **Bacilli** e.g. salmonella typhi and para typhi, shigella, brucella, corynebacterium diphtheria and tubercle bacilli.
  - **Spirochaetes** e.g. Treponemas e.g. T. pallidum (Syphilis).
  - **Vibrio** e.g. Vibrio cholera and Vibrio El- Tor.
- **Viruses**: e.g. measles, mumps, chicken pox, influenza, common cold, poliomyelitis, smallpox, yellow fever, dengue fever, and herpes zoster.
- **Rickettsia**: Rickettsia are usually transmitted to man by biting insects
- **Protozoa**: e.g. Malaria& Amoebic dysentery.
- **Fungi**: e.g. Ring worm of the scalp and of the body.
- **Parasites**: e.g. Bilharziasis, Elephantiasis, and Ancylostomiasis.

II. RESERVOIRS AND SOURCES OF INFECTION:

Reservoir of infection is the one in whom the causative organism lives, multiplies and depends primarily for its survival (i.e. the natural habitat of the organism). Such a reservoir may be human beings, animal or soil.

Source of infection is a person, animal, arthropod, soil or the substance from which the organism directly enters the host. The terms reservoir and source are **not** always synonymous.

A **homologous** reservoir state is the one, in which both the reservoir and the susceptible person belong to the same species. For example, chickenpox, measles, poliomyelitis (Human-Human being). A **heterologous** reservoir state is the one, in which both the reservoir and the susceptible belong to the different species. For example, salmonellosis (Cattle-human being), Rabies (Dogs-human being), etc.

Reservoirs are of three types—Human, Animal and Soil.

Human Reservoir: May be **case or carrier**.

Case: A case is an infected person having clinical features of the disease. Such a case may be mild, moderate or a severe case. **Mild** cases are more dangerous than severe cases because they are ambulatory and go on spreading the disease wherever they go. Severe cases are usually bedridden.

Cases do not act as source throughout the duration of illness. They do so only during the '**period of communicability**,' which varies from disease to disease.

Carrier: A carrier is an infected person but not having the clinical features of the disease (thus harboring the organisms) and serves as a source of infection to others in the community. Such carriers are seen in typhoid, diphtheria, gonorrhea, AIDS, hepatitis B, meningitis, salmonellosis and amoebiasis. They can be detected only by doing laboratory investigations.

Epidemiological importance:

- Carriers form the "submerged iceberg" phenomenon—far more numerous than visible cases.
- They maintain endemicity of diseases in communities.
- They can spread infection to new areas otherwise free of disease.
- Their detection and treatment are essential but challenging in community medicine.

Carriers are dangerous reservoirs of infection because :

- Carriers show **no clinical symptoms**, so they appear healthy.
- They and their contacts are **unaware of the risk**, so precautions are not taken.
- Carriers are **hard to detect**, often requiring laboratory tests.
- They cannot always be managed effectively (e.g., a typhoid carrier preparing food).

- Carriers are **more numerous than clinical cases** and move freely in the community, spreading infection.

Classification of carriers

A. According to the stage in the disease cycle (Chronologically), there are three types of carriers, namely incubatory, convalescent, and contact carriers.

- **Incubatory carrier**: Spreads infection during the incubation period, before symptoms appear (e.g., measles, mumps, diphtheria, polio, pertussis, influenza, hepatitis A & B).
- **Convalescent carrier:** Spreads infection during recovery, when clinically cured but still harboring pathogens due to incomplete treatment (e.g., diphtheria, typhoid, amoebiasis).
- **Healthy/contact carrier:** Subclinically infected, never develops disease, but acts as a source of infection (e.g., diphtheria, cholera); often seen in doctors, nurses, attendants, or family members in close contact with patients.

B. According to duration: Depending upon the duration of the carrier state, they are grouped into temporary and chronic carriers.

- **Temporary carriers:** Spread infection for a short period (days); includes incubatory, convalescent, and healthy/contact carriers.
- **Chronic carriers:** Spread infection for a long period (weeks to months or lifelong) due to persistence of organisms in body organs (e.g., gallbladder in typhoid, tonsils in diphtheria, liver in hepatitis B).

C. According to route of exit: Depending upon the route of exit, the carriers are also grouped into the following:

- **Urinary carriers**, where the focus of organisms is the kidney as in typhoid
- **Intestinal carriers**, where the focus of organisms is the intestine, as in typhoid, amoebiasis
- **Biliary carriers**, where the focus of organisms is the gallbladder as in typhoid
- **Cutaneous carriers** as in staphylococci
- **Nasal carriers** as in nasal diphtheria
- **Genital carriers** as in gonorrhea, AIDS.

Dangerous group of carriers

Certain occupational groups giving public services or care of vulnerable:

- Food handlers
- Nurses in care of children .
- Hospital personnel
- School personnel

Animal Reservoir

There are many animals, which act as reservoir and transmit the diseases to human beings. These diseases are called zoonotic diseases (anthropozoonoses).

Animal reservoir	Diseases transmitted
Cattle	Bovine tuberculosis, salmonellosis, Tetanus, brucellosis, Q-fever, Taenia-saginnata.
Horses	Tetanus
Dogs	Rabies, hydatid disease
Monkeys	Yellow fever, dengue fever.
Sheep	Anthrax, liver fluke (Fasciola hepatica)
Rodents	Plague, leptospirosis, endemic typhus.
Birds	Psittacosis, histoplasmosis, ornithosis.

Zoonotic diseases can also be classified as viral, bacterial, parasitic and rickettsial diseases.

1 .**Viral**: Rabies, yellow fever, dengue fever, Japanese encephalitis.

2 .**Bacterial**: Salmonellosis, brucellosis, leptospirosis, plague.

3 .**Parasitic**: Teniasis, hydatid disease, leishmaniasis.

4 .**Rickettsial** diseases:

a. **Typhus** group: Rickettsial zoonoses, epidemic typhus, scrub typhus: Murine typhus

b. **Spotted** fever group: Tick typhus, Rocky Mountain spotted fever, Rickettsial pox

c. Others: Q fever, trench fever.

Soil Reservoir

The soil is considered the reservoir in coccidioidomycosis, histoplasomosis and probably in blastomycosis. Soil acts as source of infection and transmit diseases like tetanus, gas-gangrene, botulism, ankylostomiasis and anthrax.

III. PORTAL OF EXIT:

The microorganisms have to leave the reservoir of infection through the portal of exit. In general, the microorganisms come out with the secretions or excretions connected with the place in which the organism is present i.e. by a natural outlet. The organisms may leave the reservoir by (unnatural) outlets as by insect bites or other rare methods as during bloodletting.

The following are the different portals of exits:

1. The alimentary canal:

Microorganisms leave the gastrointestinal tract mainly through feces (typhoid, paratyphoid, dysentery, helminth eggs) or vomitus (cholera, food poisoning), and infection may also originate in connected organs like the gall bladder in enteric fever

2. The respiratory tract:

Microorganisms exit via the respiratory tract in diseases such as measles, influenza, TB, diphtheria, pneumonia, and plague, spreading through infective droplets expelled during coughing, sneezing, talking, or even breathing; small droplets are more dangerous as they remain airborne longer, while the mouth is the main exit point, with nasal secretions often spreading indirectly via contaminated objects like handkerchiefs or fingers

3. The urinary tract:

Microorganisms may leave the body through urine in systemic infections (e.g., typhoid, undulant fever), in local genito-urinary diseases (e.g., gonorrhea, other venereal infections, urinary typhoid carriers), or via parasite eggs expelled during urination (e.g., Schistosoma haematobium).

4. Discharges from skin and/or mucous membrane lesions:

Microorganisms may leave the body through skin lesions (as in chickenpox, smallpox, impetigo, and other bacterial infections) or via discharges from mucous membranes, such as the conjunctival sac in acute and chronic conjunctivitis.

5. By insect bites:

Microorganisms in the blood can be taken up by blood-sucking insects, which after an extrinsic incubation period become infective and transmit diseases.

- Mosquitoes → malaria, yellow fever, dengue, viral hemorrhagic fevers, filariasis, West Nile fever, encephalitis.

- Lice → epidemic typhus, louse-borne relapsing fever, trench fever.
- Fleas → plague, murine typhus.
- Ticks → tick-borne relapsing fever, Rocky Mountain spotted fever, tick fever.
- Sand flies → kala azar, sand fly fever.
- Tsetse flies → trypanosomiasis

6. In-utero transmission:

Some of the micro-organisms can reach the fetus in uterus e.g. the viruses of German measles, smallpox, the spirochetes of syphilis.

IV. MODES OF TRANSMISSION:

A communicable disease is transmitted from the reservoir/source to a susceptible person either directly or indirectly, depending upon whether they are in close proximity or not.

Direct Modes of Transmission

There are **five** direct modes:

1. **Direct contact:** When the reservoir and the susceptible person are in close, physical, skin to skin contact. Example, scabies, STDs AIDS, lepromatous leprosy, ophthalmia neonatorum (these are also called as **contagious** diseases).
2. **Droplet infection**: When the susceptible person inhales the infected droplets, of 5 to 10 μ diameter, coming out of the mouth or nose from the reservoir during the act of coughing, sneezing, talking and laughing. For example, pulmonary tuberculosis, diphtheria, measles, pertussis, etc.
3. **Contact with the contaminated soil**: For example, all soil borne diseases, e.g. tetanus, anthrax, ancylostomiasis.
4. **Contact with the animal**: Classical example is rabies transmitted from a rabid animal by bite and inoculating the virus into the skin or mucous membrane.
5. **Transplacental (vertical) transmission:** Means transmission of the disease from the mother to the fetus through the placenta. For example, toxoplasmosis, rubella, cytomegalovirus, herpes virus (TORCH-agents) syphilis, and AIDS.

Indirect Modes of Transmission: There are **five** indirect modes:

1. **Vehicle route:** The vehicles which are capable of transmitting the diseases are water,

food, milk, ice-cream, vegetables, biological products like blood, tissues or organs (as in organ transplantation).

2. **Vector route**: Vector is an arthropod, capable of transmitting the disease. For example, mosquitoes, flies, fleas, lice, ticks, mites, cyclops, etc. The vectors transmit the diseases in the following ways:
   - **Mechanically**: Carrying the pathogens from the filthy substances to food substances by soiling the legs. For example, house-fly transmitting typhoid, trachoma, etc.
   - **Biologically**: When the pathogen undergoes multiplication or development from one stage to another stage or both multiplication and development, inside the body of the vector. Thus, biological mode of transmission is of three types:
     - **Propagative type**: In this type, the pathogen undergoes only **multiplication** inside the vector. For example, plague bacilli in the body of the rat flea; yellow fever virus in the body of female Aedes mosquito.
     - **Cyclopropagative type**: In this type, the pathogen undergoes not only **multiplication** but also a phase of **development** in its life cycle. For example, Plasmodium (malarial) parasites in the body of the female anopheline mosquito (Zygote → Ookinite →Ocyst → Sporozoites).
     - **Cyclodevelopmental type**: In this type, the pathogen undergoes only a phase of the **development** but not multiplication. For example, microfilariae developing into larval stage in the body of female culex mosquito; Guineaworm embryos developing into larva in the body of Cyclops.
3. **Air-borne route of transmission:** By droplet nuclei and infected dust
   - **Droplet nuclei (<5 µm):** Tiny droplets expelled from mouth or nose lose moisture, leaving pathogen residues that remain suspended in air and spread widely, causing epidemics/pandemics (e.g., influenza, SARS, measles, tuberculosis).
   - **Infected dust (>10 µm):** Larger droplets settle on surfaces and mix with dust; when disturbed (sweeping, bed-making, wind), they re-enter the air and spread infections in hospitals and wards (e.g., pulmonary TB, streptococcal and staphylococcal infections).
4. **Fomite route of transmission**: Fomites are the inanimate objects capable of transmitting the diseases. For example, clothes, linen (Scabies), clinical thermometer,

spoons, pencils, tongue depressor (Diphtheria, streptococcal infections) surgical instruments, syringes, needles, toothbrush, shaving brush, razor (Hepatitis B, AIDS), etc. Such infections are also called as 'mediate infections.

5. **Unclean hands and fingers**: These contaminate the food and drinks and transmit diseases like typhoid, streptococcal and staphylococcal infections, diarrheas and dysentery.

V. PORTALS OF ENTRY (INLETS):

Infectious agents enter the body of the susceptible through various inlets which can be classified into:

- **Natural inlets:** Include the mouth, nose, rectum, conjunctival sac, and urethra.
- **Un-natural inlets:** Include inoculation through blood sucking insects or through injections.

Pathology depends on entry site – The portal of entry often determines the disease type:

• Tuberculosis/Anthrax: respiratory entry → pulmonary form; intestinal entry → intestinal form; skin entry → dermal form.

VI. SUSCEPTIBLE HOST:

Susceptible Host

A susceptible host is a person or animal lacking immunity to a specific pathogen, making them liable to contract disease upon exposure.

- Exposure may or may not result in infection.
- If infection produces detectable pathology, the individual is considered to have an **infectious disease**.
- Infected persons include **patients, subclinical infections, and carriers**.

Factors Affecting Outcome of Infection

1. **Pathogenicity & Virulence of the Microorganism**
    - ***Pathogenicity***: ability to cause illness; higher pathogenicity → more clinical cases.
    - ***Virulence***: ability to cause severe or fatal disease; measured by case fatality rate.

2. **Antigenic Power of the Microorganism**
    - Ability to stimulate immunity and antibody formation.
    - Strong antigenic power → lifelong immunity (e.g., measles, mumps, chickenpox).
    - Weak antigenic power → reinfections common (e.g., common cold, syphilis, gonorrhea).
    - Measured by **second attack frequency** and **age-specific attack rates**.
3. **Period of Communicability (Duration of Infectious State)**
    - Time during which the pathogen can be transmitted.
    - Short (e.g., mumps) or prolonged (e.g., typhoid carriers may remain infectious for life).
4. **Ease of Communicability of the Disease**
    - Measured by **secondary attack rate** (spread among susceptible contacts during incubation period).
    - Indicates how easily a disease spreads within a community.

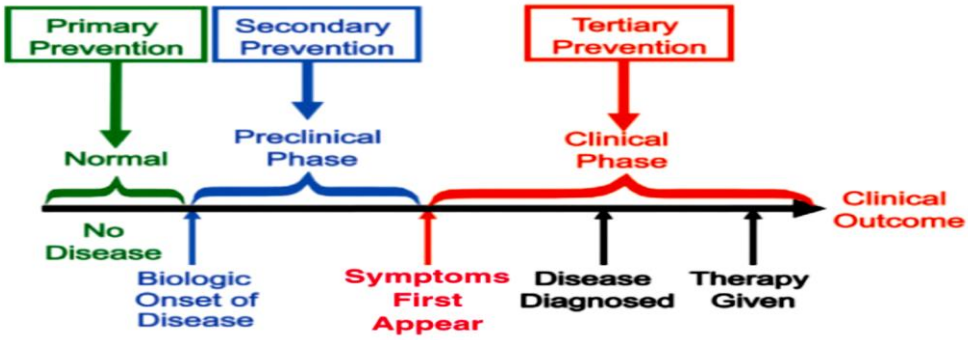
Concept of prevention

By the end of the lecture, students should be able to:

- Identify the different levels of prevention.
- Describe the principles of prevention of communicable diseases.

The objective of community-medicine is to oppose the disease process in order to preserve the health, promote the health, prolong the life and to minimize the sufferings by preventing the occurrence of the disease.

The disease can be prevented by opposing its natural history at different levels



Level of prevention: There are three levels of prevention—primary, secondary and tertiary.

1-Primary prevention :

This is the measure undertaken in the period of prepathogenesis (i.e. before the development of the disease) which removes the possibility of occurrence of the disease. Especially, it is all the more important in those diseases, for which no treatment is available, e.g. AIDS, cancer, rabies, carries tooth, etc.

Primary prevention can be adopted by two modes of intervention—namely Health promotion and Specific protection.

- **Health promotion**

This consists of 'general measures', which will strengthen the individual/host and prevents the occurrence of the disease by interrupting the interaction among the three factors of epidemiological triad.

The various measures of health promotion are:

- Health education (on personal hygiene, oral hygiene, nutrition education, life-style, etc.).
- Sex education.
- Adequate nutrition.
- Improvement in the environmental sanitation (such as control of insects, provision of protected water supply, sanitary disposal of sewage, etc.).
- Promotion of breastfeeding and proper weaning.
- Family planning and spacing of births.
- Genetic counseling (premarital and marriage counseling).
- Efficient antenatal care and postnatal care.
- Recreation facilities (sports, games, cultural activities, etc.).
- Improvement in the literacy
- Yoga exercises and meditation.

➢ **Specific protection**

This consists of 'specific measures', which prevent specific diseases. The various measures are:

- Immunization against vaccine preventable diseases
- Silver nitrate or penicillin eye drops against ophthalmia neonatorum
- Condom against AIDS
- Use of specific nutrients (Vitamin A against nutritional blindness, Iron and folic acid against nutritional anemia, iodized salt against iodine deficiency disorders)
- Masks against pneumoconiosis
- Ear plugs against noise induced deafness
- Lead apron against radiation hazards
- Barrier cream against occupational skin cancer
- Avoidance of allergens and carcinogens
- Sterilization procedures of surgical instruments
- Pasteurization of milk
- Traffic signals against road accidents
- Quality control of foods, drugs and cosmetics, etc.

These measures need to be applied in specific situation for specific groups. They are more concrete and effective.

PRIMORDIAL PREVENTION

It is also a **primary** level of prevention of the disease, but it is with reference to noncommunicable diseases, such as Obesity, hypertension, diabetes, cancer, coronary artery disease, etc. This consists of elimination or modification of **'risk factors'** of the disease. In such chronic, noncommunicable diseases, the etiological (causative) agent is not known (or not established) and the etiology is discussed in terms of 'risk factors.' They may act as contributory factors.

There are two approaches for the primordial prevention :

Population Strategy

This is directed at the whole population, irrespective of individual risk level This approach is directed towards changes in the life-style of the population by health education from the childhood itself. The results of these measures cannot be perceived immediately but are seen after several years or decades.

High-risk Strategy (Individual Strategy)

This is directed to those individuals, who are at high risk of getting the disease .

These high-risk group can be detected by screening procedures.

2 -Secondary prevention:

Secondary prevention aims to reduce the impact of a disease or injury that has already occurred. This is done by detecting and treating disease or injury at an early stage to halt or slow its progress, such as by screening, takes place during the preclinical phase of an illness, that is, after the disease has developed, but before clinical signs and symptoms have appeared, encouraging personal strategies to prevent rein jury or recurrence, and implementing programs to return people to their original health and function to prevent long-term problems. Examples include:

- Regular exams and screening tests to detect disease in its earliest stages (e.g., mammograms to detect breast cancer)
- Daily, low-dose aspirins and/or diet and exercise programs to prevent further heart attacks or strokes
- Suitably modified work so injured or ill workers can return safely to their jobs.

Early Diagnosis and Prompt Treatment

This helps in the following ways:

- Helps in recovery from the disease (restoration)
- Reduces the duration of illness in the individual
- Minimizes the suffering
- Prevents the development of complications
- Prevents further spread of the diseases in the community
- Prevents or postpones the death of the individual.

3 -Tertiary prevention:

This is the measure undertaken, when the disease process is sufficiently advanced, (i.e. in the late pathogenesis phase). Tertiary prevention can be adopted by two modes of intervention - disability limitation and rehabilitation.

Disability limitation:

This means limiting the development of further disability in the individual by giving intensive or aggressive treatment, when the patient comes in the advanced stage of the disease. The objective is to prevent the transition from impairment to handicap.

The sequence of events in a disease process are:

Disease → Impairment → Disability → Handicap

Impairment : This means defect in the structures and function of an organ or a part of the body.

Disability : This means inability to carry out certain routine, expected activities, considered normal for the age, sex etc, due to impairment.

Handicap : This means experiencing disadvantage in the life and not able to play the role, expected out of her/him, resulting from the impairment or disability.

Rehabilitation

Rehabilitation has been defined as the 'combined and coordinated use of medical, social, educational and vocational measures for training and retraining the individual to the highest possible level of functional ability"

The different aspects of rehabilitation are:

Physical (medical) rehabilitation	**Restoration of function**
Vocational rehabilitation	**Restoration of earning capacity**
Social rehabilitation	**Restoration of relationship in the society**
Psychological rehabilitation	**Restoration of personal dignity and confidence**

Examples of rehabilitation are:

- Establishing the schools for the blind
- Providing aids for crippled, such as artificial limb, crutches, wheel chair, hearing aid, etc.
- Reconstructive surgery in leprosy
- Graded exercises in paralysis
- Intraocular implantation of lens among cataract patients.

Host susceptibility, immunity and vaccination

By the end of the lecture, students should be able to:

- Define immunity and identify the types of immunity
- Differentiate between active and passive immunity
- Describe the host defense mechanisms
- Identify the different types of vaccines
- Define Sero prophylaxis and its uses
- Differentiate between chemo and Sero prophylaxis

Resistance (Nonspecific Defense Mechanisms):

It is the total natural body mechanisms which constitute barriers to the progress of invasion of infectious agents. It is not dependent on of the presence of antibodies or specifically developed tissue response. Therefore, it is not related to previous infection or artificial immunization. Natural resistance depends on the characteristics of the individual such as natural defense mechanisms, species, race, Sex, and personal habits.

Nonspecific Defense Mechanisms (resistance)

First line of defense	Second line of defense
● Skin ● Mucous membranes ● Secretions of skin and mucous Membranes	● Phagocytic White blood cells ● Antimicrobial proteins ●The inflammatory response

Immunity

The word "immunity" derives from the latin immunis, meaning exemption from military service, tax payments or other public services and is defined as "Ability of the body to recognize and defend itself against specific pathogens or antigen. It is a type of specific resistance **(Third line of defense)** usually associated with exposure to specific antigen.

TPYES OF IMMUNITY

1. Active immunity:

This develops only after exposure to inducing agents such as microbes, toxins, or other foreign substances.

Naturally acquired active immunity

It occurs when a person is exposed to a live pathogen, and develops a primary immune response, which leads to immunological memory. This type of immunity is "natural" because it is not induced by man.

The degree of post infection immunity varies and it may be:

- **Absolute** as in yellow fever where the level of immunoglobulin had never declined after contracting clinical attack of yellow fever and recovery.
- **Solid (lifelong immunity**) as in the case of small pox, chicken pox, measles, German measles, mumps.
- **Moderate (lasting for few years)** as in the case of typhoid fever, M. meningitis and pertussis
- **Transient (lasting a short period of time)** as in common cold and influenza.

Artificially acquired active immunity

It can be induced by a vaccine, a substance that contains antigen.

A vaccine stimulates a primary response against the antigen without causing symptoms of the disease.

2. **Passive Immunity**

It is the transfer of active immunity, in the form of ready-made antibodies, from one individual to another.

- **Naturally acquired passive immunity:** Passive immunity can occur naturally, when maternal antibodies are transferred to the fetus through the placenta or through breast milk .
- **Artificially acquired passive immunity:** This is a short-term immunization induced by the transfer of antibodies, which can be administered in several forms; as human or animal plasma or serum, as pooled human immunoglobulin

Immunization

It is a procedure in which immunobiological substances are administered to strengthen the defense (immune) mechanism as to protect the individual against the disease.

When this is implemented in the community at large, the susceptible population can be reduced.

There are two types of immunization—active and passive.

The differences between active and passive immunization

Active immunization	**Passive immunization**
This consists of administration of antigens.	This consists of administration of antibodies.
Immunity is produced after sometime (10-15 days)	Immunity is produced instantaneously
Tt is an active process	It is passive process
Immunity lasts longer	Immunity lasts shorter
Usually not associated with Reaction	Often associated with reactions
Usually it is given before exposure to the disease	Usually it is given after exposure to the disease
This procedure is never used for treatment purposes	This is often used as a part of treatment, as in tetanus, diphtheria
This is more useful	This is less useful

Types of vaccine: This include

1. **Live vaccines:**

Only (small pox) vaccine, made of live vaccinia virus, not variola virus; nonpathogenic, but antigenic simulating cross immunity for variola.

2. **Live attenuated Vaccines**

These are prepared from live attenuated organisms by repeated subcultures or cultivation under specified unfavorable conditions. e.g. Bacterial [BCG- Typhoid oral], Viral [(Measles, Mumps, Rubella, Oral Polio (Sabin vaccine), Yellow fever (17 D. strain), Influenza], Rickettsial [Epidemic typhus (strain E)]

3. **Inactivated (Killed) Vaccines :**

These are prepared from organisms killed by heat or chemicals. Killed vaccines are very safe but less efficacious than live vaccines. They require multiple doses which may be administered as a series of primary doses followed by regular booster doses. e.g. Bacterial [Typhoid, and paratyphoid, Pertussis, Cholera, plague], Viral [poliomyelitis (Salk vaccine), Sinopharm vaccine (corona virus)

4. **Toxoids**

These are produced from detoxicated toxins. The body produces antibodies against the toxin in response to toxoids. They offer no protection against infection. Toxoids are very safe and highly effective. E.g. Diphtheria, Tetanus

5. **Cellular fractions**

These vaccines are prepared from the extract of the bacterial cell-wall or capsule.

They are also safe, effective and some require booster doses. e.g. Polysaccharide vaccines [meningococcal, pneumococcal and, Hemophilus influenza vaccines].

6. **Subunit vaccine**

This is prepared from a component of the virus. e.g. influenza vaccine

7. **Recombinant vaccines**

These are genetically engineered recombinant DNA vaccines, i.e. the sub-unit of the virus (antigen) is inserted into the genome of another avirulent virus and vaccine is prepared. e.g. hepatitis B vaccine.

8. **Combined vaccines (mixed vaccines)**

They are so called because the preparation contains more than one antigen. e.g. pentavalent [DPT + HBV + Haemophilus influenzae type b (Hib)], DPT, DT, MMR.

9. **Tissue culture vaccines**

These are prepared by culturing the seed viruses in special cells such as chick-embryo cells, vero cells of kidney of monkeys, human embryonic lung fibroblasts, etc. e.g. HDCV (The human diploid cell rabies vaccine)

Contraindications of vaccination

- Fever, mild respiratory infection and other minor illness are not Contra indications to vaccination.
- Presence of malnutrition is also not a contraindication.
- Diarrhea is not a contraindication for OPV immunization, however the dose of OPV given during episodes of diarrhea should not be counted and another dose given at the earliest opportunity.
- Breast-feeding prior to or after OPV administration is not forbidden nor contraindication. Only avoid giving hot liquid or food for ½ hour after OPV
- Tuberculin testing is not required prior to administration of BCG till the age of 12 years.

Listening too many contraindications may leave many children unvaccinated and if ill or malnourished, they are more likely to die from a vaccine preventable disease.

Strict contraindication include:

- Very ill children who need hospital admission.
- BCG is contraindication in AIDS and immune deficiency diseases.
- Pertussis component of DPT is contraindicated under the following conditions.
    - Children with neurological disorders.
    - Severe reactions to previous DPT dose.
    - Children over the age of five years.

Causes of vaccination failure (what damage vaccine)

- **Heat**: Polio, measles, MMR, vaccines.
- **Sunlight**: BCG, DPT.
- **Chemicals:**

  Disinfecting and antiseptics, including spirits, are damaging to vaccines.

  The site of injection should be left for a while after sterilization with alcohol to dry before injecting the vaccine.
- **Time**: All vaccines lose their potency after certain time. This is recorded on the vaccine as expiry date.

- **Reconstitution**: Reconstituted vaccines like (BCG, measles, MMR) are damaged after few hours from reconstitution.

Seroprophylaxis or passively acquired artificial Immunity

Anti-sera

These are the specific immunoglobulins prepared from the plasma of immunized animals, such as horses. They are cheap and less effective. Immunity lasts for about 2 to 3 weeks only. Reactions are frequent because of animal protein. So test dose is a must. e.g. anti-tetanus serum (ATS); anti-diphtheria serum (ADS); anti-snake venom; Anti-gas-gangrene serum.

Immunoglobulins

These are ready made antibody preparations, obtained from the human beings . They produce immunity instantaneously . Therefore they are used for passive immunization.

Since they are human preparations, reactions are nil. E.g. human rabies immunoglobulin (HRI ; human tetanus immunoglobulin (HTIg) ; hepatitis B immunoglobulin (HBIg); varicella zoster immunoglobulin (VZIg).

Chemoprophylaxis

This is another type of specific measure to protect the susceptibles.

That means it is a prophylactic (preventive) chemotherapy (chemical substances, including antibiotics) . E.g.

- The use of antibiotics before surgical procedures to prevent wound infections.
- INH to the young children of the open case of pulmonary tuberculosis .
- The use of chloroquine to prevent malarial (except chloroquine-resistant strains of Plasmodium falciparum) .
- The use of tetracycline to prevent Cholera
- The use of rifampicin to prevent meningitis

Egypt National Immunization Schedule (First 2 Years of Life)

Age	Vaccine	Route	Dose
At birth	BCG	ID	0.05 ml
	OPV (Sabin, zero dose)	Oral	2 drops
	Hepatitis B (HBV, birth dose within 24 hrs)	IM	0.5 ml
2 months	OPV	Oral	2 drops
	Pentavalent (DPT + HBV + Hib)	IM	0.5 ml
	IPV (Salk)	IM	0.5 ml
4 months	OPV	Oral	2 drops
	Pentavalent (DPT + HBV + Hib)	IM	0.5 ml
	IPV	IM	0.5 ml
6 months	OPV	Oral	2 drops
	Pentavalent (DPT + HBV + Hib)	IM	0.5 ml
	IPV	IM	0.5 ml
9 months	Vitamin A	Oral	100,000 IU
12 months	OPV	Oral	2 drops
	MMR (Measles, Mumps, Rubella)	SC	0.5 ml
18 months	OPV	Oral	2 drops
	DPT booster	IM	0.5 ml
	MMR booster	SC	0.5 ml
	Vitamin A	Oral	200,000 IU

MENINGITIS

Learning Objectives

By the end of this chapter, the reader should be able to:

- Describe the epidemiology and distribution of meningococcal meningitis
- Identify the causative organism and its transmission patterns
- Recognize risk factors and vulnerable populations
- Explain clinical presentation and complications
- Outline diagnostic methods
- Discuss prevention and control strategies including vaccination

Chapter Overview

Meningococcal meningitis is a rapidly progressive infectious disease that causes inflammation of the meninges surrounding the brain and spinal cord. It is a major global public health concern because of its high fatality rate and risk of permanent neurological disability among survivors.

Global Distribution and Public Health Importance

Meningococcal meningitis occurs worldwide but is most common in the **Meningitis Belt of Sub-Saharan Africa**, a region within **Sub-Saharan Africa** located between 5° and 15° north of the Equator. This area experiences frequent large-scale epidemics.

The disease is characterized by:

- Rapid progression
- High mortality even with treatment
- Frequent severe neurological sequelae among survivors

Causative Agent

Neisseria meningitidis

Characteristics:

- Gram-negative diplococcus
- Fragile organism that dies quickly outside the body
- Can cause sporadic cases, outbreaks, or large epidemics

Major invasive serogroups:

A, B, C, E29, X, W-135, Y (classified by capsular polysaccharide antigens)

Source of Infection and Carrier State

The human nasopharynx is the natural reservoir.

Key facts:

- Asymptomatic carriers are the main source of infection
- Clinical cases contribute less to transmission
- Many infections cause mild or unnoticed nasopharyngitis

Carrier prevalence:

- Inter-epidemic periods: 4–35%
- During epidemics: up to 70–80%

A rising carrier rate is an early warning of an epidemic.

Host Factors

Age Distribution

- All ages susceptible
- Highest risk: infants (3–12 months)
- Increased susceptibility: adolescents and young adults

Immunity

Immunity develops through:

- Subclinical infection (most common)
- Clinical disease
- Vaccination

Infants receive passive maternal antibodies.

Period of Communicability

- Infectious while organisms are present in respiratory secretions
- Usually non-infectious within 24 hours of effective antibiotic therapy

Environmental and Social Risk Factors

- Dry and cold seasonal conditions
- Overcrowding
- Poor ventilation

High-risk settings:

- Schools and nurseries
- University dormitories

- Military camps
- Prisons
- Refugee shelters
- Public transport

Mode of Transmission

- Respiratory droplets from close contact
- Portal of entry: nasopharynx
- Transmission through personal items is uncommon

Incubation Period

- Typically 2–10 days
- Shorter during epidemics (usually 3–4 days)

Clinical Features

Most infections are asymptomatic, but invasive disease may develop suddenly.

Early Symptoms

- High fever
- Severe headache
- Nausea and vomiting
- Photophobia

Classic Signs

- Neck stiffness
- Altered consciousness
- Convulsions
- Bulging fontanelle in infants

Severe Progression

- Confusion
- Drowsiness
- Coma

Complications and Outcomes

- Death within 24–48 hours in 5–10% of cases despite treatment
- Permanent neurological disability in 15–20% of survivors

Common sequelae:

- Hearing loss
- Strabismus
- Paralysis

Meningococcal septicemia is less common but highly fatal.

Laboratory Diagnosis

Essential investigation: lumbar puncture

Cerebrospinal fluid findings:

- Increased pressure
- Turbidity
- Microscopic evidence of bacteria
- Culture confirmation

Specimen collection should be performed before starting antibiotics whenever possible.

Prevention and Control

General Preventive Measures

- Adequate ventilation in crowded settings
- Health education and hygiene awareness
- Surveillance for early detection during outbreaks

Treatment of Cases

- Early treatment (within first 48 hours) saves up to 95% of patients
- First-line drug: penicillin
- Alternatives (penicillin allergy): ceftriaxone or third-generation cephalosporins
- Isolation has limited impact due to high carrier prevalence

Carrier Eradication

- Rifampicin commonly used

Management of Close Contacts

Highest risk occurs within 72 hours of exposure.

Chemoprophylaxis within 24 hours using:

- Rifampicin
- Ciprofloxacin
- Ceftriaxone

- Azithromycin

Mass Chemoprophylaxis

Recommended only for closed, medically supervised communities.

Vaccination

Types of Vaccines

- Polysaccharide vaccines
- Conjugate vaccines

Serogroup Coverage

A, C, W-135, Y

Schedule

- Single 0.5 ml subcutaneous dose for persons older than 2 years
- Limited effectiveness in children under 2 years
- Does not provide strong herd immunity

Recommended for travelers to epidemic or high-risk regions.

In **Egypt**, vaccination is given to preschool children and repeated in the first year of primary school. It is contraindicated during pregnancy.

Leprosy (Hansen's Disease)

Learning Objectives

1. Understand the epidemiology of leprosy
2. Identify the etiological agent and transmission dynamics
3. Recognize clinical manifestations and immunological spectrum
4. Apply diagnostic and classification criteria
5. Evaluate prevention and control strategies

Introduction

Leprosy, also known as Hansen's disease, is a chronic infectious disease caused by *Mycobacterium leprae*. It primarily affects the skin, peripheral nerves, and the mucosa of the upper respiratory tract. The disease is characterized by long incubation periods, varied clinical manifestations, and potential for permanent disability if diagnosis and treatment are delayed. Despite effective chemotherapy, leprosy remains a public health challenge in several endemic countries due to social stigma, delayed case detection, and limited access to health services.

Epidemiology

Occurrence

Globally, more than 200,000 new cases of leprosy are reported annually. The elimination of leprosy as a public health problem is defined as a prevalence of less than 1 case per 10,000 population. While many countries have achieved this target, leprosy continues to be reported in several endemic regions.

Egypt continues to report a considerable number of new leprosy cases each year, indicating persistent transmission and the need for sustained surveillance and control measures

Etiological Agent

The causative agent of leprosy is *Mycobacterium leprae*, a strong acid-fast, rod-shaped bacillus. It is an obligate intracellular organism with extremely slow multiplication, accounting for the long incubation period and chronic course of the disease.

Reservoir

Humans are the only known reservoir of *Mycobacterium leprae*.

Mode of Transmission

Leprosy is not highly infectious. Transmission occurs mainly through respiratory droplets from the nose and mouth during close and prolonged contact with untreated infectious cases, particularly those with multibacillary disease.

Incubation Period

The incubation period is highly variable, ranging from 9 months to 20 years, with an average of approximately 5 years.

Clinical Manifestations

Skin Lesions Skin lesions may be single or multiple and exhibit considerable variation in appearance. Common types include:

- Macules (flat lesions)
- Papules (raised lesions)
- Nodules

The lesions are typically hypopigmented compared to surrounding normal skin. A cardinal feature of leprosy is **sensory loss**, especially to pinprick and light touch, within the affected skin areas.

Peripheral Nerve Involvement

Thickening of peripheral nerve trunks is a characteristic feature of leprosy. Nerve involvement may lead to:

- Loss of sensation
- Muscle weakness
- Deformities due to prolonged nerve damage

Isolated nerve thickening without associated sensory or motor impairment is not a reliable diagnostic sign.

Immunological Spectrum

Leprosy presents as a spectrum of disease determined by host immune response:

- **Tuberculoid leprosy** represents strong cell-mediated immunity.
- **Lepromatous leprosy** represents poor cell-mediated immunity.
- **Borderline leprosy** shows features of both polar forms and is immunologically unstable, tending to shift toward the lepromatous form if untreated or toward the tuberculoid form with effective treatment.

Leprosy Reactions

Acute immunological complications, termed leprosy reactions, may occur:

- **Reversal reactions** in borderline leprosy
- **Erythema nodosum leprosum (ENL)** in lepromatous leprosy

These reactions contribute significantly to nerve damage and disability.

Diagnosis of Leprosy

Diagnosis is primarily clinical and can be made by trained health workers. Laboratory investigations are rarely required.

Clinical Differences Between Polar Forms

Feature	Lepromatous Leprosy	Tuberculoid Leprosy
Skin lesions	Numerous, extensive	Single or few
Distribution	Bilaterally symmetrical	Asymmetrical
Lesion margins	Diffuse infiltrations	Sharply demarcated
Peripheral nerve involvement	Less severe	Severe
Nasal mucosa involvement	Present	Absent
Ocular involvement	Present	Absent

Cardinal Signs of Leprosy

In endemic areas, a person is diagnosed with leprosy if **one or more** of the following cardinal signs are present:

1. Skin lesion consistent with leprosy with definite sensory loss, with or without nerve thickening.
2. Positive skin smear showing acid-fast bacilli.

Skin Smear Examination

In a minority of cases, skin smears demonstrate red-stained, rod-shaped *M. leprae*, which are diagnostic of leprosy.

Classification of Leprosy

Leprosy is classified for treatment purposes into:

- **Paucibacillary (PB) leprosy**: Negative skin smears at all sites.
- **Multibacillary (MB) leprosy**: Positive skin smear at any site.

In practice, clinical criteria—number of skin lesions and nerves involved—are often used due to limited availability of reliable skin smear services.

Period of Communicability

Infectiousness is markedly reduced after initiation of effective treatment:

- Within 3 months of regular dapsone or clofazimine therapy
- Within 3 days of treatment with rifampicin

Susceptibility and Immunity:

Susceptibility depends on host cell-mediated immunity. Infection is common among close contacts, but clinical disease develops in only a small proportion.

Lepromin Test

The lepromin (Mitsuda) test measures delayed hypersensitivity to *M. leprae* antigens:

- Negative in lepromatous leprosy
- Positive in tuberculoid leprosy and many healthy individuals

Due to limited diagnostic value, World Health Organization recommends restricting its use to research purposes.

Methods of Control

A. Primary Prevention

1. **Health education** emphasizing curability, availability of effective multi-drug therapy (MDT), and reduction of stigma.
2. **BCG vaccination**, which offers partial protection, particularly against tuberculoid leprosy.
3. **Early case detection and prompt MDT**, especially for infectious MB cases.

B. Secondary Prevention

1. Notification

Leprosy is a notifiable disease in many countries.

2. Isolation

- Not required for tuberculoid leprosy
- Temporary contact isolation for lepromatous cases until MDT is established
- Hospitalization may be needed during severe reactions

3. Disinfection

Concurrent disinfection of nasal secretions and terminal cleaning are recommended.

4. Contact Surveillance

Household and close contacts should be examined initially and then annually for at least 5 years after last exposure.

5. Specific Treatment (Multi-Drug Therapy – MDT)

MDT is the cornerstone of leprosy control. Monotherapy is unethical due to inevitable drug resistance.

Drugs used:

- **Rifampicin**: Most potent bactericidal drug; administered once monthly.
- **Clofazimine**: Causes reversible skin discoloration.
- **Dapsone**: Generally safe; rare allergic reactions may occur.

WHO-Recommended MDT Regimens

- **Multibacillary leprosy**: Rifampicin + Dapsone + Clofazimine for 24 months.
- **Paucibacillary leprosy**: Rifampicin + Dapsone for 6 months.

Monthly doses of rifampicin (and clofazimine in MB cases) are given under supervision.

Disaster Implications

Interruptions in treatment during wars or disasters significantly increase disease transmission, complications, and drug resistance.

International Measures

International control efforts focus on early detection and treatment of untreated infectious cases to prevent cross-border transmission.

Conclusion

Leprosy is a chronic but curable disease whose persistence reflects gaps in early detection, treatment adherence, and social acceptance. Sustained implementation of MDT, health education, contact surveillance, and stigma reduction are essential for achieving long-term control and eventual elimination of leprosy as a public health problem

Tuberculosis

LEARNING OBJECTIVES

1. Describe the global burden of TB.
2. Identify risk factors influencing TB transmission
3. Understand modes of transmission and natural history
4. Evaluate public health strategies for TB prevention and control

WHAT IS TUBERCULOSIS?

- Tuberculosis is a worldwide, chronic communicable bacterial disease.
- Tuberculosis disease is the outcome of the fight between virulence of the organism and resistance of the body.

HISTORICAL PERSPECTIVE

- History goes back to 9000 years Discovered in Mummies of Egypt]
- 1882-Robert Koch, a German Scientist, discovered rod shaped bacteria causing tuberculosis.
- 1921-French Scientists Calmette and Guerin discovered vaccine against tuberculosis.

INTRODUCTION & PROBLEM STATEMENT:

- TB is a mycobacterium disease that is important as a major cause of disability and death in many parts of the world, but TB is curable and preventable.
- TB kills 5,000 people a day – 2-3 million each year
- One third of the world's population is infected with TB
- TB kills more young women than any other disease
- Multidrug-resistant TB (MDR-TB) remains a public health crisis and a health security threat.
- Patients with infectious pulmonary tuberculosis disease can infect 10-15 persons in a year.
- Ending the TB epidemic by 2030 is among the health targets of the United Nations Sustainable Development Goals (SDGs).

MAGNITUDE OF TUBERCULOSIS IN EGYPT

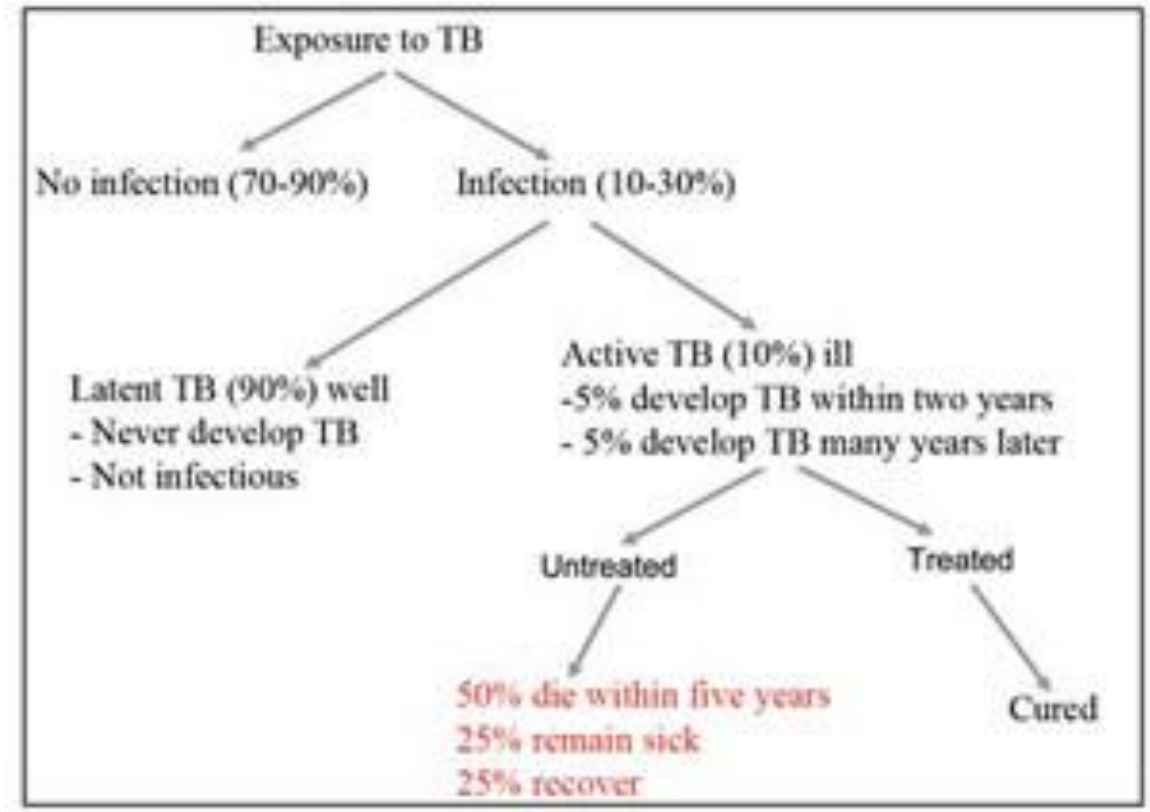
- Egypt is a low TB -prevalence country
- TB is considered an important public health problem.
- The Ministry of health & population (MOHP) has established the National tuberculosis control program in 1979.

  According to WHO Global TB Report, 2022

  For Egypt Prevalence 28 per 100 000 population
- Incidence 17 per 100 000 population
- Mortality 0.59 per 100 000 population
- form of tuberculosis (smear positive and smear negative) represents roughly 80 to 85 percent of all cases.
- The remaining 15 to 20 percent is made up by cases of extra-pulmonary tuberculosi s
- For every new smear-positive pulmonary tuberculosis case usually a case of smear- negative pulmonary or extra-pulmonary tuberculosis will also be present.
- Egypt is a low-HIV-prevalence country . Currently, less than 1 percent of adult TB patients are HIV-positive. However, continued monitoring is necessary because an increase in the incidence of HIV-TB co-infection could add to the complexity of fighting both diseases in Egypt

NATURAL HISTORY OF TUBERCULOSIS

- Exposure to TB does not lead to infection of the contact in 70–90% of cases. For those 10–30% of individuals that do become infected with *M. tuberculosis*, in about 90% of them, the body's immunity either kills the bacteria, or perhaps more often, keeps them suppressed (inactive), causing LTBI. (In HIV infected persons, TB infection progresses to disease more rapidly due to the weakening of their immunity.) In healthy individuals, only about 10% of infected persons develop active disease and become ill.
- Without treatment, 50% of patients with pulmonary TB will die within five years, but most deaths are within two years; 25% will remain sick with chronic, infectious TB which can be spread to the community. Another 25% will spontaneously recover and be healthy, due to their strong immune defenses, but they could become sick again at any time if the TB bacteria are latent.



BASIC DEFINITION OF TUBERCULOSIS CASES AND TREATMENT:

WHO has issued updated guidance on definition of cases and treatment outcomes and associated reporting framework in 2013.

Presumptive cases

It refers to a patient who presents with symptoms or signs suggestive of TB (previously known as TB suspect).

Definite cases

A patient with Mycobacterium tuberculosis complex identified from a clinical specimen, either by culture or by a newer method such as molecular line probe assay. In countries that lack the laboratory capacity to routinely identify M. tuberculosis, a pulmonary case with one or more initial sputum smear examinations positive for acid-fast bacilli (AFB) is also considered to be a ―definite‖ case, provided that there is a functional external quality assurance (EQA) system with blind rechecking.

DIAGNOSIS:

TB Diagnostics Methods

- Sputum collection and Microscopy
- Culture-based Diagnostics
- Molecular detection
- Cellular response- to detect latent and latent to active progression.
- Breath biomarker – detection
- Antigen antibody and biomarker detection
- Radiology

Clinical picture: The symptoms of pulmonary TB include a productive, prolonged cough (duration of more than 3 weeks), chest pain, and hemoptysis. Systemic symptoms of TB include fever, chills, night sweats, easy fatigability, loss of appetite, and weight loss. Approximately 15% of TB cases are extra- pulmonary. The symptoms of extra-pulmonary TB depend on the site affected. TB of the spine may cause pain in the back; TB of the kidney may cause blood in the urine.

People infected with TB bacteria have a 5–10% lifetime risk of falling ill with TB. Those with compromised immune systems, such as people living with HIV, malnutrition or diabetes, or people who use tobacco, have a higher risk of falling ill.

Physical Examination: It cannot be used to confirm or rule out TB, but it can provide valuable information about the patient's overall condition and other factors that may affect how TB is treated .

Tuberculin Skin Testing: useful for :examining a person who is not ill but may be infected with M. tuberculosis, such as a person who has been exposed to someone who has TB. The tuberculin skin test is the only way to diagnose TB infection before the infection has progressed to TB disease and determining how many people in a group are infected with M. tuberculosis.

Radiographic Examination: A postero-anterior view of the chest is the standard radiograph needed for the detection anddescription of chest abnormalities. In some instances, other views (e.g., lateral) or additional studies (e.g., CT scans) may be necessary.

Laboratory Examination: Detection of acid-fast bacilli (AFB) in stained smears examined microscopically may provide the first bacteriologic clue of TB. (definitive diagnosis)

EPIDEMIOLOGIC DETERMINANTS

Agent factors: Mycobacterium TB, the tubercle bacillus that was identified by Kock in 1882. There are 3 types that are pathogenic to man:

Human type: it is the essential causative organism of TB in man that causes all or nearly all pulmonary cases and 20-50 % of extra-pulmonary cases.

Bovine type: cattle are reservoir of infection, and infect man causing 50-80 % extra-pulmonary cases and aminority of pulmonary cases.

Avian type: Poultry are the reservoir of infection, and rarely infect man, possibly by ingestion of infected eggs, causing reported sporadic cases of extra- pulmonary TB.

Host and social factors: The risk of infection with the tubercle bacillus is directly related to the degree of exposure, and does not appear to be related to genetic or other host factors. The most hazardous period for development of clinical disease is the first 6-12 months after infection. The risk of developing disease is highest in children under 3 years old, lowest in later childhood, and high again among adolescents, young adults, the-very old and the immunosuppressed.

Mode of transmission: TB is spread from person to person through the air. When a person with pulmonary or laryngeal TB coughs or sneezes, droplet nuclei containing M. tuberculosis are expelled into the air. These tiny particles (1-5 microns in diameter) can remain suspended in the air for several hours. If another person inhales air containing droplet nuclei, transmission may occur.

The probability that TB will be transmitted depends on three factors:

1- the infectiousness of the person with TB, 2- the environment in which exposure occurred, 3- the duration of exposure.

Persons at the highest risk of becoming infected with M. tuberculosis are close contacts persons who often spend time with someone who has infectious TB. Close contacts may be family members, roommates, friends, coworkers, or others. These persons are at risk for TB infection because they are more likely to be exposed to TB. Infection rates have been relatively stable since 1987, ranging from 21% to 23% for the contacts

of infectious TB patients.

Incubation period: From infection to demonstrable primary lesion or significant tuberculin reaction, about 3-6 weeks. While the subsequent risk of progressive pulmonary or extra-pulmonary TB is greatest within the first year or two after infection, latent infection may persist for a lifetime.

MEASURES OF CONTROL AND PREVENTION

PREVENTION

1) Educate the public in mode of spread and methods of control, and the importance of early diagnosis.
2) Reduce or eliminate those social conditions that increase the risk of infection, such as overcrowding.
3) Use **preventive treatment** with isoniazid.
4) Provide public health nursing and outreach services for direct supervision of patient therapy, and arrange for the examination and preventive treatment of contacts.
5) Persons infected with HIV should be skin tested by the Mantoux method, using intermediate strength PPD started on prophylactic treatment if they are PPD positive.
6) BCG immunization of uninfected (tuberculin negative) people can induce tuberculin reactivity in more than 90% of vaccines. The protection conferred has varied markedly, that protection may persist for as long as 20 years in high incidence situations, while others have shown no protection at all.
7) Eliminate bovine tuberculosis among dairy cattle by tuberculin testing and slaughter of reactors; pasteurize or boil milk.
8) Take measures to prevent silicosis among those working in industrial plants and mines.

Preventive therapy

It has been; shown to be effective in preventing the progression of latent TB infection to TB disease.

Standard Regimens: Isoniazid is normally used alone for preventive therapy in a single daily dose of 300 mg in adults and 10 to 15 mg/kg body weight in children, not to exceed 300 mg per dose.

For persons who are at especially high risk for TB and whose adherence is questionable, directly observed preventive therapy should be considered. This method of treatment is especially appropriate in institutions and facilities where preventive therapy can be observed by a staff member.

Isoniazid can be given two times a week at a dosage of 15 mg/kg as directly observed preventive therapy. Data on the effectiveness of intermittent isoniazid preventive therapy are limited. However, the results of studies of twice-weekly therapy with isoniazid suggest that it is an effective form of preventive therapy in adults and children.

Before preventive therapy is started, clinicians should:

- Rule out the possibility of TB disease (with a medical history and chest radiograph).
- Ask for a history of treatment for TB infection or disease.
- Check for contraindications, including previous isoniazid-associated hepatic injury; history of severe adverse reactions to isoniazid, such as a drug fever, rash, or arthritis; and acute or unstable liver disease of any cause.
- Determine whether the patient is at high risk for adverse reactions

CONTROL OF PATIENTS AND CONTACTS:

Treatment of cases:

1) Promptly identify, diagnose and treat potentially infectious patients with TB disease. Establish case finding and treatment facilities for infectious cases to reduce transmission.
2) Make available medical, laboratory and x-ray facilities for; prompt examination of patients, contacts and suspects; facilities for early treatment of cases and people at high risk of infection; and beds for those needing hospitalization.
3) Directly observed therapy has been shown to be highly effective and is recommended for treatment of TB disease. Patients with TB disease should be given prompt treatment with an appropriate combination of anti-microbial drugs, with regular monitoring of sputum smears. For drug susceptible disease, a 6 month regimen consisting of isoniazid

(INH), rifampin (RIF), and pyrazinamide (PZA) is recommended for the first 2 months followed by INH and PZA for 4 months. A 4 drug initial therapy (including ethambutol (EMB) or streptomycin (SM)) is recommended if the infection was acquired in areas where an increased prevalence of INH resistance has been reported.

4) Hospital treatment is necessary only for patients with severe illness and for those whose medical or social circumstances make treatment at home impossible.

DOTS is the most effective strategy available for controlling the TB epidemic today.

- DOTS produces cure rates of up to 95 percent even in the poorest countries.
- DOTS prevents new infections by curing infectious patients.
- DOTS prevents the development of multidrug-resistant tuberculosis (MDR-TB) by ensuring the full course of treatment is followed.
- A six-month supply of drugs for DOTS costs US $11 per patient in some parts of the world. The World Bank has ranked the DOTS strategy as one of the "most cost-effective of all health interventions."

Since DOTS was introduced on a global scale in 1995, over 10 million infectious patients have been successfully treated under DOTS programmes.

THE FIVE ELEMENTS OF THE DOTS STRATEGY:

DOTS has **FIVE** key components:

Government commitment to sustained TB control activities.

Case detection by sputum smear microscopy among symptomatic patients self-reporting to health services.

Standardized treatment regimen of six to eight months for at least all sputum smear- positive cases, with directly observed therapy (DOT) for at least the initial two months.

A regular, uninterrupted supply of all essential anti-TB drugs.

A standardized recording and reporting system that allows assessment of treatment results for each patient and of the TB control programme performance overall.

Political commitment: Government commitment to sustained TB control is ***essential*** for the mobilization of resources and the sustainability of TB programmes.

Case detection and diagnosis: Sputum smear microscopy is the ***most cost- effective*** method of screening pulmonary TB suspects referring to health services. It identifies sputum smear-positive, highly infectious TB cases. TB is diagnosed using patient history, clinical examination and diagnostic tests. A sputum sample is submitted to the laboratory and the results of the microscopic exam are entered into the laboratory register. The goal is for all suspects to have a sputum smear microscopy exam and for all patients diagnosed with TB to be registered and treated.

Contact management:

PPD testing of all members of the household and other close contacts is recommended: If negative, a repeat skin test should be performed 2-3 months after exposure has ended.

- If positive: chest radiographs should be obtained.
- Preventive treatment is indicated for contacts who are positive reactors and for some initially negative reactors at high risk of developing active disease, especially young (5 years old or younger) and HIV infected close contacts, at least until the repeat skin test is shown to remain negative.

Learning Objectives

1. Explain the global burden of food- and water-borne diseases
2. Identify major causative agents and their transmission routes
3. Recognize clinical manifestations and public health significance
4. Evaluate risk factors and sources of contamination
5. Assess strategies for prevention and control**

Food borne diseases are usually infectious or toxic in nature and caused by bacteria, viruses, parasites or chemical substances entering the body through contaminated food or water. Food borne pathogens can cause severe diarrhea or debilitating infections including meningitis. Chemical contamination can lead to acute poisoning or long-term diseases, such as cancer. Food borne diseases may lead to long-lasting disability and death. Examples of unsafe food include uncooked foods of animal origin, fruits and vegetables contaminated with faeces, and raw shellfish containing marine biotoxins.

Causes:

Bacteria:

- **Salmonella, Campylobacter, and Entero-haemorrhagic Escherichia coli** are among the most common food borne pathogens that affect millions of people annually – sometimes with severe and fatal outcomes. Symptoms are fever, headache, nausea, vomiting, abdominal pain and diarrhoea. Examples of foods involved in outbreaks of salmonellosis are eggs, poultry and other products of animal origin. Food borne cases with Campylobacter are mainly caused by raw milk, raw or undercooked poultry and drinking water. Entero- haemorrhagic Escherichia coli is associated with unpasteurized milk, undercooked meat and fresh fruits and vegetables.
- **Listeria** infection leads to miscarriage in pregnant women or death of newborn babies. Although disease occurrence is relatively low, listeria's severe and sometimes fatal health consequences, particularly among infants, children and the elderly, count them among the most serious food borne infections. Listeria is found in un-pasteurized dairy products and various ready-to-eat foods and can grow at refrigeration temperatures.

- **Vibrio cholerae** infects people through contaminated water or food. Symptoms include abdominal pain, vomiting and profuse watery diarrhoea, which may lead to severe dehydration and possibly death.

Viruses:

Norovirus infections are characterized by nausea, explosive vomiting, watery diarrhoea and abdominal pain. Hepatitis A virus can cause long-lasting liver disease and spreads typically through raw or undercooked seafood or contaminated raw produce. Infected food handlers are often the source of food contamination.

Parasites:

Some parasites, such as fish-borne trematodes, are only transmitted through food. Others, for example tapeworms like *Echinococcus spp*, or *Taenia solium*, may infect people through food or direct contact with animals. Other parasites, such as *Ascaris, Cryptosporidium, Entamoeba histolytica* or *Giardia*, enter the food chain via water or soil and can contaminate fresh produce.

Prions:

Prions, infectious agents composed of protein, are unique in that they are associated with specific forms of neurodegenerative disease. Bovine spongiform encephalopathy (BSE, or "mad cow disease") is a prion disease in cattle, associated with the variant Creutzfeldt-Jakob Disease (vCJD) in humans. Consuming bovine products containing specified risk material, e.g. brain tissue, is the most likely route of transmission of the prion agent to humans.

Chemicals:

Of most concern for health are naturally occurring toxins and environmental pollutants.

- **Naturally occurring toxins** include mycotoxins, marine biotoxins, cyanogenic glycosides and toxins occurring in poisonous mushrooms. Staple foods like corn or cereals can contain high levels of mycotoxins, such as aflatoxin and ochratoxin, produced by mould on grain. A long-term exposure can affect the immune system and normal development, or cause cancer.
- **Heavy metals** such as lead, cadmium and mercury cause neurological and kidney damage. Contamination by heavy metal in food occurs mainly through pollution of air, water and soil.

The burden of food borne diseases:

The burden of food borne diseases to public health and welfare and to economies has often been underestimated due to underreporting and difficulty to establish causal relationships between food contamination and resulting illness or death.

The 2015 WHO report on the estimates of the global burden of food borne diseases presented the first-ever estimates of disease burden caused by 31 food borne agents (bacteria, viruses, parasites, toxins and chemicals) at global and regional level.

The 2018 World Bank report on the economic burden of the food borne diseases indicated that the total productivity loss associated with food borne disease in low- and middle-income countries was estimated to cost US$ 95.2 billion per year, and the annual cost of treating food borne illnesses is estimated at US$ 15 billion.

TYPHOID FEVER

Learning Objectives

1. Explain the epidemiology of typhoid fever
2. Recognize the clinical features and communicability
3. Evaluate prevention and control measures

Identification:

Typhoid fever is a potentially severe and occasionally life-threatening febrile illness. The case-fatality rate of 10%-20% observed in the pre antibiotic era can be reduced to less than 1% with prompt antibiotic therapy.

Occurrence:

Typhoid fever is still common in the developing world. An estimated 22 million cases of typhoid fever and 200,000 related deaths occur worldwide each year; an additional 6 million cases of paratyphoid fever are estimated to occur annually. Each year in the United States, approximately 300 culture-confirmed cases of typhoid fever and 100 cases of paratyphoid fever are reported. Most cases (up to 75%) are acquired while traveling internationally.

The disease may occur in different patterns (sporadic, endemic or epidemic) which are known in Egypt and many countries.

Infectious agents:

S. Typhi, the typhoid bacillus.

Reservoir:

- Humans: The chronic carrier state is most common among persons infected during middle age, especially women; carriers frequently have biliary tract abnormalities including gallstones. The chronic urinary carrier state occurs in those with schistosoma infections.

Mode of Transmission:

By ingestion of food and water contaminated by feces and urine of patients or carriers.

Incubation period:

The incubation period depends on the size of the infecting dose; from 3 days to 1 month with a usual range of 8-14 days.

Clinical picture:

Symptoms can be mild or sever with insidious onset of sustained fever, severe headache, malaise, anorexia, a relative bradycardia, splenomegaly, rose spots on the trunk in 25% of white patients, nonproductive cough in the early stage of the illness and constipation more' commonly than diarrhea in adults. Many mild and atypical infections occur. Ulceration of Peyer's patches in the ileum and cecum produces intestinal hemorrhage or perforation (about 1% of cases), especially in the untreated cases. Severe forms have been described with cerebral dysfunction. Non sweating fever, mental dullness, slight deafness and parotitis may occur. The case-fatality rate is 10%-20% and can be reduced to less than 1% with prompt antibiotic therapy.

Period of communicability:

The bacilli appear in excreta, usually from the first week throughout convalescence; variable thereafter (commonly 1-2 weeks for paratyphoid).

About 10% of untreated typhoid fever patients will discharge bacilli for 3 months after onset of symptoms, and 2%-5% become permanent carriers.

Methods of control:

A. Primary prevention:

1) Educate the public about good sanitation and personal hygiene, with special emphasis on careful hand-washing and sanitary disposal of feces.
2) Provide proper water treatment and distribution systems and sewage disposal.
2) Dispose of human feces in a sanitary manner and maintain fly proof latrines.
3) Protect, purify and chlorinate public water supplies, provide safe private supplies, and avoid possible back flow connections between water and sewer systems. For individual and small group protection, and while traveling or in the field treat water chemically or by boiling.
4) Control flies by screening, spraying with insecticides and use of insecticidal baits and traps. Control fly breeding by frequent collection and disposal of garbage, and any control measures in latrine construction and maintenance.
5) Use scrupulous cleanliness in food preparation and handling; refrigerate as appropriate. Particular attention should be directed to the proper storage of salads and other foods served cold.
6) Pasteurize or boil all milk and dairy products. Supervise the sanitary aspects of

commercial milk production, storage and delivery.

7) Enforce suitable quality control procedures in industries that prepare food and drink for human consumption. Use chlorinated water for cooling during canned food processing.

8) Limit the collection and marketing of shellfish to supplies from approved sources. Boil or steam (for at least 10 minutes) before serving.

9) Instruct patients, convalescents and carriers in personal hygiene. Emphasize hand washing as a routine practice after defecation and before preparing and serving food.

10) Encourage breast feeding throughout infancy; boil all milk and water used for infant feeding.

11) Exclude typhoid carriers from handling food and from providing patient care. Identify and supervise typhoid carriers; culture of sewage may help in locating carriers. Chronic carriers should not be released from supervision and restriction of occupation until local regulations are met, often not until 3 consecutive negative cultures are obtained from feces and urine (in schistosomiasis endemic areas).

12) Immunization: Current practice is to immunize those subject to unusual exposure to enteric infections from occupation (e.g. clinical microbiology technicians) or travel to endemic areas, those living in areas of high endemicity, and household members of known carriers by Oral live attenuated vaccine or injectable polysaccharide vaccine.

B. Secondary prevention (Control of patients, contacts and the immediate environment):

1) **Report to local health authority**: Case report obligatory in most countries.

2) Isolation: **Enteric precautions while ill;** hospital care is desirable during acute illness.

3) **Concurrent disinfection**: of feces and urine and articles soiled with excreta. In communities with modern and adequate sewage disposal systems, feces and urine can be disposed of directly into sewers without preliminary disinfection.

5) **Immunization of contacts:** Routine administration of typhoid vaccine is of limited value for family, household and nursing contacts who have been or may be exposed to active cases; it should be considered for those who may be exposed to carriers. There is no effective immunization for paratyphoid fever.

6) **Investigation of contacts and source of infection:** The actual or probable source

of infection of every case should be determined by search for unreported cases, carriers or contaminated food, water, milk or shellfish.

7) **Specific treatment:** Specific antimicrobial therapy shortens the clinical course of typhoid fever and reduces the risk for death. A fluoroquinolone, most often ciprofloxacin, is used for empiric treatment in most parts of the world. Patients treated with an antibiotic may remain febrile for 3–5 days, although the height of the fever generally decreases each day.

C. Epidemic measures:

1) Search intensively for the case or carrier who is the source of infection and for the vehicle (water or food) In which infection is transmitted.

2) Selectively eliminate suspected contaminated food.

3) Pasteurize or boil milk, or exclude milk supplies and other foods suspected on epidemiologic evidence, until safety is ensured.

4) Review integrity of water and sewage systems. Chlorinate suspected water supplies. All drinking water must be chlorinated, treated with iodine or boiled before use.

5) Routine use of vaccine is not recommended (At risk group may be vaccinated).

D. Disaster implications:

Disruption of usual water supply and sewage disposal, and of controls on food and water, transmission of typhoid fever may occur if there are active cases or carriers in a displaced population.

Efforts to restore safe drinking water supplies and excreta disposal facilities are recommended.

Selective immunization of stabilized groups such as school children, prisoners, and utility, municipal or hospital personnel can be helpful.

E. International measures:

Immunization is advised for international travelers to endemic areas, especially if travel will likely involve exposure to unsafe food and water, or close contact in rural areas to indigenous populations.

Human Immunodeficiency Virus (HIV) and Acquired Immune Deficiency Syndrome (AIDS)

Here are three clear and concise learning objectives for the topic *Human Immunodeficiency Virus (HIV) and Acquired Immune Deficiency Syndrome (AIDS)*:

Learning Objectives

1. Describe the agent, reservoir, host factors, incubation period, and major modes of transmission, distinguishing between high-risk and non-risk exposures.
2. Evaluate strategies for prevention, control, and public health response

Human Immunodeficiency Virus (HIV) infection remains a major global public health challenge. HIV targets the immune system, leading to progressive immunodeficiency and increased susceptibility to opportunistic infections and malignancies. The most advanced stage of infection is Acquired Immune Deficiency Syndrome (AIDS), a life-threatening clinical condition. This chapter provides an overview of HIV and AIDS, including epidemiological determinants, modes of transmission, clinical features, diagnostic approaches, and strategies for prevention and control.

1. Agent

Human Immunodeficiency Virus (HIV) is a retrovirus that primarily infects cells of the human immune system, especially CD4+ T lymphocytes. The virus progressively weakens immune defenses by destroying or impairing immune cell function. In early stages, infected individuals may remain asymptomatic; however, as the infection progresses, immune suppression increases susceptibility to opportunistic infections and certain cancers.

The most advanced stage of HIV infection is Acquired Immune Deficiency Syndrome (AIDS). Without treatment, the transition from initial infection to AIDS typically occurs over 10–15 years, although progression rates vary among individuals.

2. Historical Background

AIDS was first recognized as a clinical syndrome in 1981. Subsequent research identified HIV as the causative agent. Before its current nomenclature was adopted, the virus was referred to by several names, including:

- Human T-cell Lymphotropic Virus Type III (HTLV-III)
- Lymphadenopathy-Associated Virus (LAV)

3. Virology and Pathogenesis

HIV belongs to the family of retroviruses and contains RNA as its genetic material. The virus infects host immune cells and integrates its genetic material into the host genome, allowing persistent infection.

The progressive destruction of CD4+ T cells leads to:

- Impaired immune responses
- Increased susceptibility to opportunistic infections
- Higher risk of malignancies
- Multi-organ system involvement, including the central nervous system

4. Epidemiological Determinants

4.1 Agent

Human Immunodeficiency Virus (HIV), an RNA retrovirus.

4.2 Reservoir and Source of Infection

Humans are the primary reservoir. Infectious body fluids include:

- Blood
- Semen
- Vaginal secretions
- Breast milk

4.3 Host Factors

Susceptibility is influenced by:

- Risk behaviors (e.g., unprotected sexual contact, needle sharing)
- Maternal infection during pregnancy or breastfeeding
- Presence of other sexually transmitted infections
- Immune status

4.4 Incubation Period

The incubation period is prolonged and variable. It may take several years for symptomatic disease to develop.

5. Modes of Transmission

HIV is transmitted through exposure to infected body fluids. Major routes include:

1. **Sexual transmission** – unprotected vaginal, anal, or oral intercourse.
2. **Blood-borne transmission** – contaminated needles, transfusion of infected blood or blood products, or unsafe medical procedures.
3. **Mother-to-child transmission** – during pregnancy, delivery, or breastfeeding.

HIV is not transmitted through casual contact such as handshakes, shared utensils, or respiratory droplets.

6. Clinical Manifestations

6.1 Early Infection

- Often asymptomatic or mild flu-like illness
- High viral replication

6.2 Chronic HIV Infection

- Progressive immune decline
- Persistent lymphadenopathy
- Increased susceptibility to infections

6.3 AIDS (Advanced Stage)

AIDS represents the late clinical stage of HIV infection. It is characterized by severe immune suppression and life-threatening opportunistic infections or malignancies. Individuals with AIDS are prone to diseases that are typically harmless in healthy people.

7. Diagnosis

Diagnosis involves both clinical evaluation and laboratory testing.

7.1 Laboratory Diagnosis

- Antibody tests (screening)
- Confirmatory tests
- Viral load measurement
- CD4 cell count assessment

7.2 Clinical Diagnosis

Evaluation of opportunistic infections, wasting, neurological involvement, or other AIDS-defining conditions.

8. Prevention and Control

Effective prevention requires a combination of behavioral, medical, and public health interventions.

8.1 Primary Prevention

- Safe sexual practices
- Use of sterile needles and syringes
- Screening of blood and blood products
- Prevention of mother-to-child transmission
- Health education and awareness

8.2 Secondary Prevention

- Early diagnosis and treatment
- Counseling and partner notification

8.3 Tertiary Prevention

- Antiretroviral therapy to control disease progression
- Management of opportunistic infections
- Long-term care and support

9. Public Health Importance

HIV/AIDS continues to impose a substantial burden on health systems worldwide. Its impact extends beyond medical consequences to include social, economic, and demographic effects. Comprehensive prevention programs, early detection, and sustained treatment access remain essential for control.

10. Summary

HIV infection is a chronic viral disease that progressively weakens the immune system and may culminate in AIDS if untreated. Understanding epidemiological determinants, transmission pathways, clinical progression, and prevention strategies is essential for effective control.

Advances in diagnosis and treatment have significantly improved outcomes, but prevention remains the cornerstone of public health response.

VIRAL HEPAIITIS B

Learning Objectives

1. Explain the epidemiology and transmission of Hepatitis
2. Evaluate prevention and control strategies

(Type B hepatitis, Serum hepatitis, Homologous serum jaundice, Australia antigen hepatitis, HB)

PROBLEM STATEMENT:

Hepatitis B is a viral infection that attacks the liver and can cause both acute and chronic disease.

WHO estimates that in 2015, 257 million people were living with chronic hepatitis B infection (defined as hepatitis B surface antigen positive).

In 2015, hepatitis B resulted in an estimated 887 000 deaths, mostly from cirrhosis and hepatocellular carcinoma (i.e. primary liver cancer).

As of 2016, 27 million people (10.5% of all people estimated to be living with hepatitis B) were aware of their infection, while 4.5 million (16.7%) of the people diagnosed were on treatment. According to latest WHO estimates, the proportion of children under five years of age chronically infected with HBV dropped to just under 1% in 2019 down from around 5% in the pre-vaccine era ranging from the 1980s to the early 2000s.

Endemicity consider high in those areas where the prevalence of chronic infection is 8% or more, intermediate if prevalence is 2-7% and low if prevalence < 2%.

Hepatitis B prevalence is highest in the WHO Western Pacific Region and the WHO African Region, where 6.2% and 6.1% of the adult population is infected respectively. In the WHO Eastern Mediterranean Region, the WHO South-East Asia Region and the WHO European Region, an estimated 3.3%, 2.0% and 1.6% of the general population is infected, respectively. And in the WHO Region of the Americas, 0.7% of the population is infected.

EPIDEMIOLOGICAL DETERMINANTS:

Agent factors: The hepatitis B virus (HBV), a hepadnavirus, is a 42-nm partially

double-stranded DNA virus composed of a 27 nm nucleocapsid core (HBcAg), surrounded by an outer lipoprotein coat containing the surface antigen (HBsAg).

Reservoir and source of infection: man is the only reservoir, cases and carriers

Host factors:

Period of communicability**:** All persons who are HBsAg positive are potentially infectious. The infectivity of chronically infected individuals varies from highly infectious (HBeAg positive) to sparingly infectious (anti-HBe positive).

Susceptibility and resistance: Susceptibility is general. Protective immunity follows infection if antibody to HbsAg (anti-HBs) develops and HBsAgis negative.

Risk factors includes:

1) Age; the outcome of infection is age-dependent. The development of chronic hepatitis B infection is inversely related to age.
2) High risk groups whether occupational (physicians, health care and laboratory personnel) or (blood recipients, homosexuals, percutaneous drug abusers, recipients of solid organ transplants and immune compromised patients).
3) Hepatitis B and HIV infection: 1% of HIV infected persons are infected with hepatitis B virus.

5. Mode of transmission:

Body substances capable of transmitting HBV include : blood and blood products;saliva ;cerebrospinal fluid ;peritoneal, pleural, pericardial and synovial fluid;amniotic fluid ;semen and vaginal secretions and any other body fluid containing blood; andunfixed tissues and organs .

Transmission occurs by percutaneous (IV, IM, SC or intradermal) and permucosal exposure to infective body fluids .

Because HBV is stable on environmental surfaces for at least 7 days, indirect inoculation of HBV can also occur via inanimate objects .

Fecal-oral or vector-borne transmission has not been demonstrated .

Major modes of HBV transmission include

Parenteral transmission:

Transmission associated with injecting drug use can occur though

Transfer of HBV infected blood by sharing syringes and needles either directly or through contamination of drug preparation equipment .

Nosocomial exposures that have resulted in HBV transmission include transfusion of blood or blood products, hemodialysis, acupuncture and needle sticks or other injuries from sharp instruments sustained by hospital personnel .

IG, heat treated plasma protein fraction, albumin and fibrinolysin are considered safe.

Perinatal transmissions

Common, especially when HBV infected mothers are also HBeAg positive .The rate of transmission from HBeAg positive mothers is more than 70% ,The rate of transmission from HBeAg negative mothers is less than 10% .

Sexual transmission

From infected men to women is about three times more efficient than that from infected women to men.

Anal intercourse, insertive and receptive, is associated with an increased risk of infection .

Incubation period: Usually 45-180 days, average 60-90 days.

DIAGNOSIS:

Clinical picture: show general picture of hepatitis. Chronic disease and complication may occur.

Laboratory diagnosis:

A laboratory diagnosis of hepatitis B (HBV) infection is dependent upon the detection of hepatitis B surface antigen in serum. The distinction between acute and chronic infection relies on the detection of other serological markers. Serum-based assays can now detect and quantify the viral DNA.

Hepatitis B virus markers

Uses :

1- To establish the diagnosis of HB 2-To study the Epidemiology of HBV 3-To screen the blood and plasma donors 4-For evaluation of active and passive immunization for protection against HBV.

HBV markers include.

1- **HBS Ag.:**

* It is the first marker identified.

* It was found in Blood as spherical or tubular form (18-22nm).

* It is the protein which makes the virus coat.

* It is detectable in early phase following infection (viraemia) or in chronic carrier state. Chronic carrier is the person whose serum is repeatedly HBs Ag positive, over a period of 6 months or longer.

2- **Anti-HBs:** (antibodies against surface antigen)

* If HBs Ag disappears, it is usually followed after few months by the appearance of anti HBS antibodies which often remain for life. It is the last antibody to appear

* Presence of anti HBS indicates immunity 3- **anti-HBc:** (antibodies against core Ag)

* It is the first antibody to appear.

* It is not protective and plays no role in the immune pathogenesis.

* It indicates previous infection by HB virus (either the person is still diseased or became cured and immune)

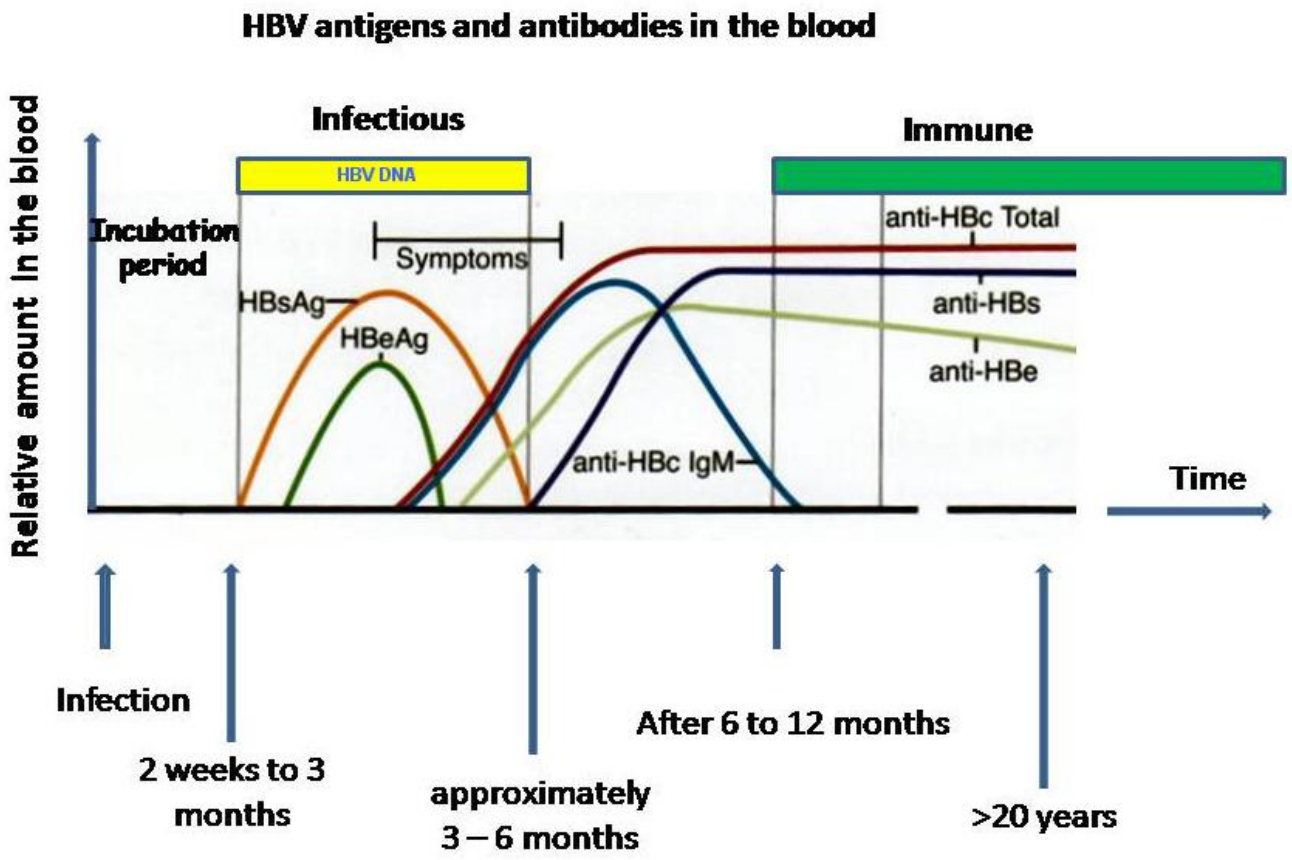
1-**HBe Ag.** its presence indicates:

*The virus is actively replicated.

*The patient is highly infectious

*The patient is more liable to complications

2- **Anti-HBe**: (Antibodies Against e antigen) its presence medicates the virus is slowly replicated, the patient is less infectious the patient is less liable to complication.



PRIMARY PREVENTION:

Active vaccination:

Effective hepatitis B vaccines have been available since 1982. Two types of hepatitis B vaccines have been licensed. Both have been shown to be safe and highly protective against all subtypes of HBV. The first type is prepared from plasma from HBsAg positive persons; it is no longer produced in the USA but is still used widely elsewhere. The second type is made by recombinant DNA (rDNA) technology.

a) In all countries, routine infant immunization should be the primary strategy to prevent HBV infection. In addition, immunization strategies can be targeted to high-risk groups, which account for most cases among adolescents and adults.

b) Testing to exclude people with preexisting anti-HBs or anti-HBc is not required prior

to immunization.

c) Immunity against HBV is believed to persist for at least 15 years after successful immunization.

d) Pregnancy is not a contraindication for receiving hepatitis B vaccine.

Schedule and dose:

The vaccine is given in a dose of 10 µg (1ml) intramuscularly at 0, 1, 6 months and a booster one might be revaccinated 5-7years later.

Persons at high risk who should routinely receive pre-exposure hepatitis B immunization include the following:

a) Sexually active men and women, including those who are diagnosed as having recently acquired other STDs, and people who have a history of sexual activity with more than one partner in the previous 6 months.

b) Men who have sex with men.

c) Sexual partners and household contacts of HBsAg positive persons.

d) Inmates of correctional facilities, including juvenile detention facilities, prisons and jails.

e) Healthcare and public safety workers who perform tasks involving contact with blood or blood contaminated body fluids.

f) Clients and staff of institutions for the developmentally disabled.

g) Hemodialysis patients.

h) Patients with bleeding disorders who receive blood products.

i) International travelers who plan to spend more than 6 months in areas with intermediate to high rates of chronic HBV infection.

Other measures:

1- Adequately sterilize all syringes and needles (Including acupuncture needles) or preferably use disposable equipment whenever possible.

2- In blood banks, all donated blood should be tested for HBsAg by positive tests (RIA or EIA); reject as donors all persons with a history of viral hepatitis, those who have a history of injecting drug use or show evidence of drug addiction.

3- Maintain surveillance for all cases of post-transfusion hepatitis, keep a register of all people who donated blood for each case. Notify blood banks of these potential carriers so that future donations may be identified promptly.

CONTROL OF PATIENTS AND CONTACTS:

Immunization of contacts: Products available for post-exposure prophylaxis include HBIG and hepatitis vaccine

a) Infants born to HBsAg positive mothers should be given a single dose of HBIG (0.5 ml IM) and vaccine within 12 hours of birth. The first dose of vaccine should be given concurrently with HBIG at birth.

b) After percutaneous (e.g., needle stick) or mucous membrane exposures to blood that contains or might contain HbsAg.

c) After sexual exposure to a person with acute HBV infection, a single dose of HBIG (0.06 ml/kg) is recommended.

Specific treatment: No specific treatment is available for acute hepatitis B. Alpha interferon and lamivudine have been licensed for treatment of chronic hepatitis B in the USA. Studies have shown that alpha interferon is successful in arresting viral replication in about 25%-40% of treated" patients.

VIRAL HEPAIITIS C

Learning Objectives

3. Explain the epidemiology and transmission of Hepatitis C
4. Evaluate prevention and control strategies

(Parenterally transmitted non-A non-B hepatitis [PT-NA-NB], Non-B transfusion associate: hepatitis, post-transfusion non-A non-B hepatitis, HCV infection)

Problem statement:

- Hepatitis C is a liver disease caused by the hepatitis C virus (HCV): the virus can cause both acute and chronic hepatitis, ranging in severity from a mild illness lasting a few weeks to a serious, lifelong illness.

- Hepatitis C is a major cause of liver cancer.
- The hepatitis C virus is a bloodborne virus: the most common modes of infection are through exposure to small quantities of blood. This may happen through injection drug use, unsafe injection practices, unsafe health care, transfusion of unscreened blood and blood products, and sexual practices that lead to exposure to blood.
- Globally, an estimated 71 million people have chronic hepatitis C virus infection
- A significant number of those who are chronically infected will develop cirrhosis or liver cancer.
- WHO estimated that in 2016, approximately 399 000 people died from hepatitis C, mostly from cirrhosis and hepatocellular carcinoma (primary liver cancer).
- Antiviral medicines can cure more than 95% of persons with hepatitis C infection, thereby reducing the risk of death from cirrhosis and liver cancer, but access to diagnosis and treatment is low.
- There is currently no effective vaccine against hepatitis C; however, research in this area is ongoing.

Epidemiological determinants

Agent:

The hepatitis C virus is an enveloped RNA virus classified as a separate genus (Hepacavirus) in the Flaviviridae family. At least six different genotypes and greater than 90 subtypes of HCV exist.

Host factors:

Reservoir: Humans

Period of communicability: From one or more weeks before onset of the first symptoms; may persist in most persons indefinitely. Peaks in virus concentration appear to correlate with peaks in ALT activity.

Mode of transmission- HCV is primarily parentally transmitted, sexual transmission has been documented to occur but is far less efficient or frequent than the parentral route.

People at high risk of HCV infection include:

- people who inject drugs;
- people in prisons and other closed settings;
- people who use drugs through other routes of administration (non-injecting);
- men who have sex with men (MsM);
- recipients of infected blood products or invasive procedures in health- care facilities with inadequate infection control practices ;
- children born to mothers infected with HCV;
- people with HIV infection;
- prisoners or previously incarcerated persons; and
- people who have had tattoos or piercings.

Incubation period: Ranges from 2 weeks to 6 months; commonly 6-9 weeks. Chronic infection may persist for up to 20 years before the onset of cirrhosis or hepatoma.

Diagnosis:

Clinical picture: show general picture of hepatitis. Chronic disease and complication may occur. Because new HCV infections are usually asymptomatic, few people are diagnosed when the infection is recent. In those people who go on to develop chronic HCV infection, the infection is also often undiagnosed because it remains asymptomatic until decades after infection when symptoms develop secondary to serious liver damage.

Laboratory diagnosis:

HCV infection is diagnosed in 2 steps:

1. Testing for anti-HCV antibodies with a serological test identifies people who have been infected with the virus.
2. If the test is positive for anti-HCV antibodies, a nucleic acid test for HCV ribonucleic acid (RNA) is needed to confirm chronic infection because about

30% of people infected with HCV spontaneously clear the infection by a strong immune response without the need for treatment. Although no longer infected, they will still test positive for anti-HCV antibodies.

Prevention:

There is no effective vaccine against hepatitis C; prevention of HCV infection depends upon reducing the risk of exposure to the virus in health-care settings and in higher risk populations.

Primary prevention interventions recommended by WHO:

- safe and appropriate use of health care injections;
- safe handling and disposal of sharps and waste;
- provision of comprehensive harm-reduction services to people who inject drugs including sterile injecting equipment and effective and evidence- based treatment of dependence;
- testing of donated blood for HBV and HCV (as well as HIV and syphilis);
- training of health personnel;
- prevention of exposure to blood during sex;

Control:

- For people infected with the hepatitis C virus, WHO recommends:
- education and counselling on options for care and treatment;
- immunization with the hepatitis A and B vaccines to prevent coinfection from these hepatitis viruses and to protect their liver;
- early and appropriate medical management including antiviral therapy; and
- regular monitoring for early diagnosis of chronic liver disease.

Tetanus

Learning Objectives

1. Explain the epidemiology and transmission of tetanus
2. Recognize the clinical manifestations and complications
3. Understand diagnosis, susceptibility, and immunity
4. Evaluate prevention and control strategies

Introduction

Tetanus is an acute, potentially fatal, vaccine-preventable disease caused by a potent neurotoxin produced by *Clostridium tetani*. Despite the availability of an effective vaccine, tetanus remains a public health problem in many developing countries, particularly among neonates and inadequately immunized adults. Unlike most other vaccine-preventable diseases, tetanus is not transmitted from person to person but occurs following contamination of wounds with spores present in the environment.

Epidemiology

Occurrence and Distribution

Tetanus occurs worldwide but is most frequently encountered in densely populated regions with hot, humid climates and soil rich in organic matter. The disease has become rare in developed countries due to high immunization coverage and improved sanitation. Globally, the incidence of tetanus has markedly declined with the expansion of DPT vaccination programs.

In 2014, approximately 11,367 tetanus cases were reported worldwide. In 2013, Egypt reported nine cases of tetanus, reflecting the impact of vaccination programs while highlighting the persistence of residual risk among non-immunized populations.

Etiological Agent

Clostridium tetani is a gram-positive, anaerobic, spore-forming bacillus. The organism characteristically develops terminal spores, giving it a drumstick or tennis-racket appearance. While the vegetative form is sensitive to heat and oxygen, the spores are highly resistant to heat, drying, and many antiseptics, and may survive autoclaving at 121°C for 10–15 minutes.

Pathogenesis

C. tetani produces two exotoxins:

1. **Tetanolysin**, whose exact role in disease remains unclear.
2. **Tetanospasmin**, a highly potent neurotoxin responsible for the clinical manifestations of tetanus.

Tetanospasmin is among the most powerful toxins known. The estimated minimum lethal dose for humans is approximately 2.5 nanograms per kilogram of body weight. The organism typically enters the body through contaminated wounds. Under anaerobic conditions, spores germinate and produce toxin, which disseminates via blood and lymphatics and acts on multiple sites within the central nervous system.

The toxin blocks the release of inhibitory neurotransmitters, leading to unopposed muscle contraction, rigidity, and spasms. Autonomic nervous system dysfunction may also occur.

Reservoir

The primary reservoirs of *C. tetani* are soil and the intestinal tracts of animals and humans. The organism is excreted in feces and persists in the environment for prolonged periods.

Mode of Transmission

Transmission occurs through contamination of wounds with tetanus spores. Both apparent and inapparent wounds may serve as portals of entry. Minor wounds now account for a greater proportion of cases, likely because major wounds are more frequently managed appropriately. Tetanus may follow burns, deep puncture wounds, crush injuries, otitis media, dental infections, animal bites, abortion, and childbirth.

Incubation Period

The incubation period ranges from 3 to 21 days, with an average of approximately 8 days. A shorter incubation period is associated with increased disease severity and higher mortality.

Clinical Manifestations

Tetanus presents in several clinical forms:

Local Tetanus

An uncommon and relatively mild form characterized by persistent contraction of muscles near the site of injury. It may last for weeks and occasionally progresses to generalized tetanus. Mortality is low (approximately 1%).

Cephalic Tetanus

A rare form, usually associated with head injuries or otitis media. Cranial nerves, particularly the facial nerve, are affected.

Generalized Tetanus

The most common form, accounting for about 80% of cases. It typically follows a descending pattern, beginning with trismus (lockjaw), facial stiffness, risus sardonicus, neck rigidity, dysphagia, and generalized muscle rigidity. Severe spasms may lead to opisthotonos, respiratory failure, and autonomic instability manifested by fever, sweating, hypertension, and tachycardia. Recovery may take several months.

Neonatal Tetanus

A form of generalized tetanus occurring in newborns born to non-immune mothers. Infection usually occurs through the umbilical stump when cut with unsterile instruments. It remains a major cause of neonatal mortality in developing countries.

Complications

Major complications include laryngospasm, respiratory failure, fractures due to sustained muscle contraction, autonomic dysfunction, nosocomial infections, aspiration pneumonia, pulmonary embolism, and secondary sepsis. Case fatality rates vary widely, ranging from 10% to 90%, with higher mortality among infants, elderly individuals, and unvaccinated persons.

Diagnosis

Diagnosis is clinical, based on characteristic signs and symptoms. There are no specific laboratory tests for tetanus. Isolation of *C. tetani* from wounds is neither sensitive nor specific and is not required for diagnosis.

Communicability

Tetanus is not contagious and does not spread from person to person. It is unique among vaccine-preventable diseases in this respect.

Susceptibility and Immunity

Susceptibility is universal. Active immunity is induced by tetanus toxoid and persists for at least 10 years after complete immunization. Passive immunity follows administration of tetanus immune globulin (TIG). Recovery from tetanus does not confer immunity; therefore, active immunization is required even after disease.

Prevention and Control

Primary Prevention

- Health education on the importance of complete immunization.
- Promotion of safe wound care.
- Environmental sanitation and community development.
- Mechanization of agriculture and reduction of exposure to animal waste.

Specific Prevention

Active Immunization

Tetanus toxoid is available as:

- DTaP or DTP (children)
- DT (children under 7 years)
- Td (adults)
- Monovalent tetanus toxoid (TT)

World Health Organization recommends a five-dose childhood schedule, with primary immunization in infancy and booster doses during childhood and adolescence.

Passive Immunization

Human tetanus immune globulin (HTIG) or equine antitoxin provides temporary protection and is used in post-exposure prophylaxis.

Post-Exposure Prophylaxis

Management depends on wound type and immunization status and includes wound cleaning, debridement, antibiotics, tetanus toxoid booster, and TIG when indicated.

Secondary Prevention

- Prompt notification to health authorities.
- Administration of TIG and antibiotics such as metronidazole.
- Airway management and supportive care.
- Initiation of active immunization during treatment.

Tetanus Neonatorum

Neonatal tetanus remains a serious public health issue in areas with inadequate maternal immunization and unsafe delivery practices. In 2013, approximately 49,000 neonatal deaths worldwide were attributed to neonatal tetanus, representing a 94% reduction since the late 1980s. Prevention relies on:

- Immunization of women of childbearing age.
- Skilled attendance at delivery.
- Strict aseptic care of the umbilical stump.

A total of five doses of tetanus toxoid provides protection throughout the reproductive years.

Conclusion

Tetanus is a preventable yet potentially fatal disease that reflects gaps in immunization coverage and safe health practices. Strengthening vaccination programs, improving maternal and neonatal care, and enhancing community awareness remain central to tetanus elimination, particularly in developing countries

Epidemiology of Zoonosis

Learning Objectives

1. Explain the epidemiology and determinants of rabies
2. Recognize modes of transmission, incubation, and clinical features
3. Understand diagnosis, treatment, and case management

   Evaluate prevention and control strategies

A zoonosis is an infectious disease that has jumped from an animal to humans. Zoonotic pathogens may be bacterial, viral or parasitic, or may involve unconventional agents and can spread to humans through direct contact or through food, water or the environment. They represent a major public health problem around the world due to our close relationship with animals in agriculture, as companions and in the natural environment. Zoonosis can also cause disruptions in the production and trade of animal products for food and other uses.

Zoonosis comprise a large percentage of all newly identified infectious diseases as well as many existing ones. Some diseases, such as HIV, begin as a zoonosis but later mutate into human-only strains. Other zoonosis can cause recurring disease outbreaks, such as Ebola virus disease and salmonellosis. Still others, such as the novel coronavirus that causes COVID-19, have the potential to cause global pandemics.

RABIES

Rabies is acute 100% fatal, viral infection of CNS caused by Lyssa virus type 1, it is primarily zoonotic disease and the only infectious diseases that is always fatal and death occurs mainly due to respiratory failure.

PROBLEM STATEMENT

Rabies occurs in more than 150 countries and leads to thousands of deaths every year, mostly in Asia and Africa.

40% of people who are bitten by suspect rabid animals are children under 15 years of age.

Every year, more than 15 million people worldwide receive a post-bite vaccination.

This is estimated to prevent hundreds of thousands of rabies deaths annually.

Rabies is a neglected disease of poor and vulnerable populations whose deaths are rarely reported and where human vaccines and immunoglobulin are not readily available or accessible.

Dogs are the source of up to 99% of all rabies transmissions to humans.

Rabies elimination is feasible by vaccinating dogs.

EPIDEMIOLOGIC DETERMINANTS:

Agent: Rabies virus, a rhabdovius of the genus Lyssa virus.

Source of infection: to man saliva of a rabid animal is the source. Many wild and domestic Canidae, including dogs, foxes, coyotes, wolves and Jackals; also skunks, raccoons, mongooses and other biting mammals act as reservoir.

Host and environmental factors:

Host factors: All warm blooded animals including man are susceptible to rabies. Rabies in man is a dead- end infection.

Susceptible host in enzootic areas include 1-24 years old persons

Laboratory staff working with rabies virus, veterinians, dog handlers, hunters and field naturalists.

Environmental factors: Rabies is an enzootic and epizootic disease of worldwide importance, some countries have rabies free status by vigorous illumination campaigns.

Mode of transmission:

- **Bite route:** Virus laden saliva of a rabid animal is introduced by a bite or scratch (or, very rarely, into a fresh break in the skin or through intact mucous membranes).
- **Non bite route**: Organ (corneal) transplants taken from persons dying of undiagnosed CNS disease have resulted in rabies in the recipients. Respiratory transmission documented in human deaths due to occupational exposure.

Incubation period: Usually 3-8 weeks. Depending on:

1- The severity of the wound, 2- Site of the wound in relation to the richness of the nerve supply and its distance from the brain 3- Amount and strain of virus introduced 4- Protection provided by clothing and other factors.

DIAGNOSIS:

Clinical picture: Onset begins by headache, fever, malaise and indefinite sensory changes often referred to the site of a preceding animal bite. Excitability and aerophobia are frequent symptoms. The disease progresses to paresis or paralysis; spasm of swallowing muscles leads to fear of water (hydrophobia); delirium and convulsions follow.

Laboratory diagnosis: clinical diagnosis of hydrophobia can be made on the basis of history of bite by a rabid animal and characteristics signs and symptoms. Rabies can be confirmed in patients early by antigen detection using immunofluorescence of skin biopsy and by virus isolation from saliva and other secretions. Neutralizing antibosies are not usually detectable in serum or CSF before the eighth day. Treatment:

TREATMENT:

No specific treatment, Case management includes:

1) Isolation in quite room away from stimulation. 2)Relieve of relative anxiety and pain by the use of sedatives and morphine.

3)Ensure hydration and diuresis. 4) Intensive therapy in the form of respiratory and cardiac support.

Prevention

Control or eradication of rabies in animals:-

1) **Registration and immunization:** Register, license and immunize all dogs in enzootic countries; immunize all cats. Educate pet owners and the public that restrictions for dogs and cats are important
2) **Killing of stray dogs and cats:** collect and euthanize ownerless animals and strays. In Egypt the ratio of stray dogs to owned dogs is estimated to be 15:1. Accordingly, the control measures are aimed mostly to eliminate stray dogs.
3) **Active surveillance:** Maintain active surveillance for rabies in animals. Laboratory capacity should be developed to perform FA testing on all wild animals involved in human or domestic animal exposures and all domestic animals clinically suspected of having rabies.

Protection of susceptible host:

1- Pre exposure prophylaxis:

Vaccination of high risk group are: The vaccine can be given in three 1.0 ml (IM) doses on days 0, 7 and 21 or 28: This regimen has been so satisfactory that routine post immunization serologic testing is not routinely recommended but may be advisable for groups at high risk of exposure or immunodeficient persons.

If risk of exposure continues, either single booster dose is given, or preferably serum is tested for neutralizing antibody every 2 years, with booster doses given when indicated.

2- Post exposure management:

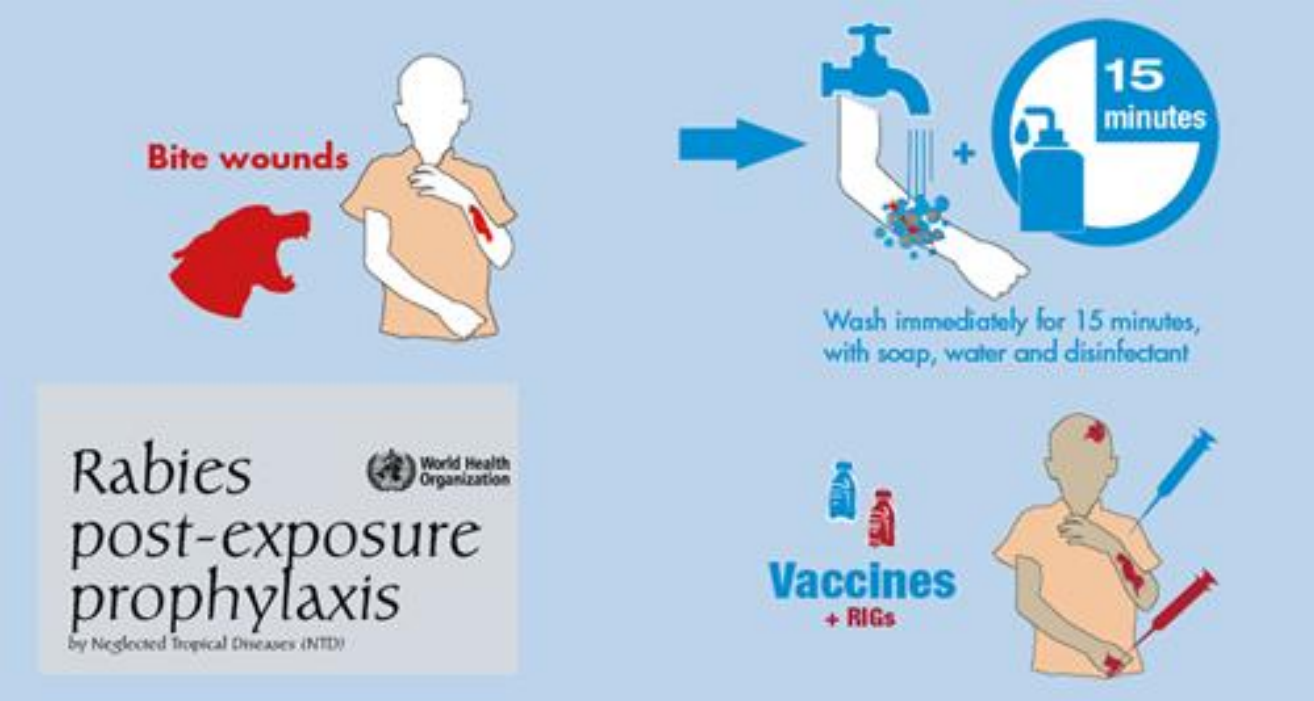
It includes: 1- wound management 2- administration of HRIG 3- administration of HDCV

Checklist for Treatment of Animal Bites:

1. Clean and Hush the wound immediately (first aid).
2. Thorough wound cleansing under medical supervision.
3. Rabies immune globulin and/or vaccine as indicated.
4. Tetanus prophylaxis and antibacterial treatment when required.
5. No sutures or wound closure advised unless unavoidable.

Wound management: The most effective rabies prevention is immediate and thorough cleaning with soap or detergent and Hushing with water all wounds caused by an animal bite or scratch. The wound should not be sutured unless unavoidable for cosmetic or tissue support reasons. Sutures, if required, should be placed after local infiltration of antiserum they should be loose and not interfere with free bleeding and drainage.

Specific immunologic protection: Immunologic prevention of rabies in humans is provided by administration of human rabies immune globulin (HRIG) as soon as possible after exposure to neutralize the virus at the bite wound site, and then by giving vaccine at a different site to elicit active immunity. HRIG should be used in a single dose of 20 lU/kg. Half should be infiltrated into and around the bite wound if possible, and the rest given IM.


Bite wounds
15 minutes
Wash immediately for 15 minutes, with soap, water and disinfectant
Rabies post-exposure prophylaxis
World Health Organization
by Neglected Tropical Diseases (NTD)
Vaccines
+ RIGs

Health care-associated infections

By the end of the lecture, students should be able to:

- Define and identify the HAIs.
- Know the magnitude of the HAIs problem
- Describe the epidemiology of HAIs
- Identify the general and specific prevention and control measures for HAIs.

Health care-associated infections, or infections acquired in health-care settings are the most frequent adverse event in health-care delivery worldwide. Hundreds of millions of patients are affected by health care-associated infections worldwide each year, leading to significant mortality and financial losses for health systems. The prevalence of HAI is expected to be about 7 to 12 percent in the developed countries and 25 to 40 percent developing countries.

Definition of health care-associated infections

Health care-associated infections, or "nosocomial" and "hospital" infections, affect patients in a hospital or other health-care facility, and are not present or incubating at the time of admission. They also include infections acquired by patients in the hospital or facility but appearing after discharge, and occupational infections among staff.

Causes of Health care-associated infections

Health care-associated infections is usually due to failure to observe aseptic precautions while carrying out the hospital procedures such as surgical operations, IV infusions, catheterization, dressings of the wounds, lumber puncture, giving injections, etc.

Predisposing factors:

- Development of the resistance by the organisms to the commonly used drugs
- Overcrowding of the hospitals
- Poor environment of the hospitals, both inside and outside
- Decreased resistance and increased susceptibility of the vulnerable groups of patients such as those suffering from tuberculosis, leprosy, diabetes, severe PEM, anemia, cardiac patients, old age of the patients, those who have undergone major surgery, those who are on steroids, cytotoxic drugs, etc.

Etiological agents:

Fifty percent of HAIs are due to staphylococci and about 45 percent are due to gram-negative bacilli such as E. coli, Klebsiella, Proteus, Pseudomonas, Salmonella, Shigella, etc. Remaining 5 percent are caused by viruses, protozoa, rickettssiae, fungi, etc.

Among the viruses, emergence of HBV and HIV has worsened the situation.

Predominantly, there are four types of HAIs:

- **Urinary tract infections**, which may be symptomatic and asymptomatic
- **Infections of the lower respiratory tract**
- **Postoperative infections**—any surgical wound resulting in purulent discharge, is considered as HAI
- **Systemic infections**, revealed by positive blood culture.

Sources and Reservoirs of HAIs: Health care-associated can be derived from:

1. **The patient's own flora**: **Self-infection (autoinfection).** Under normal conditions, the organisms concerned is not pathogenic. But due to the hospital procedure, nonpathogen assumes pathogenic property and results in an infection.

2. **The flora of another patient: Cross-infection**. In such cases the microorganism concerned is transmitted:

a. By direct contact between patients (patient's hands, saliva droplets)

b. By air (dust from fabrics carrying a patient's flora).

c. By the staff:

– Who collects the microorganisms directly on their hands or clothes and transmit them to another patient.

– Who harbors the microorganisms on the mucosa of their own respiratory and intestinal tracts, where they are reproduced and transmitted (i.e. carriers).

3. Environmental sources:

a. **Air**: Hospital air and dust usually harbors more bacteriae, which are often pathogenic and multidrug resistant.

b. **Surfaces** contaminated by patient's secretions, excretions, blood and body-fluids, animals and insects.

c. **Inanimate objects:**

- **Contaminated by the patients**: Hospital equipment, such as thermometer, linen, bed, table, cot, sanitary installations, etc. medical equipment such as endoscopes, catheters, needles, lancets, spatula, vesical probes, etc.
- **Contaminated by the hands of any hospital staff** in any part of the hospital (kitchen, laundry, treatment room, etc.)
- **Contaminated by visitors**
- **Contaminated by staff** who are ill or are carriers of microorganisms
- **Contaminated by food or contaminated water**
- **Contaminated by animals and insects.**

Thus, man occupies a central position:

a. As **reservoir** and source of microorganisms

b. As **disseminator** (communication routes)

c. As **recipient** or target, thus becoming a new reservoir.

All patients in the hospital are potential recipients of cross-infection.

Those who are severely ill, chronically ill and on steroid therapy are all highly susceptible. So cross-infection is greater in intensive care units, postoperative wards, pediatric wards and geriatric wards.

Age incidence: HAI can occur in any age group. But it is high in pediatric and geriatric age groups.

Sex incidence: HAI is equal in both the sexes.

Mode of Transmission: Health care-associated spread by the various routes as follows:

a. By contact: Close physical contact among the patients themselves by touch or with the doctor or staff.
b. By droplet infection
c. By infected dust: Which are released and spread into the atmosphere while sweeping the wards or while bedmaking, etc.
d. By fomites (inanimate objects mentioned already)
e. By parenteral or percutaneous route while carrying out the hospital procedures (Iatrogenic).

Prevention and Control of Nosocomial Infections

1. **Elimination of the Reservoirs or Sources**

- **Care of the staff:** Hospital staff suffering from any infectious disease (respiratory or alimentary or any septic skin lesions) should not attend to their duties until complete recovery.
- **Care of the patients:** All acute infectious cases must be admitted in the isolation ward only.
- **Treatment**: This should be given correctly and completely with specific antibiotics.
- **Nursing care:**
    - **Barrier nursing**: This means that the hospital staff must act as barriers to prevent cross-infection by wearing gowns, gloves, masks, etc. while touching or examining certain acute infectious cases.
    - **Task nursing**: This means posting special nurses for special duties. For example, a nurse attending to feeding of a premature child should not attend to the toilet of the child. Such care is essential in intensive care unit, premature baby-ward, etc.

2. **Breaking the Channel of Transmission**

1. **Contact transmission** can be prevented or broken by disinfection of hands with soap and water or dettol lotion soon after touching the patient suffering from contagious diseases.

2. **Isolation** of the infectious cases in the isolation ward.

3. **Concurrent disinfection** of infectious patients' body discharges like urine, saliva, sputum, excreta, etc. and also his clothes, utensils and other fomites must be carried out as long as the patient is in the hospital.

4. **Droplet** mode of transmission can be prevented by:

– Avoiding overcrowding in the wards

– Enough bed spacing of at least 12' between the patients

– Adequate lighting

– Adequate ventilation, specially cross-ventilation in the wards

5. **Control of infected dust**: By vaccum cleaning of the floor followed by wet-mopping of the floor preferably with phenyl.

6. **Transmission by various hospital procedures** can be prevented by observing strict aseptic precautions. Postoperative dressings should be made by meticulous care, caution and technique.

7. **Hospital refuse** like waste bandage cloth, dressings, plaster, etc. are all best disposed off by incineration. However, the needles and syringes are disinfected before disposal.

8. **Hospital premises** should always be kept clean both inside and outside.

3. Protection of Susceptible Persons

1. Highly susceptible persons should be admitted preferably in the isolation wards.
2. They are protected with vaccinations, e.g. tetanus, hepatitis B, gas gangrene, etc.
3. Universal blood and body fluid precautions to be adopted by all the health care workers.

4. Administrative Measures

There should be a Hospital Infection Control Committee for surveillance of HAIs, to draw guidelines for different high risk procedures, early detection and control of outbreak of HAI and also to formulate policies (regarding admission of infectious cases, isolation facilities, disinfection procedures, etc.) and to implement them.

General epidemiology of non-communicable diseases

By the end of the lecture, students should be able to:

- Identify what are chronic diseases.
- Know the Magnitude of the problem
- Identify both the human and financial burden of chronic diseases.
- Identify both modifiable and non-modifiable risk factors of chronic diseases.
- Know the preventive measures of chronic diseases.

Definition of chronic diseases

Chronic diseases and conditions have been variously defined. An **EURO symposium in 1957** gave the following definition . "An impairment of bodily structure and/or function that necessitates a modification of the patient's normal life, and has persisted over an extended period of time". **Another definition by EURO symposium in 1965 is** "a chronic illness is one of long duration in which the patient is permanently incapacitated to a more or less marked degree" **The Commission on Chronic Illness in USA** has defined "chronic diseases" as "comprising all impairments or deviations from normal, which have one or more of the following characteristics:

- are permanent
- leave residual disability
- are caused by non-reversible pathological alteration
- require special training of the patient for rehabilitation
- may be expected to require a long period of supervision, observation or care"

Definition of non-communicable diseases

Non-communicable diseases are the diseases that are not transferred from an infected person to another via any means and are mostly caused by factors like improper lifestyle and eating habits.

Non-communicable (NCDs) include diseases

- cardiovascular,
- renal,
- nervous and mental diseases,
- musculo-skeletal conditions such as arthritis and allied diseases,

- chronic non-specific respiratory diseases (e.g., chronic bronchitis, emphysema, asthma),
- permanent results of accidents,
- senility,
- blindness,
- cancer,
- diabetes,
- obesity and
- various other metabolic and degenerative diseases and chronic results of communicable diseases.

Magnitude of the Problem

Chronic non-communicable diseases (NCDs) are increasing in importance among adults in both developed and developing countries. Cardiovascular diseases and cancer are currently the leading causes of death in developed nations.

The global prevalence of chronic diseases is rising, and this trend is expected to continue due to:

- **Increased life expectancy:** More people are living to older ages and becoming more vulnerable to chronic conditions.
- **Lifestyle changes:** Rapid shifts in behavior and lifestyle are contributing to a higher risk of NCDs.
- **Improved medical care:** Advances in healthcare allow people with chronic diseases to live longer, increasing the number of individuals living with these conditions.

The impact of chronic diseases on the lives of people is serious when measured in terms of loss of life, disablement, family hardship and poverty, and economic loss to the country. Developing countries are urged to take preventive measures to avoid future "epidemics" of NCDs associated with socioeconomic change. Over 60% of global deaths are due to NCDs. Among NCDs, the highest mortality is caused by cardiovascular diseases, followed by cancers, chronic respiratory diseases, digestive diseases, and diabetes.

Burden of chronic diseases

Human Burden

Mortality figures alone may **overstate** the impact of chronic diseases. **Years of potential life lost (YPLL)** is a better measure for deaths occurring at younger ages. Many chronic diseases now begin earlier in life, making mortality a **less meaningful indicator** of total burden. Quality-adjusted life years [QALY] and disability adjusted life years [DALY]) provide a clearer picture of

the true impact on life quality and disability. Overall, the **human cost of chronic diseases is enormous and rising**.

Financial Burden

In economic terms, **death costs less** because medical spending stops. Living with chronic disease is **very expensive** due to long-term care needs. Over **75% of U.S. Medicare spending** goes toward chronic disease care. Better treatments prolong life but also **increase total healthcare costs**. New technologies and drugs improve outcomes but raise expenses. **Indirect costs** as reduced productivity, absenteeism, and presenteeism are very high and may exceed direct costs.

Common Elements in Pathogenesis (Risk factors):

Non-modifiable risk factors

- Age
- Sex
- Genetic factors
- Ethnicity

Modifiable risk factors

- **Tobacco**: accounts for over 7.5 million deaths every year (including from the effects of exposure to second-hand smoke) and is projected to increase markedly over the coming **years.**
- **Insufficient physical activity:** Approximately 3.2 million people die each year due to physical inactivity.
- **Harmful use of alcohol:** Approximately 3.3 million die each year from the harmful use of alcohol, accounting for about 4% of all deaths in the world.
- **Unhealthy diet:** high salt consumption is an important determinant of high blood pressure and cardiovascular risk. High consumption of saturated fats and trans-fatty acids is linked to heart disease.
- **Raised blood pressure:** It is a major risk factor for cardiovascular disease.
- **Overweight and obesity**: Risks of heart disease, stroke and diabetes increase steadily with Increasing body mass index (BMI). Raised BMI also increases the risk of certain cancers.
- **Raised cholesterol:** it increases the risk of heart disease and stroke.

- **Cancer associated infections:**
    - The principal infectious agents are human papillomavirus, Hepatitis B virus, Hepatitis C virus and Helicobacter pylori.
    - These infections are largely preventable through vaccinations and measures to avoid transmission, or treatable.
- **Environmental risk factors :**
    - occupational hazards,
    - air and water pollution, and
    - possession of destructive weapons in case of injuries

Prevention of Chronic Diseases

- Of particular interest to epidemiologists is the strong body of evidence suggesting that 80% of chronic disease is potentially preventable by means already available, and even genetic risk factors for chronic disease development and progression are modifiable by the effective application of lifestyle interventions.
- Some prevention and control tasks for non-communicable diseases:

1. **Tobacco avoidance**: Avoid tobacco in all its forms, including exposure to secondhand smoke.
2. **Healthy diet:**
    - **Eat properly.**
        - **Reduce** your consumption of saturated fat sweets, fried foods, red and processed meats, and full-fat dairy products, which may increase the risk of colon cancer and a more aggressive form of prostate cancer.
        - **Increase** your consumption of fruits, vegetables, and whole grains.
        - **Eat no more than a teaspoon of salt per day**. Too much sodium can raise your blood pressure.

3. **Exercise: Exercise regularly.**

- Physical activity has been linked to a reduced risk of colon cancer. Exercise also appears to reduce a woman's risk of breast and possibly reproductive cancers. Exercise will help protect you even if you don't lose weight.

4. **Avoid obesity:** Obesity increases the risk of many forms of cancer.

5- **Avoid alcohol** as it increases the risk of cancers of the mouth, larynx (voice box), esophagus (food pipe), liver, and colon; it also increases a woman's risk of breast cancer. Smoking further increases the risk of many alcohol-induced malignancies.

6- **Avoid unnecessary exposure to radiation.**

7- **Avoid exposure to industrial and environmental toxins** such as asbestos fibers, benzene, aromatic amines, and polychlorinated biphenyls (PCBs).

8- **Avoid infections that contribute to cancer**, including hepatitis viruses, HIV, and the human papillomavirus. Many are transmitted sexually or through contaminated needles.

9- **Reduce stress**, some stress is inevitable in this fast-paced world. But if you’re stressed-out day after day, it can increase blood pressure and damage your artery walls.

10- **Get good quality sleep**, a lack of sleep can do more than leave you yawning; it can harm your health. People who don't get enough sleep have a higher risk of obesity, high blood pressure, heart attack, diabetes and depression.

11- **Get regular health screenings**, Blood pressure, Cholesterol levels, Type II diabetes and cancer screening.

CORONARY HEART DISEASE (CHD)

By the end of the lecture the student will be able to :

- Define CAD
- Know the Magnitude of the problem of each disease.
- Identify both modifiable and non-modifiable risk factors of CAD.
- Know the preventive measures of each disease (hypertension, CAD and stroke).

Definition: It is the impairment of the function of the heart due to inadequate blood flow to the myocardium, as a result of obstruction in the coronary circulation (atherosclerosis, plaque, thrombus).

Coronary artery disease (CAD) is manifested in any of the following forms:

- Angina pectoris of effort
- Myocardial infarction
- Irregularities of the heart
- Cardiac failure
- Cardiac arrest

Magnitude of the problem:

Coronary artery disease (CAD) is the leading cause of death in all the developed countries, accounting for 25 to 30 percent of total deaths. In the developing countries, there has been an increasing trend in the incidence of CAD because of changes in the lifestyle and behavior pattern of the people, so much so CAD accounts for nearly 15 percent of all deaths. Therefore, CAD is considered as 'modern epidemic'.

The incidence is 2 to 3 times greater in urban areas than rural areas. It is one and a half times more among men than among women. Incidence is maximum in the age group of 50 to 60 years.

Risk Factors: These are grouped into nonmodifiable and modifiable risk factors.

Non-modifiable Risk Factors

- **Age** (>50 years, peak at 50–60 years)
- **Male sex**
- **Family history**
- **Genetic predisposition** (cholesterol, LDL regulation)

Modifiable Risk Factors

- Smoking (major risk, dose-dependent)
- Hypertension (accelerates atherosclerosis)
- High serum cholesterol (>220 mg/dl)
- Diabetes mellitus (2–3 times risk)
- Obesity (linked to LDL, hypertension, diabetes)
- Physical inactivity (low HDL, increase obesity/ hypertension)
- Hormones (oral contraceptives, hyperestrogenemia)
- Type A personality (stress-prone)
- Alcohol (heavy use)
- Environmental factors: soft water, chronic noise (>110 dB)
- Drugs (fenfluramine, phentermine misuse)
- High serum homocysteine (>15.5 µmol/L)
- Other biochemical factors: hyperuricemia, increase ferritin, increase ceruloplasmin

Prevention of Coronary Artery Disease

Primary Prevention:

The purpose of primary prevention is to limit the incidence of disease by controlling causes and risk factors.

Primary prevention depends on widespread changes that reduce the average risk in the whole population.

1. **Dietary Changes**
   - Consumption of **saturated fats** should be less than 10 percent of total energy intake.
   - The average intake of **cholesterol** should be less than 300 mg/day/adult.
   - The **serum cholesterol** level should be less than 200 mg/dl.
   - The consumption of **carbohydrates** must be proportionately increased.
2. **Smoking changes:** The goal is to achieve 'Smoke-free' society and many countries are at it. To achieve this, it requires educational activities, legislative measures and other programs.

3. **Blood pressure:** Studies have shown that a small reduction in the blood pressure in the population would produce a large reduction in the incidence of CAD. This involves a multifactorial approach based on prudent diet (reduced salt intake and avoidance of high alcohol intake), regular physical activity and control of weight.
4. **Physical activity:** Regular physical exercise will go a Longway in preventing obesity, hypertension and thus indirectly coronary artery disease.
5. **Discovery of genetic predisposition:** many diseases leading in time to early arteriosclerosis can be discovered and dealt with, thereafter for early and successful prevention of disease e.g. familial hyper cholestrolaemia, a lipoprotein b and a, hypertriglyceridemia, high blood pressure in infant and young children who were shown to become hypertensive in adult age and obesity.
6. **Mental health promotion**: relaxed lifestyle, with satisfactory recreation and avoid stress and worry.
7. **Control of oral contraception.**
8. **Prevention and control of pathological risk factors**, specially diabetes, hypertension and obesity.
9. **Health education** to adopt group sports and exercises to maintain comfortable working condition, good family relations and personal satisfaction.

Secondary Prevention

This is the action which stops the progress of the disease in its early stage and prevents complications. This consists of prevention of recurrence of CAD by cessation of exposure to risk factors, e.g. giving up smoking after an attack of myocardial infarction, followed by bypass surgery, prompt treatment with antihypertensives after detecting hypertension, similarly starting antidiabetic drugs after detection of diabetes mellitus, etc.

DIABETES MELLITUS

By the end of the lecture, students should be able to:

By the end of the lecture the student will be able to:

- Define DM
- Identify types of DM
- Know the Magnitude of the problem of DM
- Describe the epidemiological factors affecting DM.
- Identify both modifiable and non-modifiable risk factors of DM.
- Describe complications of DM
- Know the preventive measures of both types of DM

Diabetes in Greek means siphon (passing water); Melitus in Latin means honey (sweet).

It is a metabolic syndrome, clinically characterized by polyuria, polyphagia, polydypsia, hyperglycemia and glycosuria, due to absolute or relative deficiency of the hormone insulin (either by action or by secretion or both), that controls the metabolism of carbohydrate, protein, fat and electrolytes.

The underlying cause of DM is deficiency of insulin, which is absolute in IDDM and relative or partial in NIDDM.

CLASSIFICATION OF DIABETES MELLITUS

Primary:

Type 1 (Insulin dependent diabetes mellitus) (IDDM).

- Absolute insulin deficiency
- Juvenile onset (<30 years)
- Sudden onset, thin build, autoimmune association, HLA-linked
- Acute complications, family history usually absent

Type 2 (Noninsulin dependent diabetes mellitus) (NIDDM)

- Relative insulin deficiency
- Maturity onset (>30 years)
- Gradual onset, obesity-related, lifestyle-linked
- Chronic complications, family history common

Secondary Diabetes

- Pancreatic pathology (e.g., pancreatitis, tumors, cystic fibrosis)
- Other causes: hemochromatosis, trauma, autoimmunity
- Hormonal excess (catecholamines, GH, glucagon, glucocorticoids)
- Long-term drug use (corticosteroids, thiazides, OCPs)
- Liver disease, genetic syndromes

Magnitude of the Problem

- Diabetes mellitus is an 'Iceberg disease'. It is an ubiquitous malady of the world today, affecting about 150 million people worldwide.
- As of 2025, about 589 million adults (aged 20–79 years) worldwide are living with diabetes.
- 4th leading cause of death in the USA.
- Rising in younger populations due to urbanization, industrialization, lifestyle changes.
- Impaired Glucose Tolerance (IGT): Intermediate state, prevalence 3.6–9.1%.

Diagnostic criteria (OGTT):

- Fasting blood sugar >125 mg/dL
- 2-hour post-glucose load >180 mg/dL

Complications

Acute

- Hypoglycemia
- Hyperglycemia & ketoacidosis
- Infections (fungal, urinary, anaerobic, mycobacterial)

Chronic

- Macrovascular: Atherosclerosis → cerebrovascular & coronary artery disease
- Microvascular: Retinopathy, nephropathy, neuropathy
- Foot ulcers & infections
- Increased risk of ischemic heart disease (3.5–4 times higher)

- Often coexists with hypertension, dyslipidemia, obesity, metabolic syndrome

Risk factors of DM

Non-modifiable risk factor

1. Age: Diabetes can occur in any age group. Type 1 is common among younger age group below 30 and type 2 among middle aged and elderly. Thus, age constitutes a risk-factor for DM specially type 2. Younger the age, worse is the prognosis.

2.Sex: Type 1 DM is common among men and type 2 among women.

3.Genetic Factors

Type 2 shows 90 percent concordance genetic component whereas type 1 shows only 50 percent. Maturity onset diabetes of the young (MODY) is an autosomal dominant hereditary disease. If both the parents are such diabetics, the children have 100 percent chance of developing it.

Type 1 has HLA (human leukocyte antigen) linked genetic predisposition, whereas type 2 is not associated with HLA.

Modifiable risk factor

1. **Diet:** A high consumption of sugar, saturated fat intake, and conservative diet 'consisting of butter, potatoes and whole milk' is associated with an increased risk of type 2 DM.

Studies have also shown that 'Prudent diet' consisting of fruits and vegetables and diet containing nonstarch polysaccharides (NSP) like cereals have a potential protective effect on the development of DM.

2. **Sedentary lifestyle** with lack of exercise, associated with over nutrition, and obesity, is a risk factor for DM type 2.

3. Obesity - Particularly central adiposity (waist circumference or waist: hip ratio) is a greater risk factor than BMI specially for development of type 2 DM. Obesity increases the insulin resistance and/or reduces the number of insulin receptors on target cells. Obesity plays no role in type 1 DM.

4. Stress and strain: Such as pregnancy, surgery, trauma and such other stressful situations may 'bring out' the disease.

5. Alcoholism: Excessive intake of alcohol can increase the risk of type 2 DM by damaging the liver and pancreas by promoting obesity.

Other factors:

1. Socioeconomic class: Prevalence of DM is no longer related to socioeconomic status.

2. Immunological factors: Type 1 DM is a slow autoimmune disease, occurring over many years.

3. Viral infections: The important viruses that can result in the development of DM are Rubella, Mumps, Human coxsackie B4, EMC virus, Mengo virus 2T and Rheo virus type I. They tend to result in type 1 DM.

4. Pregnancy: Can cause gestational diabetes; repeated pregnancies increase risk of permanent DM.

Prevention and Care

Primary Prevention

Through elimination or modification of risk factors. The main points are:

- Improvements in the nutritional habits (such as avoiding sweets and fatty foods, alcohol, diabetogenic drugs like oral pills, steroids, regular and adequate intake of proteins, high intake of dietary fibers)
- Maintenance of body weight by doing moderate exercise.
- Correction of obesity
- Avoiding over nutrition and alcohol
- Changing lifestyle
- Regular exercises
- Maintenance of normal body weight
- Avoidance of oral contraceptives and steroids
- Reduction of factors promoting atherosclerosis such as hypertension, smoking, cholesterol level, etc.
- Yoga exercises and meditation goes a long way in the prevention and control of diabetes mellitus.

Secondary Prevention

It is by early diagnosis and prompt treatment of DM. Early diagnosis is done by screening procedures.

The aims of treatment are:

- To maintain normal blood glucose level
- To maintain normal body weight.

The principles of treatment are:

- Diet alone
  - small balanced meals more frequently
  - No fast and no feast.
  - More of raw vegetables, less of cereals.

• Diet and oral antidiabetic drugs, i.e. restriction of the diet associated with oral hypoglycemic drugs for type 2 DM.

• Diet and insulin, i.e. restriction of diet and administration of insulin injection daily for Type 1 DM.

The ultimate objective of the management of DM is to prevent complications.

Tertiary Prevention

• **Disability limitation**: This is to limit the development of further disability in a diabetic person, who comes late with complications. This consists of giving an intensive treatment in diabetic clinics.

• **Rehabilitation:** This is given for those who have become disabled and handicapped due to complications resulting in blindness, amputation of leg, etc. They are rehabilitated physically, mentally, socially, psychologically and vocationally.

CANCER

By the end of the lecture the student will be able to :

- Define cancer
- Know the Magnitude of the problem of cancer
- Describe the epidemiological factors affecting cancer.
- Identify the most prevalent types of cancer.
- Identify both modifiable and non-modifiable risk factors of DM.
- Know the preventive measures of cancer

Cancer is a most fearful disease, next to AIDS and Rabies. It is characterized by the following features:

- Abnormal and uncontrolled growth of the cells.
- The presence of aberrations in the nucleus.
- Ability to invade the surrounding tissues and even distant organs (metastasis) later.
- Eventual death of the person, if the tumor has progressed beyond a certain stage at which it can be successfully removed.
- Cancer can occur at any site, at any time, in any individual.

Magnitude of the problem

In 2022, there were about 20 million new cancer cases worldwide and 9.7 million deaths. Cancer is one of the leading causes of morbidity and mortality globally. By 2050, new cases are projected to rise to 35 million, driven largely by population growth and aging

The most common cancer in 2025 (in terms of new cases of cancer) were:

- breast (2.3 million cases
- lung (2.2 million cases)
- colon and rectum (1.9 million cases)
- prostate (1.4 million cases)
- skin (non-melanoma) (1.2 million cases); and
- stomach (1.1 million cases).

The most common cancer in male

- Lung

- Prostate
- Colorectal
- stomach
- liver cancer

The most common cancer in Female

- breast
- colorectal
- lung
- cervical
- thyroid cancer

The most common causes of cancer death in 2025 were:

- lung
- colon and rectum
- liver
- stomach and
- breast.

Age and sex incidence of cancer:

Age incidence: Usually cancer occurs among elderly people. In the developing countries, the prevalence is more among young people compared to developed countries, because developing countries are 'demographically young.' Some cancers exhibit a bimodal age curve, being common in young adults and elderly persons, e.g. Leukemia, Hodgkin's disease and kidney cancer. When cancers occur among the members of the same family, they tend to occur at an earlier age. Age of onset of Cancer cervix is related to age at marriage, earlier the marriage, earlier is the development of this cancer.

Sex: Cancers are more frequent among men than among women except cancers of reproductive organs and to some extent that of thyroid and gallbladder. Cancers of the reproductive system constitute 50 percent of all cancers among women; whereas in men they constitute only 15 percent.

Causes cancer of cancer

Cancer arises from the transformation of normal cells into tumour cells in a multi-stage process that generally progresses from a pre-cancerous lesion to a malignant tumour. These changes are the result of the interaction between a person's genetic factors and three categories of external agents, including:

- physical carcinogens, such as ultraviolet and ionizing radiation;
- chemical carcinogens, such as asbestos, components of tobacco smoke, aflatoxin (a food contaminant), and arsenic (a drinking water contaminant);
- biological carcinogens, such as infections from certain viruses, bacteria, or parasites.

Cancer develops gradually as a result of a complex mix of factors related to environment, lifestyle, and heredity. Scientists have identified many risk factors that increase the chance of getting cancer. They estimate that about 80 percent of all cancers are related to the use of tobacco products, to what we eat and drink, or, to a lesser extent, to exposure to radiation or cancer-causing agents (carcinogens) in the environment and the workplace. Some people are more sensitive than others to factors that can cause cancer.

Major Risk Factors for Cancers

1. Tobacco

- Leading cause of cancer deaths worldwide.
- Strongly linked to lung cancer (RR 10–20), plus cancers of larynx, oral cavity, pharynx, esophagus, pancreas, kidney, bladder.
- Passive smoking increase lung/laryngeal cancer risk.
- Smokeless tobacco lead to oral/head & neck cancers.

2. Alcohol

- Causes oral, esophageal, liver, breast cancers; probable link to colon/rectum.
- Accounts for about 3% of cancers (4% men, 2% women).

3. Occupational Exposures

- About 25 chemicals confirmed carcinogens (e.g., asbestos, nickel, cadmium, radon, benzene).

4. Environmental Pollution

- Air, water, soil contaminants (asbestos, chlorination by-products).

- Contributes to about 1–4% of cancers in developed countries.

5. Food Contaminants

- Aflatoxins (fungal toxins in maize/groundnuts) lead to hepatocellular carcinoma.
- Pesticides/organics linked to pancreatic, breast, lymphoma, leukemia.

6. Radiation

- Ionizing radiation leads to leukemia, breast, thyroid cancers.
- Electromagnetic fields leads to childhood leukemia (possible).

7. Sunlight (UV radiation)

- Causes skin cancers.

8. Chronic Infections

- Account for about 18% of cancers worldwide.
- Examples: HBV/HCV (liver), Human Papilloma Viruses (HPV) (cervix), Epstein-Barr virus (EBV) (lymphoid), H. pylori (stomach), Schistosoma (urinary bladder).

9. Diet & Nutrition

- About 30% of cancers linked to diet.
- Western diet increase risk of colorectal, breast, prostate, endometrial cancers.
- Fruits/vegetables protective (decrease risk (protect against digestive tract cancers).
- High salt/preserved foods increase risk of gastric cancer.
- Red/processed meat increase risk of colorectal cancer.
- High fat intake increase risk of colorectal cancer.
- Protective micronutrients: vitamins A, C, E, folate**.**

10. Obesity & Physical Inactivity

- Western diet and sedentary lifestyle increase risk of colon, breast, prostate, endometrial cancers.
- Strong risk factor for endometrial and postmenopausal breast cancer.

11. Genetic Susceptibility

- Familial clustering (breast, ovary, colon, melanoma).
- Inherited syndromes (e.g., BRCA1, retinoblastoma).
- Accounts for about 4% of cancers.

12. Reproductive Factors & Hormones

- Female hormones influence breast, ovarian, endometrial cancers.

- Oral contraceptives: slight increased risk of breast cancer risk but decrease risk of ovarian/endometrial risk.
- Hormone replacement therapy: increase risk of breast/endometrial cancer risk.

Cancer prevention: By elimination of or protection against factors known or believed to be involved in carcinogenesis

Primary prevention:

- **Health promotion:**
  - Encourage research for etiology, risk factors diagnosis.
  - Health education for public.
  - Monitoring for cancer registry.
- **Specific protection:** Cancer risk can be reduced by
  1. not using tobacco;
  2. maintaining a healthy body weight;
  3. eating a healthy diet, including fruit and vegetables and low in fat;
  4. doing physical activity on a regular basis;
  5. avoiding harmful use of alcohol;
  6. getting vaccinated against HPV and hepatitis B if you belong to a group for which vaccination is recommended;
  7. avoiding ultraviolet radiation (which primarily results from exposure to the sun);
  8. reducing exposure (as far as is possible) to ionizing radiation (through occupational or medical diagnostic imaging); and
  9. reducing exposure to outdoor air pollution and indoor air pollution.
  10. using low estrogen contraceptives when needed.
  11. treatment of pre-cancerous lesions.

Secondary prevention:

Early Diagnosis and Treatment: Cancer mortality can be reduced if cases are detected and treated early. **Early detection** can done by:

Identification of high risk groups

- Occupation.

- Family history.
- Environmental factors.
- Certain risk factors.
- Pre-cancerous condition.

Screening and case finding: by

1. **A self examination:** All persons should be instructed to self-examine themselves after the age of 30 and go to the specialist if the following warning signs are noticed.
   - A lump or hard area in the breast or testes
   - A change in a wart or mole
   - A persistent change in digestive or bowel habits
   - Persistent indigestion or difficulty in swallowing
   - Persistent cough or hoarseness
   - Excessive or irregular menstrual blood loss
   - Blood loss from any natural orifice
   - A swelling or sore that does not get better
   - Unexplained loss of weight.
2. **Clinical examination:** Doctor should have an high-index of suspicion and detect about 75 percent of cancer cases by thorough clinical examination because they occur in those body sites that are readily accessible.
3. **Screening by:**
   a. **Exfoliative cytology** i.e. study of cells shed from the surface (such as bronchial, gastric or uterine mucosa) into secretion. The finding of malignant cells in body secretions helps in preclinical or early diagnosis. Pap test and cytopipette irrigation techniques detect cervical cancers at a stage when they are 100 percent curable.
   b. X-ray and ultrasound technique to diagnoses different cancers at different sites.
   c. Thermography test to find skin temperature elevation at the cancer site.
   d. Endoscopy such as bronchoscopy, colonoscopy, gastroscopy and proctosigmoidoscopy.

Mass screening of all men above 50 and women above 30, at least in selected groups, should be undertaken.

Treatment: Cancer is treated with surgery, radiation therapy, chemotherapy, hormone therapy, or biological therapy. Patients with cancer are often treated by a team of specialists, which may

include a medical oncologist (specialist in cancer treatment), a surgeon, a radiation oncologist (specialist in radiation therapy), and others.

Tertiary Prevention

Disability limitation: This is done by giving intensive treatment to prevent the development of further disability.

Rehabilitation: This is given for those who have undergone surgery such as amputation of leg, colostomy, mastectomy, etc. and are rehabilitated with a prosthesis and training, later placed in a suitable job.

Primary health care

By the end of the lecture the student will be able to:

- Define Health for all & PHC
- List the Components of PHC.
- Identify instrument of primary health care.
- List Function of the PHC

Concept and definition of Health for all

Health for All (HFA) simply means health services for all, i.e. basic, essential, utilitarian services to be provided to all, irrespective of caste, creed, community and ability to pay for it.

Health for All was defined as, 'Attainment by all the people of the world, a level of health, which will enable or permit every individual to lead socially and economically a productive life' (useful life). The concept of primary health care has been accepted by all countries to achieve the goal Health for all.

Definition of primary health care (PHC) (WHO 1978):

"Is the essential health care based on practical, scientifically sound and socially accepted methods and technology made universally accessible to individuals and families in the community through their full participation and a cost that the community and the country can afford to maintain at every state of their development in the spirit of self reliance and self determination".

- It forms an integral part both of country's health system, of which it is the central function and main focus, and of overall social and economic development of the community.
- It is the first level of contact of individuals, the family and the community with the national health system bringing health care as close as possible to where people live and work, and constitute the first element of a continuing health care process."
- Primary health care is equally valid for all countries, developed and developing (it may take varying forms in each of them).
- A package containing promotive (1st three elements), preventive (2nd three elements), and curative (last two elements) service components were formulated to be provided (the 8 elements of primary health care.)

- The details of delivery of each element would depend on the actual priority community needs, the stage of socio-economic and health development and the available resources.

Elements of Primary Health Care (i.e. Components)

Promotive elements

- Health Education concerning prevailing health problems and the methods of preventing and controlling them.
- Promotion of food supply and proper nutrition.
- Provision of comprehensive maternal and childcare.

Preventive elements

- Immunization of children against major communicable diseases.
- Prevention and control of locally endemic diseases.
- Provision of adequate supply of safe water.

Curative last two elements

- Appropriate treatment of common diseases and injures.
- Provision of essential drugs.

PHC principles

Equity:

- mean services for all and more services to the needy and vulnerable.
    - Vulnerable groups are to be identified and reached through outreach program.
- Accessibility has to be improved by:
    - Increasing number of facilities.
    - Improving transport conditions.
    - Organizing outreach services.

Appropriate technology:

- appropriate means that besides being scientifically sound is also acceptable to those who apply it and to those for whom it is used.
- Examples include:
    - ORT in diarrheal disease control.
    - Domiciliary treatment in TB control.

    - Breast-feeding in spacing.
    - Weighing for growth monitoring.

Multi-sectoral approach:

- Individual sectors cannot function in isolation. Mutual roles and coordination should be considered.
- Examples:
    - The role of municipalities in environmental sanitation.
    - Provision of portable water and disposal of wastes.
    - Ministry of education in school health.

Community participation:

- Community participation: social awareness and community self-reliance are the key factor in human development.
- Examples include:
    - Involving people in PHC activities for planning, implementation, and evaluation of services.
    - Provide a source of free labour.
    - Financial contribution.

Advantages of community participation

- It is a cost-effective method of providing health services.
- People begin to view health more objectively. So they are more likely to accept the care.
- There will be greater commitment of the people resulting in the success of health care services.
- Health awareness becomes an integral part of village life.
- Health workers get greater support for their activities.
- People become more soft reliant in taking care of their health.
- Health care services become more relevant to the health needs of the people.
- There is less dependence on the Government.
- Quality of the health care improves.

Strategies for Primary Health Care Implementation

The term strategy is derived from military vocabulary meaning' the art of projecting and directing the large military movement and operation of campaign'.

- By instituting measures for horizontal expansion and insure efficiency in health services delivery in general.
- By developing good rapport between the health centres and the beneficiaris:
    - by organizing out-reach activities and
    - reaching the people through social and religious institutions (like school, mosque, etc.) and
    - explore/study the social problems related to health.
- By motivating the people to accept health centres as an institution of delivering comprehensive health care and not merely the dispensers of drugs and to motivate them to opt for preventive and primitive services with the participation of the community itself.
- By consolidating and strengthening the national control programmes (malaria, billharazia); by integrating preventive and curative services.
- By promoting health awareness in the people through organized health education programmes, institutional and out-reach; towards better utilization of health services and to participate with the health centres in solving health and environmental problems.
- By developing effective coordination with other health service delivery institutions like hospital; by evolving a suitable referral system.
- By coordinating with the educational and training institutions (universal, health institutes etc.) for the introduction of PHC in academic curriculum and to involve them in the continued educational programmes for the professionals of health centres.
- By coordinating with health related government sectors such as; ministers of agriculture, municipalities, education for improvement of the health institution related to services.
- By providing better quality of services to the high-risk group of the community such as; pregnant mothers, preschool children and others.

Instruments in delivery of PHC:

- Community surveys and diagnosis.
- Family health records.

- Plan of action.
- Team approach.
- Information system.

Functions of the Primary Health care

1. Improvement of environmental sanitation with special emphasis on safe water supply
2. Food supply and proper nutrient
3. MCH services including family planning
4. Prevention and control of communicable diseases (Immunization)
5. Prevention and control of endemics and implementation of relevant National Health Programs
6. Health education
7. Treatment of disease and injuries
8. Training programs for health workers, health assistants, local dais, etc.
9. Collection and reporting of vital statistics
10. Basic laboratory services
11. Referral services

Basic Nutrition Concepts

By the end of the lecture the student will be able to:

- Understand important definitions and concepts in nutrition
- Recognize Food guide pyramid.
- Identify General principles of diet planning
- Identify Nutrients and their main categories

Key Definitions

- **Nutrient**: is a substance contains elements needed to supply the body with energy, elements needed for growth, repair, maintenance of cells and their biological processes together with resistance to infection.
- **Nutrition**: The process of utilization of the consumed food by the body to be used for nourishment.
- **Balanced diet**: is the diet which provides the body with sufficient amount of foodstuffs selected in proper proportions (50-60% CHO, 25-30 % Fat, 15-20% protein, all vitamins and minerals). It has to satisfy the body requirements both quantitatively and qualitatively to prevent malnutrition.
- **Food fortification**: Addition of nutrient element which is responsible for the prevention of malnutrition to a food substance commonly consumed by all population.
- **Therapeutic food**: (supplementary food) is a food substance which is rich in a certain nutrient element or its precursor, given to the mal-nourished person to correct nutritional deficiency.
- **Reference man/woman**: Healthy adults (20–39 yrs, 65 kg male, 55 kg female) with defined activity patterns.
    - Calorie needs: 3000 kcal/day (man), 2200 kcal/day (woman).

Importance of nutrition : it is essential for:

- Efficiency of work and physical fitness.
- Mental well-being and efficiency.
- Reproduction i.e. healthy and successful pregnancy depends on adequate diet of the mother.

- Longevity: animal experiments show that adequate diet has a favourable effect on length of life.
- Resistance against infection.

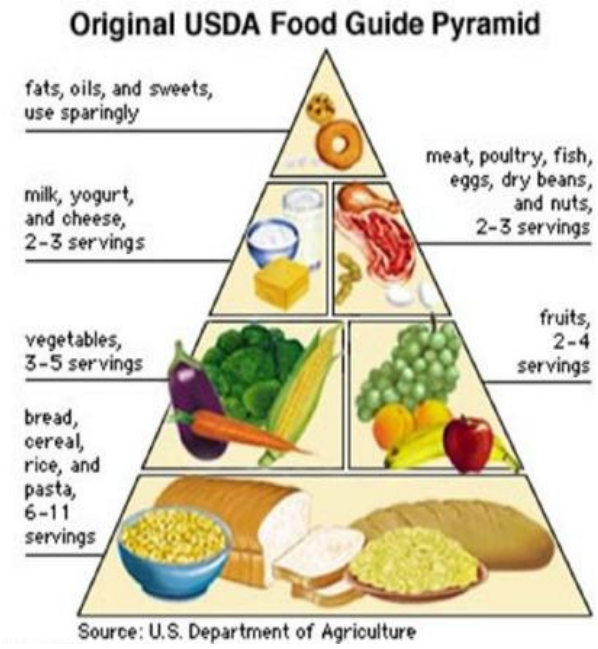
Deficiency in nutrition has been proved to be responsible for deficiency diseases such as protein energy malnutrition, anaemia, rickets etc.

Food Guide Pyramid

The Food Guide Pyramid is a visual representation of how to structure a balanced diet for health and nutrition.

Structure of the Pyramid (old or classic pyramid) (1992 USDA model):

- **Base (largest portion):** Grains (bread, rice, pasta, cereals) – recommended as the foundation of daily intake.
- **Next level:** Fruits and Vegetables – essential for vitamins, minerals, fiber, and protection against disease.
- **Middle level:** Protein sources (meat, poultry, fish, beans, eggs, nuts) and Dairy (milk, yogurt, cheese) – important for growth, repair, and bone health.
- **Top (smallest portion):** Fats, oils, and sweets – to be consumed sparingly.



New pyramid/MyPlate: The MyPlate model emphasizes whole grains, encourages filling half the plate with fruits and vegetables, and promotes moderate protein (fish, poultry, eggs, dairy), limited red meat while limiting fats and sweets for optimal health.

Diet Planning

Diet planning is concerned with construction of a diet by various food stuffs to fulfil the body requirements. Quantitative planning requires the consultation of food composition tables and thorough revision of locally available food.

General Principles of Diet Planning

- **Balanced diet:** Nutrients interact; variety ensures adequate intake in proper proportions.
- **Group feeding:** Diets for communities (institutions, camps) should follow recommended dietary allowances (RDA).
- **Allowance for losses:** Nutrient losses occur during cooking, processing, and storage; allowances must compensate for these.
- **Socioeconomic factors:** Consider food availability, cost, preparation methods, storage, habits, and regional/seasonal conditions.

Construction of the Diet: depend on

- **Individual factors:** Age, sex, weight, developmental stage, health status, and type of work must be considered.
- **Daily unit**: Each day's menu should meet RDA for all essential nutrients.
- **Number of meals:** At least 3 principal meals daily; skipping meals is discouraged.
- **Adequacy:** Ensure all conditions for a nutritionally adequate diet are met**.**
- **Food groups: Plan using 3 basic groups:**
    - Energy foods (carbohydrates, fats)
    - Body-building foods (proteins)
    - Vitality foods (vitamins, minerals)

Essential nutrients

There are six classes of nutrients required for the body to function and maintain overall health. They are subdivided into: Macronutrient and Micronutrients

Macronutrient: These are carbohydrates, lipids , proteins, water.

Micronutrients: these are: vitamins, and minerals.

Macronutrients are the source of energy, normally, **55%-65 %** of daily energy use is from and carbohydrates, **25%-30% from fat** and **15% from protein** (0.8-1 gm/ kg).

The required amounts of the essential nutrients differ by age, sex and the state of the body.

Protein Requirements

- **Composition**: Made of amino acids (C, H, O, N, sometimes S); nitrogen distinguishes proteins from energy foods.
- **Functions**:
  - Growth and tissue building
  - Tissue repair
  - Formation of antibodies, enzymes, hormones, hemoglobin
  - Provide ~10% of caloric needs
  - Maintain osmotic pressure via plasma proteins
- **Types**:
  - **High biologic value (complete)**: Contain all 8 essential amino acids (animal sources: milk, egg, fish, poultry; some plant mixtures).
  - **Low biologic value (incomplete)**: Missing one or more essential amino acids (plant sources: legumes, cereals).
- **Sources**:
  - Animal: fish , meat , egg , milk .
  - Plant: legumes richest; cereals , leafy greens , poor staples (sweet potatoes).
- **Requirements**:
  - Adults: ~1 g/kg/day increase for pregnant, lactating women and children

Carbohydrate Requirements

- **Composition**: C, H, O; formed by photosynthesis.
- **Types**:
  - Monosaccharides: glucose, fructose, galactose
  - Disaccharides: sucrose, maltose, lactose
  - Polysaccharides: starch, glycogen, cellulose
- **Functions**:
  - Main fuel for muscular exercise
  - Needed for complete fat combustion (absence → ketosis)
  - Indigestible carbs (fiber) aid intestinal tone and prevent constipation
- **Sources**: Cereals, tubers, legumes, fruits, sugars, honey, molasses.

- **Requirements**:
    - Adults: ~50–60% of total calories (≈400–500 g/day).

Fat Requirements

- **Composition**: Esters of fatty acids + glycerol; saturated (palmitic, stearic) or unsaturated (oleic, linoleic, linolenic, arachidonic).
- **Functions**:
    - Supply energy (1 g = 9.3 kcal; ~25–30% of calories).
    - Carry fat-soluble vitamins (A, D, E, K).
    - Provide essential fatty acids (linoleic, linolenic, arachidonic).
    - Precursors for vitamin D (cholesterol, ergosterol).
    - Insulation, cushioning of organs.
    - Delay gastric emptying → satiety.
    - Make food palatable.
    - Burned completely only with carbohydrates (absence → ketosis).
- **Sources**:
    - Animal: butter, cream, ghee, ice cream.
    - Plant: vegetable oils (sesame, olive, cottonseed, linseed), margarine, nuts.
- **Requirements**:
    - ~1–1.25 g/kg/day (~25% of calories, ≈750 kcal).

Mineral Requirements

Definition: Inorganic elements essential for body structure and function; required in small amounts.

Functions:

- Build and maintain bones and teeth (e.g., calcium, phosphorus, fluorine).
- Form essential components of hemoglobin, enzymes, and hormones (e.g., iron, iodine, zinc).
- Regulate fluid balance, nerve transmission, and muscle contraction (e.g., sodium, potassium, magnesium).

Types:

- **Macrominerals** (needed in larger amounts): calcium, phosphorus, sodium, potassium, magnesium, sulfur.
- **Trace minerals** (needed in minute amounts): iron, iodine, zinc, copper, fluorine, selenium, manganese.

Sources:

- Found in a wide variety of foods—milk, meat, fish, cereals, legumes, vegetables, fruits, and drinking water.

Requirements: Vary by age, sex, physiological (growth, pregnancy, lactation), and health status.

Deficiency/Toxicity:

- **Deficiency** → anemia (iron), goiter (iodine), rickets/osteoporosis (calcium), dental caries (fluorine).
- **Excess** → toxicity (e.g., fluorosis from excess fluoride, hypertension from excess sodium).

Vitamins Requirements

Definition: Organic micronutrients essential for normal body functions; must be obtained from diet since the body cannot synthesize most of them.

Classification:

- **Fat-soluble:** Vitamins A, D, E, K. Stored in body fat and liver; excess can accumulate → toxicity risk.
- **Water-soluble:** Vitamin C and B-complex (B1, B2, B3, B5, B6, B7, B9, B12), Not stored; excess excreted in urine → deficiency risk if intake is low.

Functions:

- Regulate metabolism and enzyme activity.
- Support growth, immunity, blood formation, vision, bone health, and energy release.

Requirements: Expressed as Recommended Daily Intake (RDI/DRI); vary by age, sex, and physiological state (pregnancy, lactation).

Deficiency/Toxicity:

- **Deficiency** → specific disorders (e.g., rickets from vitamin D deficiency, scurvy from vitamin C deficiency).
- **Excess** (especially fat-soluble vitamins) → toxicity (e.g., hypervitaminosis A).

Malnutrition

By the end of the lecture the student will be able to:

- Define malnutrition, its direct and indirect health effects.
- Identify ecological and predisposing factors that lead to malnutrition
- Identify common disorders such as obesity, nutritional anaemia (iron deficiency, megaloblastic, pernicious), and explain their causes, clinical features, diagnostic criteria, and management.

Definition: Malnutrition refers to excessive and unbalanced—as well as deficient—intake of essential nutrients.

Malnutrition may be:

1. Over nutrition: a state of nutrition style in which nutritional intake exceeds the body needs.

2- Under-nutrition in developing countries: About two thirds of the world's population live in countries with low and lower middle incomes.

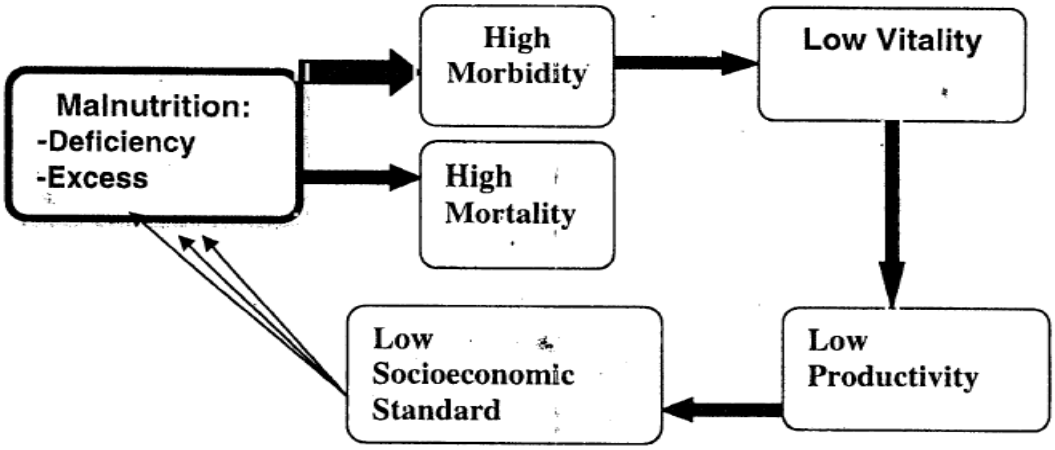
Effects of Malnutrition

Direct effects: Clinical and sub-clinical deficiency diseases, obesity.

Indirect effects:

- High infant, preschool, and school-age morbidity/mortality.
- High maternal mortality and poor pregnancy outcomes.
- Reduced vitality, weakened immunity, higher infectious and non-communicable disease prevalence.
- Lower productivity and socioeconomic development.

The Vicious Circle



Prevention Strategies of malnutrition

A. Family Level

- Nutrition education.
- Promotion of breastfeeding.
- Focus on vulnerable groups.

B. Community Level

- Socioeconomic development.
- Sanitation and improved health conditions.
- Infectious disease control.
- Food supplementation and fortification.

C. National Level

- Agricultural and animal husbandry modernization.
- Family planning for population stabilization.
- Nutrition intervention programs:
- Salt iodization, fluoridation of water.
- Supplementation for children.
- Iron/folic acid for pregnant & lactating women.
- Immunization, sanitation, infectious disease management.

D. International Level

Collaboration with FAO, UNICEF, WHO to support national governments in prevention and control programs

SOME NUTRITION DISORDERS

OBESITY

Obesity defined as 20% or more over ideal body weight. It is characterized by

- Multiple biochemical disturbances.
- Increased insulin resistance, glucose intolerance, changes in fatty acids metabolism, and alterations in other hormonal and enzymatic process an in cell physiology

Predisposing factors:

The cause of obesity is an excess intake of energy. It is clear that some persons are predisposed to a positive energy balance while others remains slims and even find it difficult to gain weight; specific metabolic derangement that alter the efficiency of energy utilization may exist. There are several factors, that predispose a person to obesity.

1. ***Genetic and environmental factors***
2. ***Endocrine disorders*** as: hypothyroidism. Hypopituitarism, Cushing's syndrome, ovarian failure, polycystic ovary syndrome. hypoglycemia, and hypothalamic lesion.
3. ***Behavioral factors:*** response to both external and internal stimulus. Internal stimulus: Gastric motility. External stimulus: Small, taste foods every hour.
4. ***Number of fat cells.***
5. ***Sedentary lifestyle***

Classification:

BMI (WHO - Classification)

BMI 25-30 = Overweight (grade 1 obesity)

BMI >30-40 = Obese (grade 2 obesity)

BMI >40 =Very obese (morbid or grade 3 obesity)

According to waist measurement

	MALES	FEMALE
LEVEL 1	> 94cm	> 80cm
LEVEL2	> 102cm	> 88cm

Level 1 is the maximum acceptable waist circumference irrespective of the adult age and there should be no further weight gain.

Level 2 denotes obesity and requires weight management to reduce the risk of type 2 diabetes & CVS complications.

Associated diseases of obesity:

Obesity increases the risk of many physical and mental conditions. These include:

- Diabetes mellitus type 2
- high blood pressure
- high blood cholesterol, and high triglyceride levels
- angina pectoris and myocardial infarction.
- osteoarthritis
- obstructive sleep apnea
- alveolar hypoventilation
- gall bladder disease
- surgical risk and
- social stigmatization

Management (line therapy of obesity):

- Dieting (food selection)
- Weight loss programs
- Exercise and physical activity
- Behavior modification
- Drug therapy
- Surgery
- Acupuncture

Dieting (food selection)

Low calorie food: Patients should emphasize large quantities of vegetable salads and fruits, modest quantities of meats, milk and cheese and minimal amounts of the least nutritious food as sweet and fats. Vitamin, mineral, and protein content will be adequate without supplementation at level as low as approximately 900 kcal/day.

Foods not allowed: Fat fish (tuna and salmon), whole milk cream, butter sweetened fruit, cakes sauces, olives, nuts, carbonated beverages.

Unlimited food: lemon, tea, coffee.

NUTRITIONAL ANAEMIA

It is the condition that results from failure of the blood forming organs to maintain normal haemoglobin concentrations, due to inadequate supply of nutrient essential for haemoglobin formation.

The most important nutrients in haemoglobin formation are protein, iron, folic acid. Vitamin 612, copper, Zink and Vitamin C. Deficiency occurs when there is imbalance between nutrients absorbed and the amount required by the body; this may be due to:

- Diminished absorption which may occur as a result of inadequate intake or defective absorption.
- Increased requirement, result from rapid cell multiplication as in growth and pregnancy or increased loss from the body as in haemorrhage and parasitic infection.

Criteria for the diagnosis of anaemia:

- Determination of haemoglobin level.
- Determination of the packed cell volume (haemotocrit) as well and the plasma iron saturation.

1. Iron deficiency Anaemia

Definition:

Iron deficiency anaemia is an anaemia due to inadequate intake of iron.

It is characterized by the production of smaller, thinner red blood cells which are deficient in haemoglobin

Clinical manifestation

Clinical features

- Symptoms are variable but always include fatigue to extreme exhaustion.
- In some severely depleted patients; sore tongue, diarrhoea.
- Pallor is present in more severe cases

Laboratory Diagnosis

- Hypochromic cells with thin rims of haemoglobin, fragmented cells and elongated red cell.
- Serum iron levels are generally reduced to levels below 30 mg (normal 70 -30 mg)

Bone marrow: Hyperplastic bone marrow

Treatment

There are many preparation as: ferrous sulphate, ferrous gluconate and ferrous fumarate given as:

- Oral (tablets or liquid), or
- parenteral: indicated only in those rare situations in which the patient can find no oral preparation that can be tolerated or in which the absorption of oral iron is impaired because of diarrhoea or gastro intestinal shunt. It can be given IM or IV (but the IM is painful). The dose of all parenteral iron preparation depends on the amount necessary to raise the haemoglobin to the desired levels.

2. Megaloblastic Anaemia

Is due to either deficiency of folic acid and/or vitamin BI2 but on vitamin B12 deficiency there is a combined atrophy of the posterior nerve which make walking trouble to the patient and is called pernicious anaemia.

Treatment:

- In simple megaloblastic anaemia, w.e give folic acid.
- In pernicious anaemia, vitamin B12 injections.

Assessment of Nutritional Status

By the end of the lecture the student will be able to :

- Define nutritional assessment
- Understand methods of nutritional assessment
- Describe the characteristic of food balance sheet for Egypt

Definition of nutritional assessment: Nutrition Assessment is a dynamic process of evaluation of the nutritional status of individuals, groups or the community, to identify the major malnutrition problem (s) so as to set plans for prevention and control of malnutrition.

Methods of Nutritional Assessment

Direct Methods (individual level)

1. **Clinical appraisal**
    - General exam: hair, mouth, gums, nails, skin, eyes, tongue, muscles, bones, thyroid.
    - Two or more deficiency signs → stronger diagnostic value.
    - **Advantages**: inexpensive, non-invasive, large groups.
    - **Limitation**: nonspecific, cannot quantify deficiency → needs biochemical tests.
2. **Anthropometric measurements**
    - Height, weight, waist, hip, skin-fold thickness.
    - Detects body composition changes; evaluates under- and over-nutrition.
    - Children: growth charts, velocity compared to standards.
    - Adults:
        - **BMI** = weight (kg)/height (m^2). High BMI → diabetes, cardiovascular risk.
        - **Waist circumference**: better predictor of mortality.
            - Level 1 risk: >94 cm (men), >80 cm (women).
            - Level 2 obesity: >102 cm (men), >88 cm (women).
        - **Hip circumference**: widest point around hips/buttocks.
        - **Waist/Hip Ratio (WHR)**: high risk if >0.95 (men), >0.80 (women).
    - Advantages: objective, reproducible, inexpensive, sensitive.
3. **Laboratory & biochemical investigations**
    - Detect early metabolic changes before clinical signs.

- Key tests: CBC, serum ferritin, blood glucose, lipid profile, vitamin D, thyroid profile, stool analysis, serum albumin.
- Advantages: precise, reproducible.
- Limitations: costly, time-consuming, not feasible for large populations.

4. **Dietary intake assessment**
    - **Methods:**
        - **24-hour recall**: quick, easy, but short-term.
        - **Food frequency questionnaire**: ~100 items; inexpensive, representative; limitations: long, serving size errors.
        - **Dietary history**: detailed, interviewer-based, accurate.
        - **Food diary**: recorded at consumption; reliable but difficult to maintain.
        - **Observed food consumption**: precise, research-oriented, rarely used clinically.

Indirect Methods (community or National Level)

- **National food consumption**
    - Formula: (Food produced + imported – exported) ÷ population ÷ 365 = average per capita/day.
- **Food balance sheet**
    - Shows average daily per capita intake of food items (meat, milk, fish, cereals).
    - Analyzed into nutrients/energy using food composition tables.
    - Reflects national food supply, not individual status.
    - ***Egypt's food balance sheet*:**
        - Cereals form bulk of diet (main source of energy, protein, iron, B-vitamins).
        - Energy intake exceeds requirements.
        - Low animal food consumption.
        - Protein mainly from plant sources.
        - Iron intake from cereals → low absorbability.

- **Vital statistics**
  - Indirect indicators: infant/neonatal/perinatal mortality, child mortality, life expectancy.
  - Morbidity data: hospital statistics (PEM, rickets, anemia, keratomalacia).
- **Ecological factors**
  - Cultural: food habits, infant feeding, traditions in food preparation/distribution.
  - Food production: crops, storage, importation.
  - Socioeconomic: family size, income, occupation, education.
  - Disease prevalence: infectious/parasitic diseases → malnutrition.
  - Health/education services: facilities, personnel, preventive/curative services, media/communication.

Environmental health

By the end of the lecture, students should be able to:

- Identify Environmental health, and its components.
- Recognize sources of environmental risks
- Global strategy for environmental health
- Identify criteria for healthful Housing & Housing standards
- Summarize housing and its relation to health
- Understand the nature, health effects and sources of environmental risks for Air
- Recognize effect of air pollution on health
- Identify global effect of air pollution
- Understand the nature, health effects and sources of environmental risks for water
- Summarize process of purification

Environment is the external factor/factors present around man and has got an influence on the health of the human being. According to ecologists, health is a state of dynamic equilibrium between man and his environment and when this equilibrium is disturbed, ill-health (disease) occurs.

Environment health definition: The term "environmental health" comprises the aspects of human health, including quality of life, that are determined by physical, chemical, biological, social, and psychosocial factors in the environment. Environmental health is the study and management of environmental conditions that affect the health and well-being of humans.

Purpose of Environmental health

The purpose of environmental health is to preventing human injury and illness and promoting well-being by

- identifying and evaluating environmental sources and hazardous agents and
- limiting exposures to hazardous physical, chemical, and biological agents in air, water, soil, food, and other environmental media or settings that may adversely affect human health

Environment components: Such an environment has been divided into four components:

Physical environment : This consists of non-living things and certain physical forces/ energy present around man . These are water, air, soil, housing, radiation, light, noise, vibration, refuse, wastes, etc.

Biological environment: This consists of living things around man . These are plants, animals, rodents, insects and microbes like bacteriae, viruses, rickettsiae, parasites, fungi, etc.

Social environment: This consists of occupation, literacy, income, religion, standard of living, lifestyle, availability of health services, etc.

Cultural environment: This consists of knowledge, attitude, beliefs practices, traditions, culture, customs, habits, etc

Environmental sanitation

The term environmental sanitation is defined by WHO as, 'the control of all those factors in man's physical environment, which exert a deleterious effect on physical development, health and survival'.

Objectives environmental sanitation:

- Improve quality of life.
- Prevent disease via clean homes, neighborhoods, communities.
- Promote hygiene (food, housing, workplaces).
- Control pests (insects, rodents).
- Encourage healthy behavior at individual, family, community levels.

Environmental Hazards include

1. **Physical**: heat, noise, vibration, radiation, pressure.
2. **Biological:** pathogens, parasites, allergens (pollens), natural toxins (plants, fish).
3. **Chemical**:
    - *Irritants*: corrosives, caustics → tissue damage, infections, cancer risk.
    - *Fibrotic agents*: silica (silicosis), coal dust (black lung), cotton dust (byssinosis), asbestos (carcinogenic).
    - *Asphyxiants*: CO, HCN, H_2S → interfere with oxygen uptake.
    - *Allergens*: e.g., formaldehyde → immune reactions.

    - *Neurotoxins*: insecticides (DDT, malathion, parathion).
    - *Mutagens*: DNA damage → heritable effects.
    - *Carcinogens*: cause malignant tumors.
    - *Teratogens*: cause developmental abnormalities in embryos.
4. **Psychological** – stress → anxiety, cardiovascular disease, ulcers.
5. **Diet-related** – nutrient deficiencies (e.g., iodine), malnutrition, poor eating habits → interact with infections.

Anthropogenic (Human-Caused) Damage

- Air pollution.
- Water pollution.
- Soil pollution & erosion.
- Deforestation.
- Desertification.

WHO standard for environmental global Strategy

- Improving the performance of environmental public health programs.
- Strengthen environmental public health surveillance.
- Foster an environmentally responsible and resilient health sector.
- Promoting environmentally healthy and resilient communities.

These strategies provide the operational framework to strengthen each environmental health domain (Thematic Areas)

Thematic Areas

- Air Quality and Health
- Climate Change and Health
- Chemical Safety
- Water and Sanitation

Housing and rural health

Criteria for Healthful Housing & Standards

- **Adequate living space** – prevent overcrowding.
- **Safe indoor temperatures** – avoid extremes of heat and cold.
- **Structural safety** – minimize injury hazards (stairs, flooring, electrical safety).
- **Accessibility** – housing suitable for people with disabilities or functional impairments.
- **Sanitation & water supply** – safe drinking water, hygienic waste disposal, proper kitchen and bathroom facilities.
- **Ventilation & air quality** – reduce indoor pollutants, ensure airflow.
- **Protection from vectors & chemicals** – minimize exposure to pests, toxic substances, and unsafe building materials.

Housing and Its Relation to Health

- **Physical health**: Poor housing contributes to respiratory diseases (from dampness, poor ventilation), injuries (unsafe structures), and infectious diseases (from inadequate sanitation and crowding).
- **Mental health**: Overcrowding, insecurity, and poor-quality housing increase stress, anxiety, and depression.
- **Social well-being**: Adequate housing supports dignity, privacy, and community resilience.
- **Public health impact**: Improved housing reduces disease burden, enhances quality of life, and supports equity in health outcomes.

Rural housing

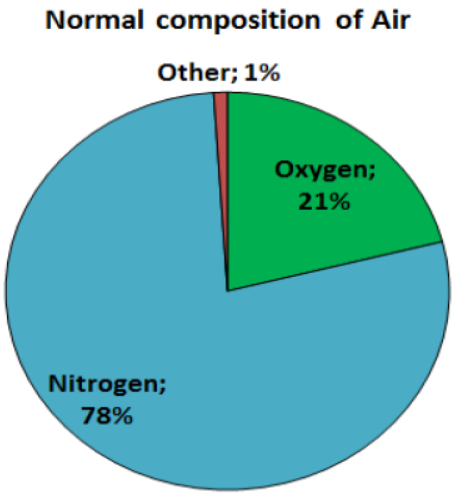
- Poor housing quality (overcrowding, lack of sanitation, unsafe structures) increases infectious disease, injuries, and respiratory illness.
- In rural settings, housing often reflects broader inequities limited infrastructure, unsafe water, and exposure to environmental hazards.
- Improving housing standards directly strengthens rural health outcomes and reduces inequities.

Standards of Rural Housing

- Housing must provide adequate space to prevent overcrowding and disease spread.
- Houses should ensure structural safety, being stable and weather-resistant.
- Homes must maintain safe indoor temperatures and good ventilation for healthy air.
- Every household should have safe water supply and sanitation facilities.
- Housing must protect against vectors and harmful chemicals.
- Rural housing should be accessible and equitable, serving vulnerable groups and near essential services.
- Housing design should integrate with local climate, culture, and resources using sustainable materials.

Air pollution

Air pollution is change in normal air constituents by introduction of chemicals, particulates, biological materials, or other harmful materials into the Earth's atmosphere, possibly causing disease, or damage to human being and other living organisms such as food crops, or the nature.



Under ordinary conditions, the composition of outdoor air is remarkably constant. This is brought about by certain self-cleansing conditions:

1.Wind: wind dilutes and sweeps away impurities by its movement.

2.Sunlight: it plays its role by oxidizing impurities and killing bacteria.

3.Rain: It removes the suspended and gaseous impurities.

4.Plant life: It generates oxygen.

Indicators of Air pollution:

- Sulfur dioxide
- Carbon monoxide
- Nitrogen dioxide
- Ozone
- Suspended particulates
- Carbon black.

The level of such pollutants are changeable according to the season as they are affected by air temperature and humidity.

Sources of air pollution:

- Natural Sources –Volcano, forest fire, dust storms, oceans, plants and trees
- Sources initiated by human beings:
    - Industrial sources: Factories of iron and steel, paper, cement, fertilizers, thermal power plant, petroleum refineries, etc.
    - Incineration or open firing.
    - Domestic sources: Burning of firewood, kerosene oil, coal, etc.
    - Vehicular sources: Motor vehicles, railways, ships, aeroplanes, etc.
    - Miscellaneous: Tobacco smoking, nuclear explosions, forest-fires, volcanoes, burning of refuse, dust-storm, ocean spray, etc.

Effects of air pollution

A. Health Effects

1. Immediate or short term effect:

- Irritation to the eyes, nose and throat,
- Upper respiratory infections such as bronchitis and pneumonia.
- Other symptoms can include headaches, fatigue, nausea, and allergic reactions.
- Aggravation of the medical conditions of individuals with asthma and emphysema.
- In the great "Smog Disaster" in London in 1952, four thousand people died in a few days due to the high concentrations of pollution.

2. Long term effect:

- Chronic bronchitis with persistent cough and expectoration.
- Asthma due to presence of allergens.
- Kidney diseases due to inhalation of nephrotoxic fumes.
- Heart diseases secondary to increase of pulmonary pressure.
- Cancers due to presence of initiator and promoter carcinogens.
- Abortions and other reproductive hazards as congenital anomalies.
- Depression and other psychological disorders.

Global effects due to air pollution

1. **Acid Rain** – Caused by sulfur dioxide and nitrogen oxides; damages vegetation, structures, and can affect areas far from the pollution source.
2. **Greenhouse Effect & Global Warming** – From fossil fuel combustion; leads to rising sea levels, melting ice, ecosystem imbalance, and biodiversity loss.
3. **Ozone Layer Depletion** – Allows harmful ultraviolet radiation to reach Earth, increasing risks to skin, eyes, and overall health.
4. **Radioactive Disasters** – Nuclear accidents release radiation; health risks and evacuation depend on radiation type and half-life.
5. **Dioxin Pollution** – Toxic chemicals from incineration, smoking, exhaust, and smelting; linked to immune and reproductive damage, diabetes, cancer, and skin conditions (chloracne).

Prevention and control of air pollution)

- **Energy Transition**
  - Replace fossil fuels with clean energy (solar, wind, hydro, nuclear).
  - Promote energy efficiency in homes, industries, and transport.
- **Industrial Controls**
  - Enforce strict emission standards for factories and power plants.
  - Use cleaner production technologies and pollution filters.
- **Transport Measures**
  - Expand public transport systems.
  - Encourage electric vehicles, cycling, and walking.
  - Reduce reliance on private cars.
- **Urban & Rural Planning**
  - Establish industries away from dense populations.
  - Increase green spaces and urban forests to absorb pollutants.
- **Household Interventions**
  - Promote clean cooking fuels (LPG, electricity, biogas) instead of biomass.
  - Improve ventilation in homes to reduce indoor pollution.

Safe drinking water

Safe water criteria

Safe water should be

- Clear
- Pleasant to taste (i.e. free from odor and color i.e. odorless and colorless)
- Free from pathogens
- Free from harmful chemical substances
- Usable for domestic purposes.

Sources of Water:

I. Rain Water: usually collects dust, bacteria and other impurities from the atmosphere and from any surface on which it falls. Rain water contains no dissolved salts, so, it may dissolve metals. it should be kept in concrete tanks and distributed through special type of pipes coated from inside with good asphalt paint.

II. Surface Water:

This includes water from

- fresh water lakes
- Rivers, canals and streams, and
- distilled seawater.

III. Underground Water:

The underground water *may get out through*

Springs: These are natural yields of water.

Wells are artificially made to get out the underground water.

Water pollution

Definition: Water pollution occurs when harmful substances (chemicals, waste, microorganisms) contaminate rivers, lakes, oceans, or groundwater, degrading water quality and making it unsafe for humans, animals, and ecosystems.

Effects of water pollution:

- **Human health:** Causes waterborne diseases (cholera, diarrhea, hepatitis).

- **Ecosystems:** Kills aquatic life, disrupts food chains, reduces biodiversity.
- **Economic impact:** Affects agriculture, fisheries, and safe drinking water supply.
- **Environmental damage:** Leads to eutrophication, oxygen depletion, and habitat loss.

Prevention & Control of water pollution

- **Industrial regulation:** Treat wastewater before discharge; enforce emission standards.
- **Agricultural practices:** Reduce pesticide/fertilizer runoff; promote organic farming.
- **Waste management:** Proper disposal of sewage, plastics, and hazardous waste.
- **Community action:** Raise awareness, conserve water, and avoid littering in water bodies.
- **Global cooperation:** Strengthen international agreements to protect shared water resources.

Water Purification in large scale

Surface water is usually polluted and contaminated and is often unsafe and should not be used without purification. The process of purification aims at eliminating harmful, unpalatable or objectionable materials and removing the harmful bacteria. To supply large communities with potable sanitary water, large scale water purification plants are needed.

Large scale Water Purification of Surface Water:

The process of water purification in large-scale purification plants includes four main steps namely:

- water intake,
- sedimentation,
- filtration
- disinfection.

1. Water Intake

- Intake pipes are placed upstream from the city and pollution sources, ensuring future expansion is considered.
- Protected zones: 500 m upstream and 150 m downstream must remain free of pollution.
- Pipes extend to the center of the river, midway between surface and bottom, avoiding floating debris.
- Diameter: 0.6–1 m, with a screen and one-way valve to prevent the water from getting out of the purification plant.

2. Sedimentation

- Water stored in large tanks for hours (4:8) to weeks (4 week), depending on demand.
- Coagulant (alum) added: reacts with bicarbonates to form aluminum hydroxide floc, trapping suspended matter.
- Alum concentration: 14–70 ppm, adjusted to impurity levels.
- Clear supernatant water exits from the top of the tank.
- Modern method: Dorr Clarifier – a rotating pillar with iron plates creates centrifugal movement, enhancing sedimentation efficiency.

3. Filtration

- Removes remaining suspended matter and microorganisms.
- **Rapid sand filter** is most suitable for large-scale use:
    - Concrete tank (5–6 m diameter).
    - Base: branching metallic pipes with strainers, leading to exit pipe.
    - Layers: graded gravel (60–90 cm) topped with sand (60–90 cm).
    - Water head: about 100 cm above sand, controlled by inlet valve.
- **Surface film (biological layer) forms at the top sand layer:**
    - Contains suspended matter, traces of alum, algae, protozoa, and bacteria.
    - Develops in 15–30 minutes with alum traces.
    - Becomes denser over time, slowing filtration, so requires periodic washing or sand replacement.
- **Rapid sand filters reduce bacterial content to about 5% of original load.**

4. Disinfection

- This is the last step in large-scale water purification.
- As indicated above, the water coming out of the rapid sand filter still contain bacteria and are not safe for human use unless it is disinfected.
- This is usually done by chlorine gas that is added to the water from big cylinders of liquid chlorine.
- A sufficient dose of 1 - 1.2 PPM. provides 0.1 - 0.2 PPM. of' residual chlorine after a contact time of 30 minutes.

Sanitary Standards of good drinking water:

1. **Physical:**

- Colorless, odorless, clear and tasteless.
- Drinking water should be free from turbidity and color.

2. **Chemical:**

- It should not be acid or strongly alkaline. It should be neutral or moderately alkaline at pH 8.00
- It should not contain an undue amount of solids. The maximum amount of solids should not exceed 500 parts per million.
- Hardness should not exceed 20 degrees of which not more than 4 degrees are permanent hardness.

  Hardness is defined as the soap-destroying power of water. It is classified into:

  1-Carbonate "temporary" is due to the presence of calcium and magnesium bicarbonates.

  2-Non-carbonate "permanent" is due to calcium and magnesium sulphates.
- Chemical indicators of organic matter contamination should conform to the following standards:
  - It should be free from nitrogenous organic matter.
  - Compound organic matter should not more than 0.1 PPM.
  - Albuminoid ammonia, another first product of organic decomposition should be less than 0.01 PPM.
  - The Oxygen absorbed in 3 hours at 27o C (to oxidize the organic matter in the water) should be less than 3 – 4 PP.
  - Nitrites: should **never** be present since they are intermediate products of organic oxidation and their presence indicates recent and still going on pollution.
  - Nitrates are final products of organic oxidation. Their presence in water indicates past pollution. They should not exceed 1.0 PPM. in surface water and 5.0 PPM. in deep well water.
  - Nitrates may be present in deep well in higher concentration due to very ancient organic matter in the soil as in case of nitrate soil in Chilli.

3. **Biological**

- **Bacteriological:**
    - Total bacterial count:100 organisms /ml
    - Presumptive coliform: absent in 100 ml of filtered chlorinated water.
    - Ideally drinking water should not contain any microorganisms known to be pathogenic.
    - The primary bacteria is the coliform group, including both fecal and non – fecal organisms. Typical example of the fecal group is E. coil and of the non – fecal group klebssiella aerogens.
    - Bacteria that indicates fecal contamination of water;
        - Echerichia Coli: normaly no coliform organism in 100 ml water.
        - Streptococcus faecalis: poorly survive in water and its presence indicates recent contamination with fecal matter.
        - Colstridium welchi: survives for long time and if present indicates remote contamination with fecal matter
- Drinking - water should be free from any **viruses** infectious to man.
- Drinking water should not contain any pathogenic intestinal **protozoa**. Rapid and slow sand filtration is effective in removing a high proportion of pathogenic intestinal protozoa.

4. **Radiological** -Drinking water should be free from any source of radiation.

SCHOOL HEALTH

By the end of the lecture the student will be able to:

- Recognize school health program
- Identify School health team
- Enumerate Duties of school medical officer
- List Common diseases among school children

School health program is program that promotes health through schools, when delivered through schools that are themselves organized to promote health are one of the essential means through which the twin goals of health for all and education for all can be achieved.

Importance of school health program

The health services to school children (5-15 yrs) is an economical yet powerful means of raising the community health because of the following reasons:

- School-children constitute nearly 20 to 25 percent of our population (Large number)
- They are nutritionally vulnerable.
- School age is a period of rapid growth and development.
- School age is a 'formative period', physically, mentally, socially and intellectually, transforming the school-child into a promising adult (A child of today is the citizen of tomorrow).
- Health of the country, depends upon the health of the children.
- The health habits formed at this stage will be carried to the adult age, old age and even to the next generation.

School health program goals:

The overall goal of a school health program is to provide a standard, healthy environment for all students.

A comprehensive, coordinated school health program helps each child achieve and maintain an optimal health condition so that maximal physical, emotional and intellectual growth can occur.

Objectives of school health

- **Health promotion:** To keep child physically fit and mentally alert to receive the education

- **Health protection:** To protect the child from various communicable and noncommunicable disease
- **Health restoration:** By early diagnosis by periodical screening and prompt treatment of diseases and their follow-up
- **Health education:** To create health consciousness among children and also to inculcate healthy habits and practices.
- **Healthy living**: In a healthy environment of school.

Major component of the school health program

- Health Education
- Physical Education
- Health Services
- Nutrition Services
- Health Promotion for Staff
- Counseling and Psychological Services
- Healthy School Environment
- Parent/Community Involvement

School health team.

The school health team includes;

- Physicians,
- nurses,
- dentists,
- psychologists,
- counselors,
- teachers,
- dietitians, and others.

School health services:

Health services include the procedures carried out by members of the school health team (physicians, nurses, dentists, psychologists, counselors, teachers, dietitians, and others) to appraise, promote and protect the health of every child in the school that total well-being of the child-physical, social, emotional and intellectual is the primary concern.

Certain activities that are usually provided to accomplish this goal include the following:

1- Continuous observation*:*

Teachers can help them to recognize health concerns in children that may go unrecognized by others.

2- Health appraisal:

1. ***Medical examination:***

At least four physical examinations should be carried out during the school life of a child.

The health examination should provide a history of the health status of the child and identify any health problems that the child might have.

2. ***Screening programs:***

A number of different health screening activities are conducted in the elementary and middle level schools.

The most commonly found screening activities are height and weight screening, urine and stool screening, vision and hearing screening.

3. ***Common school children immunizations:***

Common school children immunizations in Egypt are:

BCG: was given at first for children in the first primary year and 3rd preparatory then due to some complications after its administration and the argument regarding its efficiency it **was stopped.**

DT: for the first and fourth primary student.

Meningitis vaccination: for children in the preschool and 1st primary.

4. ***Communicable disease control:***

Control of the communicable diseases is the responsibility shared by the home, school and community.

Absence should be encouraged when children have even minor cold symptoms, since most childhood diseases have those same signs in the early stages.

The teacher is responsible for detecting and excluding any child from the classroom who has symptoms of a communicable disease

5. ***Dental services: Dental disease begins in childhood.***

Three basic forms of activity should be included in a school dental health program:

Dental health education.

Prevention measures.

Referral with follow through procedures in cases of discovered dental problems

6. ***Health records:***

Most schools maintain cumulative health folders as a means of steering and organizing all the data accumulated for each child during school years.

7. ***First aid and emergency care:***

First aid is skilled application of accepted principle treatment on the occurrence of an accident or in case of sudden illness.

The school has three responsibilities for emergency care and disaster control:

(1) to prevent injuries from happing in school;

(2) to provide safety education;

(3) to develop a plan for handling emergencies.

8. ***Mental health:***

Counseling and Psychological Services are services provided to improve students' mental, emotional, and social health.

These services include individual and group assessments, interventions, and referrals.

9. ***Health education:***

Health education in school setting is essentially important for helping children and youth develop the knowledge and skills they will need to avoid health risks and maintain good health throughout life.

10. ***Nutrition Services:***

Access to a variety of nutritious and appealing meals that accommodate the health and nutrition needs of all students.

11. ***Healthful School Environment:***

The importance of sanitary school environment on effective learning and providing comfort and educational facilities is also emphasized. Poor sanitation has its impact on health and health teaching as well.

Schools should have their own building design while residential buildings are not satisfactory.

Duties of school medical officer.

The physician

1. Medical examination and activities of health appraisal.
2. Attending school clinic, and school health units for diagnosis and management of morbidity, and referral for specialized service when needed.
3. First aid and emergency service.
4. Participation in prevention and control of communicable diseases, including specific protection by immunization and chemoprophylaxis when needed.
5. General inspection of school environment , to gave sanitary remarks and recommendations
6. Participation in school health education program: supervision of seminars, health references for teachers and other personnel.
7. supervising registration of health record data, and preparation of statistical indices and health reports.

Common diseases among school children

1. **Infectious diseases:**

Infectious diseases are responsible for a large proportion of child morbidity and mortality in developing countries.

These include acute respiratory tract infections, repeated diarrhea, parasitic diseases, etc.

The contributing factors are childhood malnutrition, defective immunization, environmental problems and other economic, political and logistic factors.

Prevention of infectious diseases is mainly by environmental sanitation, health education, immunization and good nutritional support.

a. **Respiratory infections:**

Respiratory infections are the most frequent occurring illness in childhood.

b. **Parasitic infestation:**

There are many intestinal parasites affecting school children.

These include Schistosoma , round worm, hook worm, entrobius ,amaebiasis and nana.

c. **Urinary tract infection:**

Approximately 3-5% of girls and 1% of boys acquire UTIs.

The prevalence of urinary tract infection in school children ranging from 0.4% in boys to 2.0% in girls (0 .4- 2.0%).

2. **Deficiency diseases:**

Nutritional disorders account for the majority of morbidity, mortality and dropouts among them. Anemia and vitamin A deficiency is the commonest among them.

3. **Skin disease:**

Low socioeconomic status, malnutrition, overcrowding, and poor standard of hygiene are important factors accounting for the distribution of skin diseases in school children in developing countries.

4. **Dental problems:**

Until very recently, it was not uncommon to see nearly two thirds of the school population affected to some degree by dental caries.

5. **Handicaps:**

Mainly arise from errors of refraction, defective hearing and heart diseases rheumatic valvular diseases or congenital is the most important

6. **Chronic illnesses:**

It was found that prevalence of chronic illness in school-aged children was up to 20%.
These chronic illnesses including diabetes millets and Epilepsy

7. **Accidents and Emergencies:**

At school children may be exposed to the risk of a variety of injuries as wounds, bleeding, fractures, broken teeth, and eye injuries and occasionally burns and others.
In the rood, wounds and traffic accidents

8. **Tobacco and drugs abuse**
9. **Reproductive health problems**

By the end of the lecture the student will be able to:

- **Define. adolescence period.**
- **Recognize adolescent health problems.**
- **Understand management and prevention of adolescent health problems.**
- **Describ Elements of adolescent health care program.**

1.1 Introduction

Adolescence is a unique developmental stage that bridges childhood and adulthood. It is characterized by rapid physical growth, emotional maturation, and expanding social roles. These changes influence health behaviors that often persist throughout life.

From a public health perspective, adolescence is a **window of opportunity** to establish healthy behaviors and prevent disease. Early intervention during this period can reduce morbidity, improve productivity, and enhance long-term quality of life.

1.2 Definition of Adolescence

Adolescence is commonly defined as the age period from **10 to 19 years**, according to international health classifications. It is often divided into three phases:

- **Early adolescence (10–13 years):** Onset of puberty and initial physical changes
- **Middle adolescence (14–16 years):** Emotional development and identity formation
- **Late adolescence (17–19 years):** Transition toward independence and adult roles

1.3 Characteristics of Adolescent Development

1.3.1 Physical Development

Adolescence is marked by rapid growth and sexual maturation. Key changes include:

- Growth spurts in height and weight
- Development of secondary sexual characteristics
- Hormonal changes affecting body composition
- Increased nutritional requirements

1.3.2 Psychological Development

Adolescents develop new cognitive abilities and emotional awareness:

- Abstract and critical thinking
- Identity formation and self-concept development

- Emotional variability and sensitivity
- Increasing autonomy and decision-making capacity

1.3.3 Social Development

Social relationships become more complex and influential:

- Greater importance of peer relationships
- Changing family dynamics
- Exploration of social roles and responsibilities
- Exposure to societal expectations and pressures

1.4 Importance of Adolescent Health

Maintaining health during adolescence is essential because:

- It establishes lifelong health behaviors
- It prevents early onset of chronic diseases
- It supports mental and emotional resilience
- It promotes educational achievement and productivity
- It reduces risky behaviors and social problems

Healthy adolescents are more likely to become healthy adults and contribute positively to their communities.

1.5 Determinants of Adolescent Health

Adolescent health is influenced by multiple interacting factors:

1.5.1 Biological Factors

- Genetic predisposition
- Pubertal timing and hormonal changes
- Nutritional status

1.5.2 Behavioral Factors

- Diet and physical activity
- Substance use
- Sleep patterns
- Risk-taking behaviors

1.5.3 Social and Environmental Factors

- Family support and communication
- School environment

- Peer influence
- Socioeconomic conditions
- Access to healthcare services

1.6 Nutritional Needs in Adolescence

Adolescents require increased intake of:

- Calories (growth spurt)
- Protein (tissue development)
- Iron (especially females)
- Calcium and vitamin D (bone growth)

Common problems:

- Obesity
- Under-nutrition
- Iron deficiency anemia

1.7 Mental Health in Adolescents

Mental health is a major component of adolescent well-being.

Common Mental Health Issues

- Anxiety disorders
- Depression
- Behavioral disorders
- Stress-related conditions

Risk Factors

- Academic pressure
- Family conflict
- Social isolation
- Digital media exposure

Stress Pathway in Adolescents

1.8 Health Promotion Strategies for Adolescents

Effective adolescent health programs should include:

- Health education and life-skills training
- Nutritional guidance and physical activity promotion
- Mental health support services
- School-based health programs
- Family and community engagement
- Accessible and youth-friendly healthcare services

1.9 Role of Public Health in Adolescent Well-Being

Public health systems play a major role in:

- Disease prevention and health promotion
- Early detection of health problems
- Policy development and protective legislation
- Community awareness and education
- Monitoring health trends among adolescents

1.1٠ Adolescent Vaccines in Egypt

Vaccination is a key preventive strategy to protect adolescents from infectious diseases and reduce long-term health complications. National immunization programs provide routine and recommended vaccines during adolescence.

National Immunization Oversight

Vaccination policies and implementation are coordinated by the Egyptian Ministry of Health and Population.

Routine and Recommended Adolescent Vaccines

Vaccine	Purpose
Tetanus and diphtheria booster (Td)	Maintains protection from childhood vaccination
Hepatitis B (if not completed earlier)	Prevents chronic liver disease
Measles, mumps, rubella (MMR booster if indicated)	Prevents viral infections
Meningococcal vaccine (risk-based)	Prevents meningitis
Human papillomavirus (HPV)*	Prevents cervical and other cancers

*HPV vaccination programs may vary by policy updates and availability.

Public Health Importance of Adolescent Vaccination

- Maintains immunity after childhood vaccines
- Prevents outbreaks in schools and communities
- Reduces long-term complications and mortality
- Supports herd immunity

Challenges in Adolescent Immunization

- Limited awareness among adolescents and parents
- Missed vaccination opportunities
- Access barriers in rural areas
- Vaccine hesitancy

Strategies to Improve Coverage

- School-based vaccination programs
- Public awareness campaigns
- Reminder and follow-up systems
- Integration with routine adolescent health visits

Key Message

Global adolescent health guidelines emphasize prevention, equity, and youth-friendly care, while national vaccination programs protect adolescents from preventable infectious diseases. Together, they form essential pillars of modern adolescent public health practice.

Table: Routine National Immunization Schedule — Egypt

A. Birth to 18 Months (Core Expanded Programme on Immunization)

Age	Vaccine	Protects Against
At birth (within 24 hours)	Hepatitis B (1st dose)	Hepatitis B infection
At birth	BCG	Tuberculosis
At birth	Oral polio vaccine (OPV zero dose)	Poliomyelitis
2 months	Pentavalent (DPT-HepB-Hib) + OPV + IPV	Diphtheria, pertussis, tetanus, hepatitis B, Haemophilus influenzae type b, polio
4 months	Pentavalent + OPV + IPV	Same as above
6 months	Pentavalent + OPV + IPV	Same as above
9 months	OPV booster	Poliomyelitis
12 months	MMR	Measles, mumps, rubella
18 months	DPT booster + OPV + MMR booster	Diphtheria, tetanus, pertussis, measles, mumps, rubella

Routine vaccination protects children from major infectious diseases and is delivered through primary health care units nationwide.

B. Preschool and School-Age Boosters

Age	Vaccine	Purpose
4–6 years (school entry)	DPT booster ± Polio	Maintain immunity
10–12 years (school health program)	Td or Tdap booster	Continued protection from tetanus and diphtheria
Adolescence (risk-based)	Meningococcal vaccine	Prevent meningitis outbreaks

Booster doses maintain immunity developed during infancy and early childhood.

C. Recommended (Non-routine / Risk-based / Program-dependent)

Vaccine	Target Group
Human papillomavirus (HPV)	Adolescent girls (program dependent)
Influenza	High-risk children and adolescents
Hepatitis A	Selected cases
Pneumococcal vaccine	Chronic disease or immunocompromised
COVID-19	According to national policy

These vaccines may be recommended depending on risk factors, campaigns, or policy updates.

Structure of National Immunization Across Childhood and Adolescence

Public Health Significance

- Protects against vaccine-preventable diseases such as tuberculosis, polio, and measles.
- Supports herd immunity and outbreak prevention.
- Delivered through thousands of primary care facilities nationwide.

1.11 Conclusion

Adolescence is a formative period that determines future health, productivity, and social stability. Addressing adolescent health needs requires a comprehensive approach that integrates physical, psychological, and social care. Strengthening adolescent health services and promoting healthy lifestyles during this stage is a fundamental investment in the future of society.

Communication and health education

By the end of the lecture the student will be able to:

- Define health education
- Identify objectives and key areas of health education.
- Explain communication components and methods used in health education.
- Discuss advantages and limitations of personal, mass, and community approaches.

Health education is a teaching process that provides basic knowledge of health and practice of proper health behaviors. It is basic item of any health program.

The definition adapted by National conference on preventive Medicine in USA, 1977 is *"health education is a process that **informs**, **motivates** and **helps people to adopt and maintain** healthy practices and lifestyles, **advocate** environmental changes as needed to facilitate this goal and conducts professional training and research to the some end"*

Objective of health education:

- **Promote healthy lifestyles** (nutrition, exercise, smoking cessation).
- **Prevent disease** through awareness of risk factors and screening.
- **Improve self-care** in chronic conditions (diabetes, hypertension, asthma).
- **Enhance compliance** with treatment and follow-up.
- **Empower families** to take responsibility for their health.
- **Support community health goals** (vaccination, maternal-child health, environmental safety).

Key Areas of health

- **Lifestyle modification**: Diet, exercise, alcohol, smoking.
- **Chronic disease management**: Diabetes, hypertension, asthma, COPD.
- **Maternal & child health**: Antenatal care, breastfeeding, immunization.
- **Mental health**: Stress management, suicide prevention, resilience building.
- **Accident prevention**: Road safety, home safety, occupational health.
- **Environmental health**: Clean water, sanitation, pollution awareness

Components of health education (elements of communications):

Education is essentially a matter of communication and the key elements in the communication process are:

1- **Communicator (educator or sender):** who is the originator of the message. A variety of personnel can give health education e.g. personal of health services, medical students, nurses, social worker, school personnel and community leaders.

2- **Message**: which must be in line with objectives, specific, precise clear, understandable and attract attention and interest of recipient.

3- **Recipients (audience):** the consumers of the messages may be individual, group or total population. For effective approach the educator must known personal and social characteristics of the recipients, attitude of recipient towards the message of education

4**- Channels of communication (Methods),** which must be carefully, selected particularly its cost and availability.

Functions of communications:

1. Establishes relationships between the persons involved in the communication act.
2. Imports new information and reinforces a person's knowledge
3. Direct the receiver in some way to change behavior, stimulate thought, provoke questions, entertain, reinforce attitude, change attitude.
4. Make the receiver adopt the idea as port of his regular behavior

Communication Barriers:

1- Social and culture gap between communicator and participants.

2- Limited receptiveness of participants.

3- Negative attitudes to communication.

4- Contradictory messages

5- Insufficient emphasis on education by the health professional.

Practice (Method of health education):

(1) **Personal approach:** Face to face education or education through spoken word. Both the educator and sender are found together, with the same place for talk and discussion.

A- Individual and family education:

Health education may be given in personal interviews. The topics for health counseling may be selected according to the relevance of situation. Health education may be given in:

1– Health appraisal (comprehensive medical exam, providing clinical services and screening tests).

- In prenatal visit.
- Target mother for family planning.
- Out reach program.
- Dots in case of TB.

2– Home visits by nurse, health work and social worker

B– Group health education:

it is an effective way of educating the community. The choice of the subject must relate directly to the interest of the group.

Method of group teaching:

- Lessons and lectures.
- Seminars (group discussions), which is a two-way communications.
- Panel discussions: group of 4-8 persons talk about the topic and discuss it in front of a group of audience.
- Symposium, which is a series of speeches on a selected subject.
- Workshop used in case of a novel experiment in education.
- Role-playing, the group members who enact their roles as they have observed or experienced them.
- Practical demonstration e.g. how to prepare ORS.
- Simulation exercises.

Advantages of personal approach:

1- More effective and impressive: educator directly faces the recipients.

2- Recipients participate in discussion (inter action between send & recipient)

3- Valuable to discuss confidential matters or problems e.g. family planning and birth control.

4- Personal approach can influence KAP (Knowledge, attitude and practice) of recipients.

Disadvantage:

It reaches a limited number of individuals, families, or confined groups.

(2) Education of general public (Mass communication):

The people are communicated for message of health education by different mass media e.g. broadcasting (radio and T.V.), written word (press as newspaper and magazines, booklets and pamphlets), cinema and theater, teaching aids (slides, film strips and scientific films) and posters. It is used for indirect teaching of certain topics and constant keeping them in the public eye e.g. avoidance of accidents and boosting immunization. Mass media can be classified into:

- Visual media: written words and posters.
- Auditory media: radio broadcasting.
- Audiovisual media. TV and cinema.

Advantages:

1- Reach millions of people everywhere in no time.

2- Suitable to provide knowledge and information, and discuss national health problems as, diarrheal diseases, malnutrition, drug abuse and drug dependence.

Disadvantages:

1- No Feed back from the recipient. It gives knowledge and instruction only.

2- Not necessarily influence and change attitude, practice and behavior of exposed recipients.

(3) Community participation:

Community non-governmental organizations (NGOS), leaders and influential groups participate in delivery of health services, especially primary health care. Health education is a particularly valuable function.

Use of campaigns:

Special health education campaigns are mounted to achieve a particular aim. e.g. use of safety belts in cars. Occasionally, shocks tactics are used to emphasize a serious subject e.g. demonstration of damaged cars exhibited beside main roads remind travelers about the result of road accidents.

Occupational health

By the end of the lecture the student will be able to :

- Identify concept and aim of occupational medicine
- Identify different health effects from different hazards
- Understand clinical features and management of these health effects
- Outline Important items of prevention and control programs for occupational health problems

Definition of occupational health:

World Health Organization (WHO), define occupational health as the promotion and maintenance of the highest degree of physical, mental and social well-being of workers in all occupations.

Aim of occupational health

- To increase the efficiency
- To increase the production
- To decrease the accidents

Objectives of occupational health

- To promote the health of the workers
- To maintain the highest degree of physical, mental and social well-being of the workers
- To prevent the diseases by elimination of factors which are inimical to their health.

Working (Occupational) Environment

It is the environmental conditions prevailing in the working place, which have a bearing or influence on the health of the worker, by three types of interactions. In other words there are three types of working environment .

- First one, 'man with machine
- Second one, 'man with environment
- Third one, 'man with man':

Role of Occupational Medicine

Most Common Activities of Occupational Medical Physicians Worldwide

1. Occupational health-risk/targeted screenings

2. Traditional occupational injury/illness management
3. Primary health care
4. Return-to-work examinations
5. Pre-employment examinations
6. Administrative duties
7. General health screenings
8. Workers' compensation
9. Health promotion/disease prevention
10. Evaluation of worksite environment
11. Health education
12. Disability management
13. Training/teaching/mentoring
14. Research
15. Emergency medical care
16. Safety assessments of workplace
17. Health benefits management
18. Evaluation of effects of worksite activities in external environment

Ergonomics

'Ergonomics' is derived from two Greek words 'Ergon' meaning work and 'Nomos' meaning principles or laws. Ergonomics is the Science of Work. Ergonomics means placing the worker in an environment (job), which is adapted to his physiological and psychological capacity. Ergonomics is essentially fitting the workplace to the worker. The better the fit the higher the level of safety and worker efficiency.

The main object of ergonomics is to achieve the best mutual adjustment between the man and the machine, which are complimentary to each other. It takes account of the worker's capabilities and limitations in seeking to ensure that tasks, equipment, information and the environment suit each worker.

The benefits of good Ergonomics are

- Improved health and safety by reducing work – related injuries and disorders;
- Improved comfort, morale and job satisfaction;

- Improved productivity and reduced workers' compensation costs and employee turnover.

Occupational health hazards

Depending upon the nature of the occupation, an industrial worker is exposed to one or more of the following five types of hazards

- Physical,
- Chemical,
- Biological,
- Mechanical and
- Psychosocial hazards),

1. **Physical hazards:** These are the hazards occurring due to following physical agents:

Health hazards due to exposure to heat: The effects are

- **Heat syncope:** It is fainting due to pooling of blood in lower limbs.
- **Heat cramps:** It is painful and spasmodic contractions of muscles due to loss of sodium and chloride
- **Heat exhaustion**: It means loss of salt leading on to circulatory failure.
- **Heat hyperpyrexia:** It is characterized by failure in heat regulating mechanism without the features of heat stroke. Temperature is about 106°F (40 °C). It may proceed to heat stroke.
- **Heat stroke:** There will be failure in the heat regulating mechanism, resulting in high temperature of the body, delirium, convulsions, partial or total loss of consciousness. Skin is dry and hot.
- **Prickly heat**
- **Burns, etc.**

Health hazards due to exposure to cold: The effects are

- Frost bite
- Reynaud's disease
- Erythromelalgia

- erythrocyanosis,
- chilblains
- trench-foot,
- gangrene, etc.

All because of cutaneous vasoconstriction.

Health hazards due to exposure to Light: The effects are

Effects occur either due to inadequate light or excessive light.

- **Inadequate light:**
  - Headache, eye strain, eye fatigue (visual fatigue).
  - Chronic effect is 'Miner's nystagmus'.
- **Excessive light:**
  - Bright light results in glaring, visual fatigue, blurring of vision, discomfort and accidents.

Health hazards due to exposure to Radiation: The effects are

1. Ionizing radiation: X-rays, radio-active isotopes like cobalt 60, phosphorus 32.

i. Acute effects: Acute radiation syndrome .

ii. Chronic effects:

- Somatic effects: Leukemia, aplastic anemia, cancer, tumor induction, pancytopenia.
- Genetic effects: Stillbirths, congenital defects, neonatal deaths, sex chromosome aneuploidy, sterility

2. Nonionizing radiation:

i. Ultraviolet radiation:

- On the skin: Darkening of the skin, thickening of the skin, erythema, cancer of the skin.
- On the eyes (Ex: Welding): Photophobia, conjunctivitis, keratitis, corneal ulcer, blindness (Welder's flash), snow blindness.

ii. Infrared radiation: Cataract

iii. Microwaves: Microwave sickness

iv. Laser radiations: Thermal burns, cataract, retinal burns.

Health hazards due to exposure to atmospheric pressure: The effects are

i. High atmospheric pressure (as in mines):

Caisson disease (compressed air illness), syncope, aches and pains in joints, air embolism, cardio-respiratory distress, paralysis.

ii. Low atmospheric pressure (as in high attitudes):

- Deafness (due to blocking of Eustachian tubes),
- pain in the ears,
- rupture of eardrums,
- expansion of gasses in the sinuses and
- body cavities,
- headache,
- pulmonary edema,
- dyspnea, etc.

Health hazards due to exposure to Electricity: The effects are

Electric shock, burns, ventricular flutter.

Health hazards due to exposure to vibration (working with pneumatic tools like drills and hammers): The effects are

- Numbness,
- white fingers,
- injuries.

Health hazards due to exposure to Noise: The effects are

Auditory effects:

- Deafness (temporary or permanent),
- tinnitus (buzzing in the ears)
- degeneration of cochlea and eighth nerve.

Nonauditory effects:

- Fatigue,

- irritability,
- nervousness,
- interference with speech and communication,
- annoyance,
- increased intracranial tension,
- hypertension,
- peptic ulcer,
- higher environmental stress.

2. **Chemical hazards:** These result from

➢ **Inhalants**: Dusts, fumes and gases.

- **Dusts**: Pneumoconioses.
- **Fumes**: Metal fume fever (This chemical intoxication results from inhalation of fumes of molten metals like arsenic, antimony, beryllium, cadmium, cobalt, lead, zinc, etc.)
- **Gasses**:
  - **Asphyxiating** gasses: CO, SO2, Cl2, H2S, methyl isocyanide gas, etc.
  - **Anesthetic** gasses: Ether, chloroform, trichloroethylene.

➢ **Irritants** (Such as dyes, acids and alkalis): Occupational dermatoses, e.g. dermatitis, folliculitis, eczema, ulcerations, cancer.

➢ **Ingestants:** Toxic hazards occurring from the metals like lead, arsenic, mercury, cadmium, manganese, chromium, etc.

➢ **Allergens**: Allergic rhinitis, bronchitis, asthma, dermatitis, urticaria, etc.

3. **Biological hazards**

These are from the animals and soil. These are common in agricultural industry.

- **From the animals:** They are called 'Zoonotic diseases'. E.g., Anthrax, bovine tuberculosis, salmonellosis, Japanese encephalitis, rabies, plague, Kyasanur forest disease, etc.
- **From the soil:** Tetanus, ankylostomiasis, gas-gangrene, malignant edema, anthrax, aspergillosis, coccidioidomycosis, mycetoma.

4. **Mechanical hazards**

   These are mainly accidents.

5. **Psychosocial hazards**

- These are due to failure of the worker to develop a healthy relationship with his co-workers, employers, management, supervisors, etc.
- They are divided into two groups.
- Psychological (Behavioral) changes:
    - Such as hostility, aggressiveness, anxiety, depression, frustration, tardiness, alcoholism, drug addiction, sickness absenteeism, etc.
- Psychosomatic ill health:
    - Such as neurosis, fatigue, propensity to peptic ulcer, hypertension, asthma, etc.

6. **Other Hazards**

- Occupational cancers:
    - Cancers of the skin, lungs, bladder and blood forming organs (i.e. Leukemia).
- Occupational dermatoses:
    - Dermatitis, eczema, folliculitis, urticaria, ulcers, etc.

Occupational diseases

Definition of Occupational diseases

Occupational diseases occur as a result of exposure to physical, chemical, biological or psychosocial factors in the workplace.

Occupational diseases are

- cause-specific,
- occurring mainly among working population,
- exposure at workplace is essential,
- notifiable and
- compensable.

However, some other diseases where work conditions or stress can play a role in their occurrence, these are **called work-related**.

These diseases are also frequently seen in the general community. In these diseases, work may be a factor in its causation or may aggravate a pre-existing condition.

Examples of work-related diseases are:

- Hypertension.
- Ischemic heart disease.
- Psychosomatic illness.
- Musculoskeletal disorders.

PNEUMOCONIOSES

(Pneumons = lungs; Konia = dust; Pneumoconiosis is singular).

These are also known as dust diseases. Pneumoconioses are a group of lung diseases occurring out of the specific occupation, caused by inhalation of insoluble dust, over a prolonged period of exposure.

Pathologically it is characterized by fibrosis of the lung parenchyma followed by its complications. Once a pathology, it is usually progressive, permanent, pulmonary pathology.

Clinically, it is characterized by persistent cough, progressive breathlessness, gradually cripples the person by reducing the working capacity due to fibrosis of the lungs followed by the features of complications such as tuberculosis, emphysema, chronic obstructive pulmonary disease (COPD), pulmonary hypertension, cor pulmonale and even carcinoma in some cases.

There is **no treatment or cure** for pneumoconioses. **Prevention** is the only intervention. Therefore, it is essential to prevent these diseases from arising.

Factors influencing pneumoconioses are:

- **Concentration of the dust in the air:** Higher the concentration, greater the health hazards. The permissible limit is 200 mcg per cu meter of air.
- **Composition of the dust:** More complicated the composition of the dust, greater the health hazard.
- **Size of the dust particles:**
    - This is the most important determining factor. Smaller the size of the dust particle, greater the tissue reaction.
        - Particles more than 10 μm, settle down on the ground and become soil dust.

- Between 10 and 5 μm, the particles are caught in the upper respiratory passage.
- Between 5 and 3 μm, they are caught in mid respiratory passage.
- Between 3 and 0.5 μm, they reach the alveoli and cause tissue reaction (fibrosis). They are most dangerous.
- Less than 0.1 μm, they are harmless because of Brownian movements they exhibit and they are absorbed.

➢ **Duration of exposure:**

- Longer the duration of exposure to the dust, greater the health hazard.
- Nearly 10 to 15 years of exposure is necessary for the development of tissue reaction.
- However, there are cases on record in which extensive fibrosis has occurred within 2 years following exposure to high concentration of dust.

➢ **Individual susceptibility (Health status):**

- Better the health and nutritional status, lesser the chances of development of pneumoconiosis early.

The common pneumoconioses

SILICOSIS

Silicosis is a nodular fibrosis of the lung due to inhalation of respirable dust containing free silica (Silicon dioxide).

Exposure: Workers in the following industries are exposed to silica dust:

- Mining.
- Glass industry.
- Pottery industry.
- Ceramic industry.
- Sandblasting and mineral polishing.
- Foundry workers.

Pathogenesis

- The inhaled dust is engulfed by macrophages.

- Because the silica is cytotoxic, the macrophages rupture and eliberate fibroblast attractive substances.
- This initiate fibrosis and formation of fine nodules that may coalesce together to from larger nodules.
- The hilar lymph nodes are enlarged with calcium deposition at the periphery giving the characteristic picture of eggshell calcification or snowstorm appearance in the X Ray .
- Egg shell appearance may also occurs in sarcoidosis and Tuberculosis.

Diagnosis of silicosis:

1-**Occupational history**: working in jobs entailing inhalation of silica dust for a period of 5-10 years.

2- **Clinical examination** demonstrates symptoms of lung fibrosis e.g. shortness of breath, dyspnea, easily fatigability.....etc cough, expectoration may be present in case of concomitant chronic bronchitis.

Local chest expectoration is not specific.

3- X ray of the lung shows nodular opacities of varying size more at the upper parts of the lungs. There may be hilar **eggshell** calcification or **snowstorm** appearance.

4- **Ventilatory function** tests show drop in forced vital capacity.

Complications:

1- Silicosis is a risk factor for TB infection

2- Progress to chronic obstructive pulmonary disease (COPD), emphysema, pulmonary hypertension, cor-pulmonale

3- Recent studies point to the relation between silica exposure and bronchogenic carcinoma.

4- Coplan's syndrome (i.e. Silicosis with rheumatoid arthritis).

Prevention

Follow the general rules. Note that:

1- **Environmental measures** are the most important line of prevention:

- Substitution of sand by coundum (Al2o3) as an artificial abrasive for sand blasting.
- Wetting the dust during preparation and shaping.

2- **Periodic medical examination** is by clinical examination and x ray each two years.

Asbestosis

Asbestos can induce different diseases in different sites of the body. So, it is called now Asbestos related diseases. Asbestos is imported from Canada, south Africa, Mexico and brazil

Occupational Exposure and uses in Egypt:

Asbestos is a unique natural fibrous material with the following advantages:

- Very resistant to friction so used in the manufacture of car brakes and clutch.
- Highly insulating so used in clothes and gloves for over workers and for insulation of roofs.
- Have a binding capacity so used in asbestos cement industry for manufacture of water pipes, ceiling tiles & roofs and walls.

Pathogenesis

- The fibers are cytotoxic to the macrophages.
- Rupture of the macrophages, liberate fibro-genic materials which induce the following:
- Diffuse fibrosis of the lung due to exposure to asbestos. The fibers may be surrounded by ferritin granules giving the so-called asbestos bodies. These are yellow brown in color, beaded and clubbed ended. They can be detected microscopically in the sputum of asbestos exposed workers. They are indicators of exposure to asbestos but not diagnose asbestosis.
- Interstitial fibrosis especially at the lower half of the lung.
- X-ray chest shows
    - ‘ground-glass’ or (honey – comb) appearance of lower two thirds of the lungs and heart shows ‘shaggy’ appearance (i.e. without clear-cut cardiac margins).
    - Pleural thickening
    - Pleural plaque: Cartilaginous thickening in the parietal pleura
    - Pleural effusion (rare)

Diagnosis of asbestosis:

1- Occupational history:

- history of exposure to asbestos fibers for a period of 5-10 years.

2- Clinical examination

- demonstrates symptoms of lung fibrosis e.g Shortness of breath, dyspnea, easily fatigability.. etc
- Osculation may reveal fine crepitations at the base of the lung due to fibrosis.

3- X ray shows:

- 'ground-glass' or (honey – comb) appearance of lower two thirds of the lungs and heart shows 'shaggy' appearance, tenting of the diaphragm, Pleural thickening, pleural effusion, pleural plaques.

4- Pulmonary function tests show

- drop in forced vital capacity

5- Sputum examination may reveal asbestos bodies.

- Other asbestos related diseases cancer larynx, and chronic Asthmatic bronchitis

Complications:

- Neoplasms: Common malignant lesion is bronchogenic carcinoma. Less common is malignant mesothelioma of pleura or peritoneum, often associated with effusion or ascites respectively
- Cor-pulmonale.
- Asthmatic bronchitis is common especially among workers of cement asbestos pipes.

Prevention:

The same lines of prevention applied with silicosis are applied here with workers exposed to asbestosis.

Bagassosis

Extrinsic allergic alveolitis due exposure to spores of the fungi thermoactinomyces sacchari found on the stored bagass in the wood and paper industry.

The **pathological** process

- begins by diseases inflammatory process in the form of granulomatous pneumonitis that occasionally undergoes organization leading to
- interstitial fibrosis and thickening of the alveolo-capillary membrane.

Clinical presentation: it is in the form of acute phase or chronic phase

Acute form:

- **Onset**: it occurs 4-6 hours after exposure.
- **Symptoms**:

- The symptoms are in the form of malaise, fever, chills, pains, unproductive cough, chest tightness and dyspnoea.
- These symptoms last from 12 hours up to a week.

- **Signs**:
  - fever, tachycardia and tachypnoea and basal crepitations.

Chronic form:

show clinical features as in other pneumoconiosis.

Investigations:

- X-ray of the chest shows 'mottling' appearance in lung fields
- Serology, pulmonary function tests, and lung biopsy.

Prevention:

The most important is the prevention of growth of the fungi on the hay or the bagasse by spraying 1% propionic acid. Otherwise follow the general rules.

Byssinosis

Asthma like manifestation occurs among workers exposed to cotton dust in textile industry in the wars of bale opening, carding combing, spinning and sometimes weaving.Although cotton industry is wide spread in Egypt, This disease is rare.

Diagnosis :

- history of exposure to cotton dust to more than 10 years.
- The worker suffers of breathlessness and chest tightness and sometimes cough on the day after a day off.
- Clinical examination: Picture of chronic obstructive airways disease.
- X ray is usually normal except some increase in bronchovascular markings.
- Pulmonary function tests show drop in forced expiratory volume especially at the shift end of the first working day after a holiday.

Treatment: as in asthma or asthmatic bronchitis.

Prevention: Follow general rules.

Mill fever

Occurs in new employees unaccustomed to dust although tolerance develops after few days of acclimatization.

Acute onset of fever, chills, cough and general malaise. They start after 5-6 hours from exposure and clear rapidly on removal of patient from the dusty environment.

Weaver's cough

Occurs in workers exposed to low grade wet cotton containing moulds and fungi. It affects all workers, old and new in an epidemic form, after 4-6 hours from exposure.

Mild dyspnea and persistent irritating cough with yellowish green sputum, acking limbs and back, headache, slight evening temperature and epistaxis in some cases.

Farmer's lung

This is a chronic disease of lungs, due to inhalation of mouldy hay or grain dust, in the agricultural fields. During winter season, the grain-dust with a moisture content of over 30 percent, bacteriae grow rapidly and produce heat of about 45°C. The heat encourages the growth of certain thermophilic actinomycete fungi of which Microspora (Micropolyspora) faeni is the main cause of Farmer's lung.

This condition is considered to be an allergic (hypersensitivity) reaction to the inhalation of antigen (dust) with the development of antibodies. The clinical features are the same as that of bronchial asthma (Eosinophilia) (Allergic bronchitis). First acute attacks are noticed. Repeated attacks cause pulmonary fibrosis and lung damage (cor pulmonale). X-ray chest shows 'fine nodular density'. Since agricultural industry is the major industry in our country, this is one of the major public health problems of our country

General Measures of Prevention & Control of Occupational Diseases

1. Engineering measures:

Engineering methods for prevention of occupational hazards are:

- **Substitution** : Substitution of hazardous substances or operation by non-hazardous ones or e.g.,
    - fiberglass in place of asbestos,
    - compression instead of hammering.

- **Isolation:**
  - isolation of the hazardous process inside the work places so that other surrounding workers are not affected e.g. isolation of radiation or noise sources.
- **Segregation:**
  - segregation of the hazardous process from the work place. e.g. incinerators.
- **Good housekeeping :** The golden rule is that there is a place for everything and **everything has a place.**
- **Good general ventilation :** It dilutes the substance in work environment.
- **Periodic environmental surveys :** These will help in detection of any leak at an early stage. Threshold limit value of a substance should not be exceeded.
- **Safe plant design :** Safety engineers should be involved in the construction of the plant. A positive approach has been found to be more effective.

2. Medical Control Measures

Medical examination

- **Pre placement medical examination:** It is done before one is employed in a particular job i.e. at the time of entry into an employment and It helps in rejecting the individuals who are unfit and also helps in placement of the worker to the right job so that there is maximum productivity and at the same time it is less detriment to worker's health. It also gives a baseline for various parameters which can be used for future comparisons.
- **Periodical medical examination:**
  - Early detection of health effects of hazardous exposure to be treated in its reversible stage.
  - Monitoring the efficiency of occupational heath services.
- **Health Education**

  Health education about the process, handling of raw material, correct use of protective devices has a lot of impact in prevention of occupational diseases.

  The aim is to assist the worker in his process of adjustment to the working, home and community environment.

  It also includes guiding the worker about various legislations available and social security schemes existing for the benefit of workers.

3. Personal Protective Equipment (PPE)

To prevent the penetration of the uncontrolled hazardous exposure in the body e.g. masks for dust or gases exposure, earmuffs of earplugs for noise exposure, leaded aprons for radiation exposure etc.

Note: protective devices must be suitable, comfortable, don't interfere with job continuity and periodically maintained otherwise it will not be used

4. Legislative Measures

In order to safeguard the health, safety and welfare of the industrial workers, laws have been framed in every country. E.g. new Egyptian Labor Law (2003)

Components of Occupational Health Services:

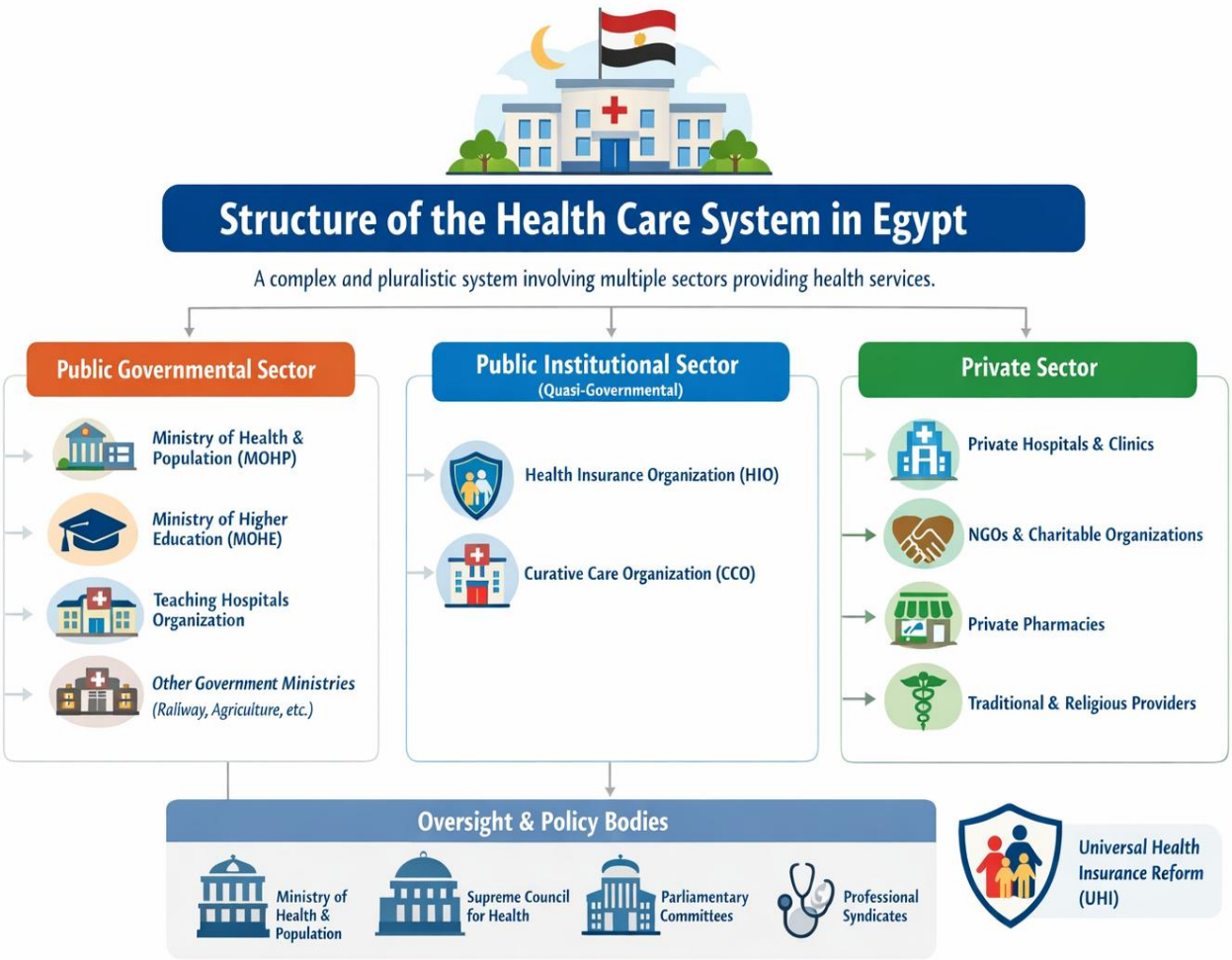
- **Medical:** carried by occupational physician(s), nurse(s), and technician(s).
  - Pre-employment medical examination.
  - Periodic medical examination.
  - First aid and management of emergencies.
  - Treatment of common diseases and causalities.
  - Health Education.
  - Care of workers nutrition.
  - Records and statistics.
  - Control of communicable diseases.
- **Environmental:** Carried by occupational hygienist(s) (engineer or chemist).
  - Periodic monitoring of different types of exposure (e.g. Air sampling, noise measurement) and assure that exposure at the work places don't exceed the allowable levels or concentration.
  - Advice how to ameliorate the working condition as illumination - ventilation - housekeeping etc....
- **Social worker specialized in work sociology science:**
  - Advice how to adopt and ameliorate the interpersonal relation-ship to be cooperative rather than competitive.
  - In developed countries psychological work stress is now the most important occupational exposure.

Health Care System in Egypt — Structure, Governance, and Challenges

By the end of the lecture the student will be able to :

- Identify the sectors of the Health System in Egypt
- Understand the organization of the Health System in Egypt
- Recognize levels of health care
- Enumerate examples of MOHP programs and health services

📘 Overview



A health system includes all organizations, institutions, resources, and people whose primary purpose is to promote, restore, or maintain health. According to the **World Health Organization**,

an effective health system requires workforce, financing, information, medical supplies, transport, communication, and governance to function properly.

The health care system in **Egypt** is complex, highly pluralistic, and administratively centralized, with multiple public and private providers and financing mechanisms. Service delivery is often fragmented, and integration across providers remains limited.

🏛️ Overall Structure of the Egyptian Health System

Health services are managed, financed, and delivered through three main sectors:

1. Public governmental sector
2. Public institutional (quasi-governmental) sector
3. Private sector

Each plays a distinct role in financing and providing care.

🏢 Public Sector

The public sector includes both governmental and institutional entities. Although they differ in operational autonomy, both function under state authority.

1️⃣ Public Governmental Sector

This sector consists of ministries and governmental bodies funded through the national budget, mainly via the **Ministry of Finance Egypt**.

Government health services operate as an integrated delivery system where financing and service provision occur within the same organizational structure.

Major governmental providers include:

- **Ministry of Health and Population Egypt**
- **Ministry of Higher Education Egypt**
- **Teaching Hospitals Organization Egypt**
- Other ministries providing health services for their employees (e.g., agriculture, education, transport)

Government facilities may also generate additional revenue through user fees in special service units ("economic departments"), known as self-funding.

2️⃣ Public Institutional (Quasi-Governmental) Sector

This sector includes semi-autonomous public organizations where government ministries retain major decision-making authority.

Key institutions include:

- **Health Insurance Organization Egypt**
- **Curative Care Organization Egypt**

Although these organizations operate with separate budgets and administrative rules, the Ministry of Health maintains strategic control.

Health Insurance Organization (HIO)

Established in 1964, HIO provides coverage to:

- Government employees
- Selected public and private workers
- Pensioners and widows
- Students and school children (Student Health Insurance Program, 1993)
- Newborns (expanded coverage by ministerial decree)

Funding sources include:

- Payroll-based contributions
- Government subsidies
- School fees
- Copayments from beneficiaries

🧭 Health System Reform and Universal Coverage

Egypt enacted major health reform legislation in 2018 to establish universal health insurance based on family medicine and primary health care.

Key principles include:

- Primary health care as the first level of contact
- Family doctor model
- Separation of financing, service provision, and accreditation
- National progress toward Universal Health Coverage (UHC)

The restructured system includes:

- Social Health Insurance Organization
- Public Provider Organization
- Accreditation and Monitoring Authority
- Continued oversight by the Ministry of Health and Population

The reform aims to ensure equitable access to quality services without financial hardship.

🏥 Private Sector

The private sector includes both for-profit and nonprofit providers, such as:

- Private clinics and hospitals
- Pharmacies
- Traditional midwives
- Nongovernmental organizations
- Religious and charitable health facilities

These institutions are registered with the Ministry of Health and Population and the Ministry of Social Affairs.

👩‍⚕️ Professional Syndicates

Professional syndicates provide health services through insurance-based schemes funded by:

- Per-capita contributions
- Beneficiary copayments

Services are limited by contractual ceilings on expenditures.

📜 Bodies Influencing Health Policy

Health legislation and policy are influenced by several advisory and legislative bodies, including:

- Parliamentary health committees
- Senate health committees
- **Supreme Council for Health Egypt**
- Professional associations

These bodies review and guide national health policy development.

⚠️ Major Challenges Facing the Egyptian Health System

The system faces multiple structural and operational challenges:

- Rapid population growth
- Dual burden of communicable and non-communicable diseases
- Limited public confidence in government facilities
- Migration of healthcare professionals due to low salaries
- Administrative bureaucracy

❗ Key System Weaknesses

Financing and Infrastructure

- Low public health expenditure (~3% of GDP)
- High out-of-pocket spending (>60% of total health expenditure)
- Deteriorated infrastructure in public hospitals
- Shortages of medicines and supplies

Human Resources and Governance

- Workforce shortages and migration
- Fragmented institutional structure
- Weak health insurance coverage
- Expansion of private sector dominance
- Underdeveloped health information systems

Administrative and Strategic Issues

- Complex organizational structure
- Over-centralization of authority
- Weak communication between departments
- Emphasis on curative over preventive care

✅ Strengths and Achievements

Despite challenges, Egypt's health system has demonstrated important progress:

- Over 95% of the population lives within 5 km of a health facility
- Expansion of family health and social insurance programs
- Strong medical education and workforce training capacity
- Local pharmaceutical industry supplying about 90% of medicines
- High immunization coverage and control of endemic diseases
- Established epidemiology and disease surveillance system
- Launch of the National Health Insurance project (2018) to expand coverage and improve efficiency

Maternal and Child Health (MCH)

Learning Objectives

By the end of this chapter, the reader should be able to:

1. Define maternal and child health and explain its public health importance.
2. Describe the concept of mother and child as a single health unit.
3. Identify components of maternal and child health care.
4. Explain stages of the maternity cycle.
5. Describe objectives and services of maternal health programs.
6. Recognize high-risk pregnancy and warning signs.

2.1 Introduction to Maternal and Child Health

Maternal and Child Health (MCH) is a major component of public health that focuses on the health of women during pregnancy, childbirth, and the postpartum period, as well as the health of infants and children.

MCH programs aim to ensure:

- Safe pregnancy and childbirth
- Healthy newborn survival
- Proper child growth and development
- Prevention of maternal and child mortality

These services are preventive, promotive, curative, and rehabilitative.

2.2 Mother and Child as a Single Health Entity

Mother and child are inseparable biologically, socially, and medically.

Reasons for their interdependence:

- Child health depends on maternal physical and nutritional status.
- Pregnancy outcome reflects maternal health.
- Some interventions protect both mother and child (e.g., tetanus immunization, nutrition).
- Both are at risk during childbirth.
- Postpartum care includes mother care, newborn care, and family planning.

2.3 Why Maternal and Child Health is a Priority

Mothers and children represent a highly vulnerable population because they experience:

- High mortality and morbidity

- Economic dependency
- Limited empowerment
- Increased exposure to preventable health risks

Most maternal and child deaths are preventable through cost-effective interventions.

2.4 Components of Maternal and Child Health Care

Maternal care spans the entire reproductive cycle.

Major Components:

1. Pre-conceptional care
2. Antenatal care
3. Intranatal care
4. Postnatal care
5. Inter-pregnancy care

The Maternity Cycle

Pre-conception → Pregnancy → Labour → Postpartum → Inter-pregnancy

2.5 Pre-Conceptional Care

Care provided before pregnancy to prepare women for healthy childbearing.

Objectives:

- Health promotion and proper nutrition
- Prevention of health hazards
- Early diagnosis and treatment of disease
- Physical and mental preparation of parents

2.6 Antenatal Care

Antenatal care begins after conception and continues until labour.

Purpose:

- Monitor pregnancy progress
- Detect complications early
- Provide health education
- Support maternal and fetal health

Services include:

- Clinical examination

- Laboratory investigations
- Immunization
- Nutritional supplementation
- Counseling

Objectives of Antenatal Care

- Maintain maternal health
- Detect high-risk pregnancies
- Provide self-care guidance
- Prepare for labour and breastfeeding
- Educate about danger signs
- Plan safe delivery

Recommended Antenatal Visits

Minimum four visits plus early registration.

2.7 First Antenatal Visit

Recommended at 8–12 weeks gestation.

Functions:

- Confirm pregnancy
- Determine expected date of delivery
- Screen for high-risk conditions
- Start supplements
- Provide immunization
- Health education

2.8 Identification of High-Risk Pregnancy

Maternal Risk Factors

- Age <18 or >35
- Multiparity
- Previous obstetric complications
- Chronic diseases
- Severe anemia
- Hypertension or diabetes

Pregnancy-Related Complications

- Bleeding
- Pre-eclampsia
- Multiple pregnancy
- Abnormal fetal presentation
- Preterm labour

2.9 Health Education During Pregnancy

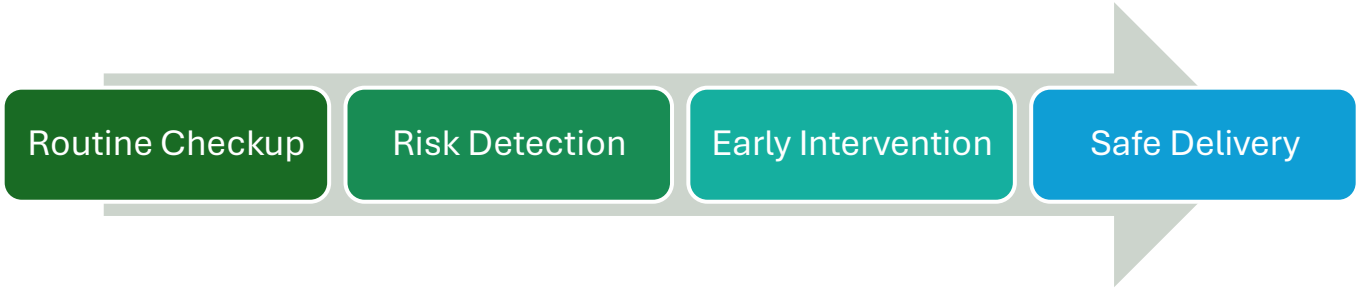
Important areas:

- Diet and rest
- Hygiene
- Exercise
- Breast care
- Weight monitoring
- Medication safety

Warning Signs Requiring Immediate Care

- Swelling of feet
- Convulsions
- Severe headache
- Visual disturbance
- Vaginal bleeding
- Severe abdominal pain

Pregnancy Monitoring Pathway



2.10 Intranatal Care (During Labour)

Objectives:

- Maintain asepsis
- Minimize injury

- Manage complications
- Ensure safe newborn care

The Five Cleans:

- Clean hands
- Clean surface
- Clean blade
- Clean cord
- Clean tie

2.11 Postpartum Care

Care provided after childbirth to ensure recovery of mother and health of newborn.

Objectives:

- Restore maternal health
- Prevent complications
- Support breastfeeding
- Provide family planning

Common Postpartum Complications

- Infection
- Hemorrhage
- Urinary tract infection
- Thrombophlebitis
- Psychiatric disorders

2.12 Inter-Pregnancy Care

Care between pregnancies to prepare for the next conception.

Objectives:

- Health promotion
- Adequate nutrition
- Birth spacing
- Family planning

Continuum of Maternal Care

Pre-pregnancy → Pregnancy → Delivery → Postpartum → Next Pregnancy

2.13 Objectives of Maternal Health Programs

- Reduce maternal mortality
- Ensure safe motherhood
- Improve newborn survival
- Provide reproductive health services
- Promote family planning

2.14 Services of Maternal Health Programs

- Antenatal clinics
- Skilled birth attendance
- Emergency obstetric care
- Postnatal follow-up
- Nutrition programs
- Health education

2.15 Summary

Maternal and child health is a cornerstone of public health. Because mother and child form a biologically connected unit, interventions must cover the entire reproductive cycle — from pre-conception to inter-pregnancy care. Effective MCH programs reduce mortality, prevent complications, and ensure healthy families and communities.

By the end of the lecture, students should be able to:

- Define management in different ways
- Identify the principles and main functions of management
- Recognize definition, leading, organizing a health team
- Identify planning health activities
- Understand implementing & evaluating health activities

DEFINITIONS OF MANAGEMENT:

- There are many definitions of management, here, we shall adopt the following definitions:

I- Management is getting things done:

Management "***is what one wants to do and then getting it done***". In other words, management "***first of all, be sure that objectives are well specified, and then that they are achieved***".

II- Management is getting things done through people

This definition does not mean only that a manager commands and people execute, although this happens in reality. Beside people (**m**anpower) as the very important resource, **m**aterials and **m**oney are so important and all of the three elements (**3ms**) compose what known as resources, without any element of them you c**an'**t achieve your objectives.

III- Management is the efficient use of resources:

- In health work, as in other human activities, the different types of resources used for achieving objectives must be carefully balanced and used efficiently.
- Efficiency "***is to do things right***".
- This means saving resources and accomplish your objectives by less resources. On the contrary, if things done wrong, the process will be repeated many times, and this will consume a lot of resources, for instance, malpractices in medicine will lead to loss of valuable resources in addition to bad health, psychological and social consequences.

IV- Management is to make decisions:

- A decision "***is a choice between two or more courses of action***".

- Decision should be **informed decision**; otherwise, things will go in wrong path.
- This means that all necessary information about the concerned issues should be accessible to decision-makers in order to make right decision.
- Main component of a system

Function of Management

1. Planning
2. Organizing
3. Leading / directing
4. Controlling

PLANNING

- Planning is a basic managerial function .It is setting goals and deciding how to best achieve them in advance.
- Planning is predetermining future and selecting appropriate goals and actions to achieve them.
- The process by which management set objectives, assess the future, and develop course of action to accomplish these objectives
- Planning requires decision making by all levels of managers

- Planning is also to decide in advance about what to do, how to do ,when to do and who is to do.
- A good planning is also required for good utilization of human and non-human resources to accomplish pre-determined goals.
- Planning requires decision making by all levels of managers
- Planning is also to decide in advance about what to do, how to do, when to do and who is to do.
- A good planning is also required for good utilization of human and non-human resources to accomplish pre-determined goals.
- Planning requires decision making by all levels of managers
- Planning is also to decide in advance about what to do, how to do, when to do and who is to do.
- A good planning is also required for good utilization of human and non-human resources to accomplish pre-determined goals.

ORGANIZING

- An important function of management.
- Also important for performing staffing ,directing and controlling functions.
- The process of arranging people and physical resources to carry out plans and accomplish the organizational goals.
- Its ongoing.

Organizing involves:

- ✓ Defining tasks required for achieving goals .What task to be done?
- ✓ Grouping the activities in logical pattern
- ✓ Determining manpower requirement
- ✓ Establishing authority and responsibility for each position .Who reports to whom?
- ✓ Assigning the activities to specific position and people
- ✓ Coordinating their activities authority relations
- ✓ Organizing efficiency and reducing the operation cost through avoiding repetition and duplication of activities.

LEADING

- Leading – A continuous process of setting objectives and trying to achieve them through the efforts of other people.
- Leadership is an important function of management.
- Leadership is guiding and influencing people to achieve goals willingly and enthusiastically in a given situation
- Leading – A continuous process of setting objectives and trying to achieve them through the efforts of other people.
- Leadership is an important function of management.
- Leadership is guiding and influencing people to achieve goals willingly and enthusiastically in a given situation.

CONTROLLING

- Controlling consists of actions and decision, manager undertaken to ensure actual result.
- It ensures the right thing is done in the right manner and at the right time.

The steps of controlling:

- Establishing standards
- Measuring actual performance
- Finding and analyzing deviations
- Corrective action

PRINCIPLES OF MANAGEMENT THAT WILL APPLY IN DIFFERENT SITUATIONS

⇒ Management by objectives

⇒ Learning from experience

⇒ Division of labor

⇒ Substitution of resources

⇒ Coordination of work activities

⇒ Functions determine structure

⇒ Delegation of authority

⇒ Management by exception

"**Management by Objectives**"

- Deciding and saying what to be accomplished is setting an objective) a goal ,a purpose ,an end ,a target .(There are many kinds of objective.
- The management principle that underlies the comparison of objectives with their achievement in order to judge effectiveness is known as“ **Learning from Experience**.”
- When there is a gap between objectives and results (or achievements), management analyzes why only the observed results were achieved and why fell short of the set objectives.
- Some causes can be easily remedied, and action is taken accordingly.
- Others cannot be removed in short term and are then called constraints.
- Management learns from this process and uses what it has learned in its further decisions for achieving its objectives. This process is sometimes called “feedback”.
- When work is divided, or distributed, among members of a group, and the work is directed and coordinated, the group becomes a team.
- In a team, and generally then there is specialization and division of labor, which each category of staff exercising its own skills towards achieving the objectives, management consists in assigning a balanced proportion of each kind of staff to the work to be done.
- The team approach is the way in which management attempts to bring about balance among the different members of the team and the work they do.

“Coordination of Activities” or “Convergence of work”

- Convergence of work means that the activities of the various people who do the work come together in the achievement of objectives
- The activities should be designed, assigned and directed in such a way that they support each other in moving towards a common goal.

  Also implies that working relations-they ways in which the members of a team interact with one another-should contribute to the success of each activity, and thus to general effectiveness.

“Substitute of Resources”

- Substitution means replacement
- One particular type of substitution of resources is labour substitution e.g. using trained ANM or volunteers for tasks formerly undertaken by professionals

“Functions Determine Structure”

- When work is clearly defined, i.e. the function and duties of individual members of the team are clearly defined and known to all, the working relations (the structure) follow.

"Delegation of Authority"

- Delegation takes place when someone with authority "lends" the authority to another person, conditionally or not, so as to enable that person to take responsibility when the need arises.
- Also ensure that the decision, once taken, is made known to all concerned. This is communication
- Decision should be communicated between those who make decisions, those who implement them, and the people affected by the decisions.
- "Shortest decision-path": deals with the issue: who should make which decision? And often when and where as well. Delegation of authority is the answer to clarify this
- In such way, decisions are made as close as possible in time and place to the object of the decision and to those affected by it.
- It saves time and work (e.g. in transmitting information) and also ensures that decisions can take full account of the circumstances which make the decisions necessary and in which they are put into effect.

"Management by Exception"

Management of exception means two things:

First: be selective. Do not become overloaded with routine and unnecessary information. Keep your mind available for critical information, on which manager will be required to act.

Second: make big decisions first. To be overloaded with petty decisions may result in more important ones being neglected or what has been called "postponing decisions until they become unnecessary."

In short, management by exception means selectivity in information and priority in decision.

IMPORTANCE OF MANAGEMENT

1. Optimum utilization of resources
2. Competitive strength
3. Cordial organizational relation

4. Motivation of employees
5. Introduction of new techniques
6. Effective management: society gets the benefits
7. Brings stability and prosperity
8. Develops team spirit
9. Ensures effective use of managers
10. Ensures smooth functioning (raises the efficiency, productivity and profitability
11. Reduces turnover and absenteeism
12. Creates sound organization

Health care team approach

Definition of health care team approach:

The provision of health services to individuals, families, and/or their communities by at least two health providers who work collaboratively with patients and their caregivers-to the extent preferred by each patient— to accomplish shared goals within and across settings to achieve coordinated, high- quality care.

- Team-based care uses multidisciplinary teams (which may involve new staff, or the shifting of tasks among existing staff).
- Teams can include physicians, and other allied health professionals, such as nurses, pharmacists, counsellors, social workers, nutritionists, community health workers, or others.
- Task shifting is the reassignment of clinical and non-clinical tasks from one level or type of health worker to another so that health services can be provided more efficiently or effectively.

For example, when medical officers are in short supply, some services can be effectively shifted to equipped and well trained non physicians such as clinical officers and nurses, while maintaining quality. This increases accessibility of health services to the community. Task shifting also can apply to laboratory functions, supply management, and pharmacy service

Advantages

- Expanded access to care (more hours of coverage, shorter wait times)
- Better patient support
- Team member collaboration

- Improved patient adherence to medications
- Better follow-up
- Improved patient knowledge
- Better quality of life
- Time saving for patient and health care team
- Cost efficient
- Improved patient and physician satisfaction

Barriers:

- Rapid staff turnover
- Retention of training
- Patient attitudes: perception by patients of being treated by non-physician health workers
- Physician attitude and reactions
- Legislation and policy

To achieve this potential, the transition to team-based primary care requires, for most practices, profound changes in the culture and organization of care, in the nature of interactions among colleagues and with patients, in education and training, and in the ways in which primary care personnel and patients understand their roles and responsibilities.

Requirements:

Policy decisions are usually made nationally, but there are methods that health center managers can use to help ensure successful implementation:

- Consult closely and coordinate with the physician.
- Train health care workers on new skills.
- Enable additional health workers to prescribe medications.
- Clearly define roles and responsibilities for different team members.
- Arrange close supervision, mentoring and support by experienced staff.
- Schedule regular clinical team meetings and good communications between staff to discuss patient cases and issues, so that they can work together to solve problems.

- Facilitate regular dialogue between staff about how to improve tasks in order to increase service efficiency and quality.
- Develop measurable processes and outcomes.

Implementation steps:

1. Engage the team
2. Determine the team composition
3. Design workflows to reflect the new model of care
4. Increase communication among the team, practice and patients
5. Use a gradual approach to implement the model
6. Optimize the care model

Engage the team:

Bring together a multi-disciplinary team of nurses, medical assistants (MAs), physicians, pharmacists, community health workers, nutritionists, administrators and information technology staff members with a leader who has enough authority within the practice or organization to empower the process. Consider involving patients on the team as well.

Determine the team composition

Design the model of care that will meet the needs of your patients and team. Consider which current team members could learn a new skillset and fulfil a new role. Your team may include a counsellor, nutritionist, nurse practitioner, physician assistant, clerical staff or others.

Ensure that the team(s) consists of physicians and supporting team members who are eager to transform the clinic into a team-based care model. They should be champions and good communicators who are willing to put in extra effort to prepare for the transition and continue to develop the new model once it is underway.

Design workflows to reflect the new model of care

- Determine your new team-based care workflows. Remember, you are creating your ideal future state, so think outside the box when designing your dream team and ideal practice. If certain aspects of your current workflow function well, feel free to incorporate them into your future state! Try not to limit yourself; consider how an already great process can be made better.

- You can design workflows for an entire clinical process (including pre-visit, visit, and post-visit), or for individual tasks that you want to assign to the team.

Increase communication among the team, practice and patients

- Start by keeping the practice aware of the team's work. Physicians and staff may feel out of the loop and disengage if they are not involved.
- Include the team's task-shifting work as a standing agenda item at team meetings and department gatherings.
- Broadcast updates in a weekly meeting or call.
- Have doctors share space with the rest of their team in a common workspace to support team communication

Use a gradual approach to implement the model

- Team-based care implementation will be a gradual process. It will take time, and every day will not be perfect.
- Be patient; know that several months may go by before the team feels like they are really comfortable with the new system.

Optimize the care model

Shared work space: Teams that sit in closer proximity communicate with greater frequency and ease. Questions can rapidly be answered, reducing the time that someone may have to wait before completing a task or responding to a patient. Everyone will be aware of the work that their teammates are doing, enabling easier task-sharing and division of work.

Communication management:

- In a team-based care model, the number of phone and email messages that are sent to the team should decrease for several reasons:
- Lab results are discussed during the visit, so the number of messages sent back and forth to discuss results or set up a call is significantly reduced.
- Patients receive additional education at the conclusion of their visit, resulting in fewer questions after the visit.

- Care coordination is enhanced. Patients will leave with their follow-up appointments, corresponding labs and diagnostics scheduled, so they should have fewer requests after leaving the office.
- Referrals to supportive services such as behavioral health or to a health educator can be made during the visit. Involving additional team members in a patient's care provides them with a point of contact for follow-up questions regarding these specific services.

References:

- **Community Medicine with Recent Advances, Third Edition: 2014**
- **ParkK's Text Book of Preventive and social Medicine 23rd edition**
- **Official website of World Health Organization**
- **Text book of public health and community medicine,First Edition,2009.**
- **Jekel's Epidemiology, Biostatistics, Preventive Medicine, and Public Health,Fourth Edition,2014.**
- **The Health system and Public Health. The Egyptian Health system, chapter 2. Pages: 7-11**
- **Occupational & Environmental Medicine, 4th Edition, Joseph Ladou**